Protocol

This trial protocol has been provided by the authors to give readers additional information about their work.

Protocol for: Fizazi K, Tran N, Fein L, et al. Abiraterone plus prednisone in metastatic, castration-sensitive prostate cancer. N Engl J Med 2017;376:352-60. DOI: 10.1056/NEJMoa1704174

This supplement contains the following items:

1. Original protocol, final protocol, summary of changes.
2. Statistical analysis plan (only one iteration)

Janssen Research & Development*

Clinical Protocol

A Randomized, Double-blind, Comparative Study of ZYTIGA® (Abiraterone Acetate) Plus Low-dose Prednisone Plus Androgen Deprivation Therapy (ADT) Versus ADT Alone in Newly Diagnosed Subjects With High-Risk, Metastatic Hormone-Naive Prostate Cancer (mHNPC)

Protocol 212082PCR3011; Phase 3

JNJ-212082 (abiraterone acetate)

*Janssen Research & Development is a global organization that operates through different legal entities in various countries. Therefore, the legal entity acting as the sponsor for Janssen Research & Development studies may vary, such as, but not limited to Janssen Biotech, Inc; Janssen Products, LP; Janssen Biologics, B.V.; Janssen-Cilag International N.V.; Janssen, Inc; Janssen Infectious Diseases BVBA; Janssen R&D Ireland; or Janssen Research & Development, L.L.C. The term "sponsor" is used throughout the protocol to represent these various legal entities; the sponsor is identified on the Contact Information page that accompanies the protocol.

EudraCT NUMBER: 2012-002940-26

Status:	Approved
Date:	31 July 2012
Prepared by:	Janssen Research & Development, LLC

Compliance: This study will be conducted in compliance with this protocol, Good Clinical Practice, and applicable regulatory requirements.

INVESTIGATOR AGREEMENT

I have read this protocol and agree that it contains all necessary details for carrying out this study. I will conduct the study as outlined herein and will complete the study within the time designated.

I will provide copies of the protocol and all pertinent information to all individuals responsible to me who assist in the conduct of this study. I will discuss this material with them to ensure that they are fully informed regarding the study drug, the conduct of the study, and the obligations of confidentiality.

Coordinating Investigator (where required):

Name (typed or printed): ____________________

Institution and Address: ____________________

Signature: ____________________ Date: ____________

(Day Month Year)

Principal (Site) Investigator:

Name (typed or printed): ____________________

Institution and Address: ____________________

Telephone Number: ____________________

Signature: ____________________ Date: ____________

(Day Month Year)

Sponsor's Responsible Medical Officer:

Name (typed or printed): Arturo Molina, MD, MS

Institution: [redacted]velopment, LLC

Signature: [redacted] Date: July 31, 2012

(Day Month Year)

Note: If the address or telephone number of the investigator changes during the course of the study, written notification will be provided by the investigator to the sponsor, and a protocol amendment will not be required.

TABLE OF CONTENTS

LIST OF ATTACHMENTS

Document No.: EDMS-ERI-40268906

SYNOPSIS

A Randomized, Double-blind, Comparative Study of ZYTIGA® (Abiraterone Acetate) Plus Low-Dose Prednisone Plus Androgen Deprivation Therapy (ADT) Versus ADT Alone in Newly Diagnosed Subjects With High-Risk, Metastatic Hormone-naive Prostate Cancer (mHNPC)

ZYTIGA® (abiraterone acetate) is a prodrug of abiraterone, an irreversible inhibitor of 17α hydroxylase/C17, 20-lyase (cytochrome P450c17 [CYP17]), a key enzyme required for testosterone synthesis. Marketing authorisation for ZYTIGA and prednisone/prednisolone was granted in April 2011 in the United States and in September 2011 in the European Union. Treatment with ZYTIGA improves survival in patients with metastatic castration-resistant prostate cancer. This study will test whether ZYTIGA also improves survival in patients with newly diagnosed metastatic, hormone-naive prostate cancer who are at an increased risk for rapid disease progression.

OBJECTIVE AND HYPOTHESIS

Primary Objective

The primary objective is to determine whether ZYTIGA in combination with low-dose prednisone and ADT is superior to ADT alone in improving survival in subjects with mHNPC with high-risk prognostic factors.

Secondary Objective

The secondary objectives are to evaluate the clinically relevant improvements as well as the safety of ZYTIGA plus low-dose prednisone and ADT compared to ADT alone and to identify microRNA (miRNA) and mRNA profiles predictive of ZYTIGA response or resistance in this patient population.

Hypothesis

The hypothesis is that addition of ZYTIGA to ADT as initial therapy (compared to ADT alone) will improve survival in men with newly-diagnosed, high-risk mHNPC.

OVERVIEW OF STUDY DESIGN

This is a multinational, multicenter, randomized, double-blind, active-controlled study designed to determine if newly diagnosed subjects with mHNPC who have high-risk prognostic factors will benefit from the addition of ZYTIGA and low-dose prednisone to ADT (LHRH agonists or orchiectomy). Approximately 1270 subjects will be enrolled in this study. The study population includes newly diagnosed (within 3 months prior to randomization) adult men with high-risk mHNPC (high-risk prognosis defined under Subject Selection). Subjects will be stratified by presence of visceral disease and ECOG performance grade of 0, 1 vs. 2 prior to randomization. Subjects must have distant metastatic disease as documented by positive bone scan or metastatic lesions on CT or MRI to be eligible. Eligible subjects may have received ADT or had an orchiectomy within 3 months of randomization. Subjects may also receive anti-androgens for up to 3 months prior to randomization but the continued use of anti-androgens is to be terminated before randomization. ZYTIGA and low-dose prednisone will be considered as study drugs.

The study will consist of a Screening Phase of up to 28 days before randomization to establish study eligibility and document baseline measurements; a Double-blind Treatment Phase; and a Follow-up Phase of up to 60 months to monitor survival status and subsequent prostate cancer therapy. Patient-reported outcomes (Euro-QoL [EQ-5D-5L], Brief Pain Inventory-Short Form [BPI-SF], Functional Assessment of Cancer Therapy-Prostate [FACT-P], and Brief Fatigue Inventory [BFI]) will be collected. Each cycle is 28 days. Treatment will continue until disease progression or the occurrence of unacceptable toxicity.

Subjects who meet all of the inclusion criteria and none of the exclusion criteria will be centrally randomized in a 1:1 ratio to the active group [ZYTIGA (1000 mg once daily) plus low-dose prednisone (5 mg once daily) plus ADT] or the control group [ADT plus placebos].

Two interim analyses in addition to the final analysis are planned for this study after observing 50% (421 events) and 65% (548 events) of the total number of required (842) events for the final analysis.

Subjects will be monitored for safety throughout the study. Adverse events including laboratory adverse events will be graded and summarized using National Cancer Institute Common Terminology Criteria for Adverse Events (NCI-CTCAE), Version 4.0. Dose modification guidelines are provided. An Independent Data Monitoring Committee (IDMC) will be commissioned for the study to perform regular safety review and at the planned interim analyses.

SUBJECT SELECTION

The study population includes newly diagnosed (within 3 months prior to randomization) men with mHNPC documented by positive bone scan or metastatic lesions on CT or MRI, ≥18 years of age, with high-risk prognostic factors, and ECOG performance grade of 0, 1 or 2. High-risk prognosis is defined as having 2 of the following 3 risk factors: (1) Gleason score of ≥8; (2) presence of 3 or more lesions on bone scan; (3) presence of visceral metastasis.

DOSAGE AND ADMINISTRATION

ZYTIGA 1000 mg (four 250 mg tablets) or placebo will be taken orally once daily continuously, concomitantly with oral low-dose prednisone 5 mg a day. ZYTIGA must be taken on an empty stomach. No food should be consumed for at least 2 hours before the dose of ZYTIGA/placebo is taken and for at least one hour after the dose of ZYTIGA/placebo is taken. Tablets should be swallowed whole with water.

All subjects will receive androgen deprivation therapy (LHRH agonists) or have had surgical castration. The choice of the LHRH agonist will be at the Investigator's discretion. Dosage will be consistent with product label and should not change during the study.

EFFICACY EVALUATIONS

Overall survival (OS) is defined as the time from randomization to date of death from any cause. Survival data will be collected throughout the Treatment Phase and during the Follow-up Phase. Once a subject has completed the Treatment Phase, OS follow-up will be performed every 4 months for up to 60 months (5 years) or study termination (whichever occurs first).

Patient reported outcomes will be measured utilizing the BPI-SF, EQ-5D-5L, FACT-P, and BFI and data will be collected throughout the study as shown in the Time and Events Schedule.

Medical resource usage data, associated with medical encounters, will be collected in the eCRF by the investigator and staff for all subjects throughout the study. For each adverse event that occurs, MRU information associated with that adverse event should be recorded whenever possible.

Efficacy Endpoints

Primary: Overall Survival

Secondary: rPFS adapted from PCWG2 criteria, time to next skeletal related event, time to PSA progression, and time to subsequent therapy for prostate cancer as well as time to initiation of chemotherapy

Exploratory: PSA response rate, Patient Related Outcome measures (EQ-5D-5L, BPI-SF, FACT-P, BFI), pain measures (time to pain progression), time to symptomatic local progression defined as occurrence of urethral obstruction or bladder outlet obstruction, and prostate cancer specific survival.

BIOMARKER EVALUATIONS

Biomarker analyses are designed to confirm previously identified markers predictive of response (or resistance) to ZYTIGA or correlated with poor prognosis. Formalin fixed paraffin embedded tumor samples obtained at the time of the original diagnosis will be collected from approximately 300 subjects to allow for molecular analyses. Serum samples will also be collected from approximately 300 subjects at multiple time points to allow for related analyses.

SAFETY EVALUATIONS

Safety evaluations include adverse events, vital signs measurements, physical examinations, and clinical laboratory tests.

STATISTICAL METHODS

The primary analysis population will use the intent-to-treat (ITT) population, which includes all randomized subjects. The ITT population will be used for the analysis of subject disposition and efficacy. The safety population includes all subjects who received at least 1 dose of study drug.

A sample size of approximately 1,270 subjects is planned.

Efficacy Analysis

Kaplan-Meier product limit method and Cox model will be used to estimate the time to event variables and to obtain the HR. Response rate variable will be evaluated using the Chi-square statistic.

Biomarkers Analysis

The associations of the above biomarkers with clinical response or survival endpoints will be assessed using appropriate statistical methods (analysis of variance [ANOVA], categorical, or survival model), depending on the endpoint. A detailed statistical analysis plan will be prepared for these exploratory studies.

Safety Analyses

The safety parameters to be evaluated are the incidence, intensity, and type of adverse events, clinically significant changes in the subject's physical examination findings, vital signs measurements, and clinical laboratory results. Exposure to study drug and reasons for discontinuation of study treatment will be tabulated.

TIME AND EVENTS SCHEDULE

PHASE		**Screening Phase**	**Treatment Phase**								**Follow-Up Phase**
CYCLE (28 days):	**Notes**	Time frame prior to randomization	Cycle 1		Cycle 2		Cycle 3		q1M from Cycle 4 to Cycle 13, then q2M until disease progression	End of Treatment (within 30 days of last dose)	Every 4 months up to 60 months or study termination
CYCLE DAY:	Study visits/procedures have a ±2-day time window. Each cycle is 28 days.		1	15	1	15	1	15	1		
Screening											
Informed consent		4 weeks									
Inclusion/exclusion criteria		4 weeks									
Prestudy anticancer therapy		4 weeks									
Medical history and demographics		4 weeks									
Randomization			X								
Study Drug Administration											
Dispense blinded study drug			X		X		X		X		
Dosing Compliance					X		X		X	X	
Safety											
Electrocardiogram, 12-lead	Serum potassium <3.5 mM should be corrected before ECG recording	4 weeks									
Physical examination	Height and weight recorded only at screening	2 weeks			X		X		X	X	
Vital signs	Blood pressure, heart rate, respiratory rate, and oral/aural body temperature at screening; afterwards only blood pressure	2 weeks	X		X		X		X	X	
Efficacy											
CT or MRI	The same modality (CT or MRI) should be used throughout the study. PET not acceptable. Imaging visits may occur up to 8 days before cycles requiring images. Unscheduled assessments if signs of disease progression are observed.	4 weeks (may use scan taken within 3 months of randomization if deemed evaluable by central reader) Central review of selected baseline scans.							q4M starting with Cycle 5	X	

PHASE		**Screening Phase**	**Treatment Phase**								**Follow-Up Phase**
CYCLE (28 days):	**Notes**	Time frame prior to randomization	Cycle 1		Cycle 2		Cycle 3		q1M from Cycle 4 to Cycle 13, then q2M until disease progression	End of Treatment (within 30 days of last dose)	Every 4 months up to 60 months or study termination
CYCLE DAY:	Study visits/procedures have a ±2-day time window. Each cycle is 28 days.		1	15	1	15	1	15	1		
Bone Scan	Unscheduled assessments if signs of disease progression are observed.	4 weeks (may use scan taken within 3 months of randomization if deemed evaluable by central reader)							q4M starting with Cycle 5	X	
ECOG		4 weeks	X		X		X		X	X	
PROs	EQ-5D-5L, BPI-SF, FACT-P, and BFI to be done before any other visit procedure.	2 days	X		X		X		Until radiographic or clinical progression	X	EQ-5D-5L only; every 4 months for a total of 12 months.
Medical Resource Utilization			X		X		X		X	X	
Survival status and subsequent therapy	May be obtained by telephone or chart review.										X
Clinical Laboratory											
Hematology	Hemoglobin,white blood cells including neutrophil count, platelet count	7 days			X		X		X	X	
Serum chemistry	Potassium, creatinine, fasting glucose, lactate dehydrogenase	7 days		X	X		X		X	X	
Liver function tests	ALT, AST, total bilirubin	7 days		X	X	X	X	X	X	X	
PSA		7 days	X		X		X		X		
Testosterone		7 days	X		X		X			X	
Ongoing Subject Review											
Concomitant therapy	*Continuous*										
Adverse events	*Continuous from time of Consent Form signature until 30 days after last study drug dose*										
Biomarkers											
FFPE tumor block for biomarker assessments	Samples obtained at time of original diagnosis from approximately 300 subjects		X								
Serum for microRNA profiling	Serum samples (8.5 mL) from approximately 300 subjects		X				X			X	

ABBREVIATIONS TO THE TIME AND EVENTS SCHEDULE:

ALT = alanine aminotransferase; AST = aspartate aminotransferase; BFI = Brief Fatigue Inventory; BPI-SF = Brief Pain Inventory - Short Form; CT = computed tomography; ECOG = Eastern Cooperative Oncology Group; EQ-5D-5L = Euro-QoL; FACT-P = Functional Assessment of Cancer Therapy-Prostate; FFPE = formalin fixed paraffin embedded; MRI = magnetic resonance imaging; PET = positron emission tomography; PROs = patient reported outcomes; PSA = prostate specific antigen; q1M = every month; q2M = every 2 months; q4M = every 4 months

ABBREVIATIONS

ADT	androgen deprivation therapy
CRF	case report form (paper or electronic as appropriate for this study)
DCF	data clarification form
DMC	Data Monitoring Committee
ECG	electrocardiogram
eDC	electronic data capture
GCP	Good Clinical Practice
HBsAg	hepatitis B surface antigen
HIV	human immunodeficiency virus
IAC	Interim Analysis Committee
ICH	International Conference on Harmonisation
IEC	Independent Ethics Committee
IHC	immunohistochemistry
IRB	Institutional Review Board
IVRS	interactive voice response system
IWRS	interactive web response system
LC-MS/MS	liquid chromatography/mass spectrometry/mass spectrometry
LFT	liver function test
LHRH	lutenizing hormone releasing hormone
mCRPC	metastatic castration-resistant prostate cancer
MedDRA	Medical Dictionary for Regulatory Activities
mHNPC	metastatic hormone-naive prostate cancer
MRU	medical resource utilization
PD	pharmacodynamic
PK	pharmacokinetic
PQC	Product Quality Complaint
PRO	patient-reported outcome(s)
USP	United States Pharmacopeia

1. INTRODUCTION

Prostate cancer is the most common noncutaneous cancer among men in Europe.[15] It is an androgen dependent disease and inhibition of testosterone is a key element in the control of prostate tumor growth. Of the 383,000 men in Europe diagnosed annually with prostate cancer,[15] approximately 15% to 30% present with metastatic disease.[16,17,9,12,27,31] On the contrary, patients presenting with newly diagnosed (M1) metastatic hormone-naive prostate cancer (mHNPC) account for approximately 4% of all prostate cancer diagnosis in the United States.[5] This disparity may result from differences in the use of PSA screening in different geographies.[33] Currently, the standard of care for patients with newly diagnosed mHNPC is androgen deprivation therapy (ADT), which consists of initiating a luteinizing hormone-releasing hormone (LHRH) agonist (medical castration) or less commonly, orchiectomy (surgical castration) with or without concurrent anti-androgens.[11] The Southwest Oncology Group (SWOG S8894) reported that 77% of men newly diagnosed with metastatic prostate cancer lived less than 5 years and only approximately 7% of men treated with hormonal therapy were alive at or after 10 years.[37] Several prognostic factors influence survival in M1 disease. Median overall survival has been reported to range from 13 months up to 75 months depending on the presence of high-risk prognostic features such as high PSA concentration at diagnosis, high Gleason score, increased volume of metastatic disease as well as the presence of bony symptoms.[21]

In this study, high-risk patients with shortest median OS will be selected based on parameters described previously. The high-risk prognostic factor of Gleason score ≥8 was selected based on data from the SWOG 9346 trial[13] which reported that it was a strong predictor for risk of death (hazard ratio 1.58 and $p < 0.0001$). Baseline PSA alone was not considered a factor for selection of the high risk patient group because it was not as predictive of survival in univariate and multivariate models compared with Gleason score. In addition, baseline PSA alone did not show high association with post-baseline PSA decreases to below 4 ng/mL, a level that has been shown to have survival benefit.[13]

The second and third high-risk prognostic factors are both related to high-volume disease (defined as 3 or more lesions by bone scan or involvement of viscera). A single-center study of 286 patients has reported overall survival was 3.1 years in those with high-volume disease compared to 7.8 years in those with low-volume disease.[21]

Because the reduction of testosterone to castrate levels has been shown to improve survival of prostate cancer patients, initiation of ADT is standard of care for patients with M1 disease. A large randomized study in 938 men as well as a systematic review in men with locally advanced or asymptomatic metastatic disease demonstrated improved overall survival in those treated early.[23,25] The same study also demonstrated that the risk of deferring treatment increases the risk of developing debilitating symptoms such as pathological fractures and spinal cord compressions.[23,1] However, ADT is not curative and patients eventually progress to castration-resistance requiring further therapy including chemotherapy. Possible mechanisms of resistance to conventional ADT include the persistence of androgen production despite medical or surgical castration resulting from adrenal sources of testosterone or the up-regulation of intra-tumor testosterone production.[42]

This study evaluates the effectiveness of adding ZYTIGA plus low-dose prednisone or prednisolone (hereafter referred to as low-dose prednisone) to ADT in men with newly diagnosed metastatic prostate cancer who are hormone-naive (mHNPC) and have a high-risk profile.

1.1. ZYTIGA® (Abiraterone Acetate)

ZYTIGA® (abiraterone acetate) is a prodrug of abiraterone, an irreversible inhibitor of 17α hydroxylase/C17, 20-lyase (cytochrome P450c17 [CYP17]), a key enzyme required for testosterone synthesis. This enzyme is found in the testes, adrenals, and prostate tumors.[4,22,28] The Marketing Authorisation for ZYTIGA and prednisone was granted in the United States in April 2011 and in the European Union in September 2011 for the treatment of metastatic castration-resistant prostate cancer in adult men whose disease has progressed on or after a docetaxel-based chemotherapy regimen.

For the most comprehensive nonclinical and clinical information regarding the efficacy and safety of ZYTIGA, refer to the latest version of the Investigator's Brochure for ZYTIGA. The term sponsor used throughout this document refers to the entities listed in the Contact Information page(s), which will be provided as a separate document.

1.2. Clinical Experience with ZYTIGA

Study COU-AA-301 was a Phase 3, multinational, randomized, double-blind, placebo-controlled study of oral ZYTIGA and oral prednisone in 1,195 subjects with mCRPC whose disease had progressed on or after 1 or 2 chemotherapy regimens, at least one of which contained docetaxel. The study conclusively demonstrated that further lowering testosterone concentrations below those achieved with standard therapy to suppress androgen production (LHRH agonists or orchiectomy) using CYP17 inhibition with ZYTIGA improves survival in patients with mCRPC.[6]

Study COU-AA-302 was a Phase 3, multinational, randomized, double-blind, placebo-controlled study of abiraterone acetate and oral prednisone in 1,088 asymptomatic or mildly symptomatic subjects with mCRPC who had not received chemotherapy. This study had co-primary endpoints of radiographic progression free survival (rPFS) and overall survival. Treatment with ZYTIGA plus prednisone decreased the risk of radiographic progression or death by 57% compared with placebo plus prednisone (HR=0.425; $p<0.0001$). The study met its nominal significance level (0.01) for the co-primary endpoint of rPFS by independent review. There was a 25% decrease in the risk of death in the ZYTIGA and prednisone group compared with the placebo plus prednisone group (HR=0.752; p=0.0097) when the IDMC unanimously recommended unblinding the treatment and allowing subjects in the placebo group to receive ZYTIGA. The median OS had not been reached for the ZYTIGA group and was 27.2 months in the placebo group. The safety profile was similar, although the duration of treatment was longer, to that observed with ZYTIGA plus prednisone in subjects in the post-docetaxel setting (COU-AA-301).

In healthy subjects after single dose administration of 1000 mg ZYTIGA, there is a substantial food effect and absorption of ZYTIGA increases greatly with increasing fat content of a meal (Study COU-AA-009). Compared to administration after an overnight fast, geometric mean maximum concentration (C_{max}) and the area under the concentration-time curve (AUC) of abiraterone increased approximately 7-fold and 5-fold, respectively, when administered following a low-fat meal (estimated 2% of calories from fat) and increased by approximately 17-fold and 10-fold, respectively, when administered following a high-fat meal (estimated 56% of calories from fat).

In the two large phase 3 randomized trials (COU-AA-301 and COU-AA-302), treatment with ZYTIGA and prednisone had an acceptable safety profile and resulted in a favorable benefit/risk ratio. The safety profile of ZYTIGA plus prednisone was distinct from that of cytotoxic agents. Adverse events usually did not interfere with administration of ZYTIGA. Toxicities associated with mineralocorticoid excess (hypertension, hypokalemia, and fluid retention) secondary to the mechanism of ZYTIGA action were amenable to medical management and resulted in infrequent dose interruptions, dose reductions, or treatment discontinuations. Liver function test (LFT) abnormalities, if they occurred, were managed with careful laboratory monitoring, treatment interruptions and retreatment only after return to baseline values. In the combined dataset of safety for studies COU-AA-301 and COU-AA-302, the most frequently reported adverse events were fatigue (43.8%), back pain (32.6%), nausea (28.4%), arthralgia (29.5%), constipation (26.1%), bone pain (24.2%), peripheral edema (26.0%), hot flush (20.6%) and diarrhea (20.5%). Most events were Grade 1 or 2 in severity.

1.3. Overall Rationale for the Study

ZYTIGA in combination with prednisone has been approved for the treatment of men with mCRPC who have received prior chemotherapy containing docetaxel. The efficacy and safety of ZYTIGA (1000 mg daily tablet dose) and prednisone (5 mg twice daily) therapy in these patients is established by the results of Study COU-AA-301 and COU-AA-302, both Phase 3, multinational, randomized, double-blind, placebo-controlled studies. Study COU-AA-301 was the first Phase 3 study to demonstrate that further lowering testosterone concentrations below that achieved with ADT using CYP17 inhibition with ZYTIGA improves survival in patients with mCRPC. The rationale for using ZYTIGA in patients with high-risk prognostic factors in mHNPC is based on the positive results of Study COU-AA-301[6] and COU-AA-302[32] and the unmet medical need for alternative treatment options for these patients. Additionally, two studies using ZYTIGA in addition to ADT in the neoadjuvant setting demonstrated higher rates of undetectable PSA and higher rates of complete pathologic response or near-complete pathologic response in patients undergoing prostatectomy for high risk localized prostate cancer, suggesting a potential therapeutic role for inhibiting extragonadal androgen synthesis prior to the emergence of castration-resistance.[38,7]

2. OBJECTIVES AND HYPOTHESIS

2.1. Objectives

Primary Objective

The primary objective is to determine whether ZYTIGA in combination with low-dose prednisone and ADT is superior to ADT alone in improving survival in subjects with mHNPC with high-risk prognostic factors.

Secondary Objective

The secondary objectives are

- To evaluate the clinically relevant improvements as well as the safety of ZYTIGA plus low-dose prednisone and ADT compared to ADT alone.
- To identify microRNA (miRNA) and mRNA profiles predictive of ZYTIGA response or resistance.

2.2. Hypothesis

The hypothesis is that addition of ZYTIGA to ADT as initial therapy (compared to ADT alone) will improve survival in men with newly-diagnosed, high-risk mHNPC.

3. STUDY DESIGN AND RATIONALE

3.1. Overview of Study Design

This is a multinational, randomized, double-blind, active-controlled study designed to determine if newly diagnosed subjects with mHNPC who have high-risk prognostic factors will benefit from the addition of ZYTIGA and low-dose prednisone to ADT (LHRH agonists or orchiectomy). Approximately 1270 subjects will be enrolled in this study. The study population includes newly diagnosed (within 3 months prior to randomization) adult men with high-risk mHNPC (high-risk prognosis defined in Section 4.1). Subjects will be stratified by presence of visceral disease (yes/no) and Eastern Cooperative Oncology Group (ECOG) performance grade (0, 1, versus 2) prior to randomization. Subjects must have distant metastatic disease as documented by positive bone scan or metastatic lesions on computed tomography (CT) or magnetic resonance imaging (MRI) to be eligible. Eligible subjects may have received ADT or had an orchiectomy within 3 months of randomization. Subjects may also receive an anti-androgen for up to 3 months prior to randomization but the continued use of the anti-androgen is to be terminated before randomization for subjects receiving an LHRH agonist.

If the LHRH agonist is not started until subjects are randomized into the study, then investigators should initiate an anti-androgen shortly before or at the start of an LHRH agonist and continue its use for at least 7 days and up to 2 weeks after the start of an LHRH agonist.[26,11] This is to address the tumor flare that may be associated with the initiation of LHRH agonist. ZYTIGA and low-dose prednisone will be considered as study drugs.

The study will consist of a Screening Phase of up to 28 days before randomization to establish study eligibility and document baseline measurements; a Double-blind Treatment Phase; and a Follow-up Phase of up to 60 months to monitor survival status and subsequent prostate cancer therapy. Patient-reported outcomes (Euro-QoL [EQ-5D-5L], Functional Assessment of Cancer Therapy-Prostate [FACT-P], Brief Pain Inventory - Short Form [BPI-SF], and Brief Fatigue Inventory [BFI]) will be collected. Each cycle is 28 days. Treatment will continue until disease progression or the occurrence of unacceptable toxicity (see Section 10).

Subjects who meet all of the inclusion criteria and none of the exclusion criteria will be centrally randomized in a 1:1 ratio to the active group [ZYTIGA (1000 mg once daily) plus low-dose prednisone (5 mg once daily) plus ADT] or control group [ADT plus placebos]. Selection of ADT is Investigator's choice provided that the dosing (dose and frequency of administration) is consistent with product label. Each subject will be reviewed by the Sponsor before randomization to ensure that the eligibility criteria have been met.

Two interim analyses in addition to the final analysis are planned for this study after observing 50% (421 events) and 65% (548 events) of the total number of required (842) events for the final analysis.

Formalin fixed paraffin embedded (FFPE) tumor samples collected at the time of diagnosis, and serum samples collected at multiple time points during the study, will be obtained from approximately 300 subjects for biomarker evaluations. FFPE tumor samples will be used for biomarker studies designed to confirm previously identified predictive biomarkers and prognostic markers. These results will be used to inform the development of similar anti-androgen therapies and to identify patients at higher risk of progression for selection in future studies.

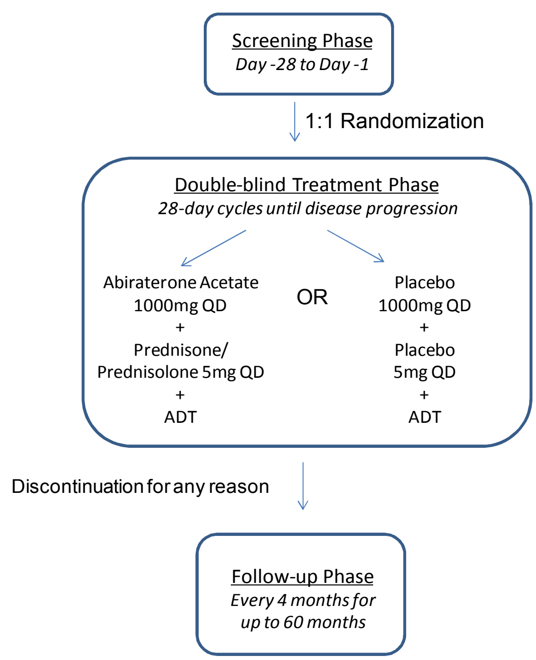
Subjects will be monitored for safety throughout the study. Adverse events (AEs) including laboratory adverse events will be graded and summarized using National Cancer Institute Common Terminology Criteria for Adverse Events (NCI-CTCAE), Version 4.0. Dose modifications will be made as required according to dose modification rules. An Independent

Data Monitoring Committee (IDMC) will be commissioned for the study to perform regular review of the safety data and at the planned interim analyses.

In the event of a positive study result at either of the interim analyses or at the final analysis, all subjects will have the opportunity to enroll in an open-label Extension Phase of this protocol. The Extension Phase will allow subjects to receive active drug (ZYTIGA plus prednisone) until ZYTIGA becomes commercially available in the subject's country. Subjects will be monitored for long-term survival and serious adverse events. Subjects who were previously receiving placebo in the Treatment Phase will have liver function and potassium monitoring completed per label recommendations for the first 3 months of ZYTIGA dosing in the Extension Phase. Those subjects who were previously receiving ZYTIGA will forego the additional liver function monitoring

A study flowchart is provided in **Figure 1**.

Figure 1: Study Flowchart



ADT=androgen deprivation therapy (may include surgical castration); QD=once a day

3.2. Study Design Rationale

ZYTIGA in combination with prednisone has been approved for the treatment of men with mCRPC who have received prior chemotherapy containing docetaxel. The efficacy and safety of ZYTIGA (1000 mg daily tablet dose) and prednisone (10 mg daily) therapy in these patients is established by the results of Study COU-AA-301, a Phase 3, multinational, randomized, double-blind, placebo-controlled study that demonstrated that further lowering testosterone concentrations below that achieved with ADT using CYP17 inhibition with ZYTIGA improves survival in patients with mCRPC. In addition, results have been reported from two randomized phase 2 trials conducted in the neoadjuvant setting before a prostectomy was performed in patients with high-risk localized disease. In these studies, the addition of ZYTIGA to ADT compared to ADT alone demonstrated an improved PSA response, a higher rate of undetectable PSA concentration, and a higher rate of complete or near complete pathologic responses.[38,7] These results support the addition of ZYTIGA to ADT in patients with hormone-naive disease, prior to the emergence of castration-resistance. Both studies used once-daily prednisone 5 mg instead of twice-daily as in Study COU-AA-301 and the lower prednisone dose sufficiently managed mineralocorticoid side effects. Inhibition of androgen biosynthesis with ZYTIGA and prednisone is predicted to improve overall survival among patients with high-risk prognostic factors in mHNPC. Finally, the use of continuous ADT is supported by recent data from men with newly diagnosed hormone-sensitive metastatic prostate cancer presented by Hussain et al. In this large trial of 1535 eligible patients who were randomized to receive intermittent ADT or continuous ADT, intermittent ADT was not proven to be noninferior to continuous ADT.[14]

Blinding, Control, Study Phase/Periods, Treatment Groups

This is a randomized, double-blind, active-controlled study. Randomization will be used to minimize bias in the assignment of subjects to treatment groups, to increase the likelihood that known and unknown subject attributes (eg, demographic and baseline characteristics) are evenly balanced across treatment groups, and to enhance the validity of statistical comparisons across treatment groups. Blinding will also enhance the validity of PRO data.

Biomarker Collection

The goal of the biomarker study is to confirm biomarkers from other studies that are predictive of response to ZYTIGA. The markers will be used in the development of future studies of anti-androgen therapies. The second goal is to confirm prognostic biomarkers from previous reports that were correlated with poor clinical outcome. These latter biomarkers are important for selection of “high risk” patients with earlier disease stage who could be treated in future clinical trials.

Patient Reported Outcomes (PROs)

The goal of collecting patient reported outcomes data is to explore patient benefits in pain, fatigue, and functional status. In both Studies COU-AA-301 and COU-AA-302, significant delay in progression for pain and functional status was found for ZYTIGA plus prednisone compared with prednisone plus placebo. Patient reported outcomes will be measured utilizing the EuroQol EQ-5D-5L, Brief Pain Inventory-Short Form (BPI-SF), Functional Assessment of Cancer Therapy–Prostate (FACT-P), and Brief Fatigue Inventory (BFI). Data will be collected throughout the study as well as during the Follow-up Period (up to 12 months) as specified in the Time and Events Schedule.

Medical Resource Utilization and Health Economics Data Collection

Medical resource usage data (MRU), associated with medical encounters, will be collected in the eCRF by the investigator and staff for all subjects throughout the study. For each adverse event that occurs, MRU information associated with that adverse event should be recorded when possible. Protocol-mandated procedures, tests, and encounters are excluded. The following data may be used to conduct exploratory economic analyses:

- Number and duration of medical care encounters, including surgeries, and other selected procedures (inpatient and outpatient)
- Duration of hospitalization (total days length of stay, including duration by wards; eg, intensive care unit)
- Number and character of diagnostic and therapeutic tests and procedures
- Outpatient medical encounters and treatments (including physician or emergency room visits, tests and procedures, and medications)

4. SUBJECT SELECTION

The inclusion and exclusion criteria for enrolling subjects in this study are described in the following 2 subsections.

4.1. Inclusion Criteria

Each potential subject must satisfy all of the following criteria to be enrolled in the study:

1. Willing and able to provide written informed consent
2. Men age 18 years and older
3. Newly diagnosed metastatic prostate cancer within 3 months prior to randomization with histologically or cytologically confirmed adenocarcinoma of the prostate without neuroendocrine differentiation or small cell histology.
4. Distant metastatic disease documented by positive bone scan or metastatic lesions on CT or MRI.
5. Two of the following high-risk prognostic factors:
   - Gleason score of ≥8
   - Presence of 3 or more lesions on bone scan
   - Presence of measurable visceral (excluding lymph node disease) metastasis (RECIST 1.1)
6. Eastern Cooperative Oncology Group (ECOG) performance status grade of 0, 1 or 2 (Attachment 1)

7. Adequate hematologic, hepatic, and renal function:
   - hemoglobin ≥9.0 g/dL independent of transfusions
   - neutrophils ≥1.5 x 10^9/L
   - platelets ≥100 x 10^9/L
   - total bilirubin ≤1.5X upper limit of normal (ULN) [except for subjects with documented Gilbert's disease in which case total bilirubin not to exceed 10X ULN]
   - alanine (ALT) and aspartate (AST) aminotransferase ≤2.5X ULN
   - serum creatinine <1.5X ULN or calculated creatinine clearance ≥50 mL/min
   - serum potassium ≥3.5 mM
8. Ability to swallow study medication tablets
9. Agrees to use a condom and another effective method of birth control if he is having sex with a woman of childbearing potential or agrees to use a condom if he is having sex with a woman who is pregnant

4.2. Exclusion Criteria

Subjects who meet any of the following criteria will be excluded from participating in the study:

1. Active infection or other medical condition that would make prednisone use contraindicated
2. Any chronic medical condition requiring a higher systemic dose of corticosteroid than 5 mg prednisone per day
3. Pathological finding consistent with small cell carcinoma of the prostate
4. Known brain metastasis
5. Any prior pharmacotherapy, radiation therapy, or surgery with curative intent for metastatic prostate cancer. The following exception is allowed:
   - Up to 3 months of ADT with LHRH agonists or orchiectomy with or without concurrent anti-androgens prior to subjects' randomization is permitted
   - Subjects may have one course of palliative radiation or surgical therapy to treat symptoms resulting from metastatic disease (e.g., impending cord compression or obstructive symptoms) if it was administered prior to randomization. Radiation or surgical therapy could not have been initiated 4 weeks after the start of ADT or orchiectomy.
6. Uncontrolled hypertension (systolic blood pressure ≥160 mmHg or diastolic BP ≥95 mmHg). Subjects with a history of hypertension are allowed provided blood pressure is controlled by anti-hypertensive treatment
7. Active or symptomatic viral hepatitis or chronic liver disease

8. History of adrenal dysfunction

9. Clinically significant heart disease as evidenced by myocardial infarction, or arterial thrombotic events in the past 6 months, severe or unstable angina, or New York Heart Association (NYHA) Class II-IV heart disease or cardiac ejection fraction measurement of <50% at baseline (Attachment 2)

10. Atrial fibrillation, or other cardiac arrhythmia requiring pharmacotherapy

11. Other malignancy (within 5 years), except non-melanoma skin cancer

12. Administration of an investigational therapeutic or invasive surgical procedure (not including surgical castration) within 30 days of Cycle 1 Day 1 or currently enrolled in an investigational study

13. Any condition or situation which, in the opinion of the investigator, would put the subject at risk, may confound study results, or interfere with the subject's participation in this study

NOTE: Each subject will be reviewed by Sponsor before randomization to ensure that the eligibility criteria have been met.

5. TREATMENT ALLOCATION AND BLINDING

Treatment Allocation

Subjects will be randomly assigned to the active or control group in a 1:1 ratio after the investigator has verified that all eligibility criteria have been met. Randomization will take place across all study sites using a centralized Interactive Web/Voice Response System (IWRS/IVRS). A country by country randomization scheme will be performed using permuted block randomization. Subjects will be stratified by presence of visceral disease and ECOG performance grade.

At randomization, the IWRS/IVRS will assign a unique subject identification number to each subject. The subject's identification number will be used on all study-related documents including electronic case report forms (eCRFs). A treatment number will also be assigned to each subject. This treatment number is the link between a subject's eCRF and blinded treatment group assignment. Subject identification numbers will not be reused. Subjects withdrawn from the study will not be replaced. All subjects must commence treatment within 72 hours (3 calendar days) of randomization.

Blinding

The investigator will not be provided with randomization codes. The codes will be maintained within the IWRS/IVRS, which has the functionality to allow the investigator to break the blind for an individual subject.

- Under normal circumstances, the blind should not be broken until completion of the study or the IDMC recommendation for unblinding is accepted by the Sponsor.
- Unblinding can be performed if the investigator determines that specific emergency treatment/course of action would be dictated by knowing the treatment status of the subject. In such cases, the investigator may in an emergency determine the identity of the treatment by contacting the IWRS/IVRS. It is required that the investigator contact the sponsor or its designee to discuss the particular situation, before breaking the blind. Telephone contact with the sponsor or its designee will be available 24 hours per day, 7 days per week. In the event the blind is broken, the sponsor must be informed as soon as possible. The date, time, and reason for unblinding must be documented by the IWRS/IVRS, in the appropriate section of the eCRF, and in the source document. The documentation received from the IWRS/IVRS indicating the code break must be retained with the subject's source documents in a secure manner. Subjects who have had their treatment assignment unblinded should be discontinued from the Double-blind Treatment Phase and entered in the Follow-up Phase.
- Unblinding will be performed in the event that the IDMC recommends unblinding and subjects are to be crossed-over to receive ZYTIGA.
- Unblinding may be performed if the subject discontinues from the study because of disease progression and the investigator feels this information is necessary to determine the next course of therapy.
- In general, randomization codes will be disclosed fully only when the study is completed and the clinical database is closed.

6. DOSAGE AND ADMINISTRATION

6.1. Planned Dose

ZYTIGA 1000 mg (four 250 mg tablets) or matching placebo should be taken orally once daily, concomitantly with oral low-dose prednisone 5mg or prednisone placebo. Placebo will be provided as a tablet formulation and will be matched in size, color, and shape to ZYTIGA and prednisone to maintain the study blind. Tablets should be swallowed whole with water. ZYTIGA must be taken on an empty stomach. No food should be consumed for at least 2 hours before the dose of ZYTIGA/placebo is taken and for at least one hour after the dose of ZYTIGA/placebo is taken. If a ZYTIGA or placebo dose is missed, it should be omitted and will not be made up.

All subjects will receive and remain on a stable regimen of androgen deprivation therapy (LHRH agonists) or have had surgical castration. The choice of the ADT will be at the Investigator's discretion and is to be selected prior to randomization. Dosing (dose and frequency of administration) will be consistent with product label and should not change during the study.

6.2. Dose Modification and Management of Toxicity

In clinical studies in subjects with mCRPC, the most common adverse events related to ZYTIGA monotherapy have included fatigue most likely attributable to the underlying disease; and hypertension, fluid retention/edema, and hypokalemia due to mineralocorticoid excess caused by compensatory ACTH drive. In this study, low-dose prednisone is expected to mitigate these effects through abrogation of the ACTH drive. Dose increase from 5 to 10 mg/day is permitted if hypokalemia persists despite optimal potassium supplementation and adequate oral intake, or if any of the other mineralocorticoid effects persist (see Section 6.4).

Following prolonged therapy with corticosteroids, subjects may develop Cushings syndrome characterized by central adiposity, thin skin, easy bruising, and proximal myopathy. Withdrawal of the corticosteroid may result in symptoms that include fever, myalgia, fatigue, arthralgia, and malaise. This may occur even without evidence of adrenal insufficiency.

In the event where dose-reduction is used for adverse event management, 2 dose reductions are allowed. At each dose reduction, 1 tablet of ZYTIGA/placebo will be removed, eg, 4→3 tablets, and 3→2 tablets. Any return to protocol dose level (4 tablets) after dose reduction must follow documentation of adverse event resolution and a discussion with the sponsor.

6.3. Guidelines for Abnormal Liver Function Test

For subjects who develop liver function test abnormalities (ALT and/or AST greater than 5X ULN but not exceeding 20X ULN or total bilirubin greater than 3X ULN but not exceeding 10X ULN), study drug should be interrupted. Treatment may be restarted at a reduced dose of 750 mg once daily following return of liver function tests to the subject's baseline or to AST and ALT less than or equal to 2.5X ULN and total bilirubin less than or equal to 1.5X ULN.

For subjects with ALT and/or AST greater than 20X ULN or total bilirubin greater than 10X ULN, study drug should be interrupted. The decision to restart treatment at a reduced dose will be made in consultation with the sponsor medical monitor on an individual basis.

For subjects who resume treatment, serum transaminases and bilirubin should be monitored at a minimum of every 2 weeks for 3 months and monthly thereafter.

If liver function test abnormality recurs at the dose of 750 mg once daily, re-treatment may be restarted at a reduced dose of 500 mg once daily following return of liver function tests to the subject's baseline or to AST and ALT less than or equal to 2.5X ULN and total bilirubin less than or equal to 1.5X ULN.

If liver function test abnormality recurs at the reduced dose of 500 mg once daily, study drug should be discontinued.

6.4. Guidelines for Hypertension, Hypokalemia and Fluid Retention/Edema Due to Mineralocorticoid Excess

The study drug should be used with caution in subjects with a history of cardiovascular disease. The study drug may cause hypertension, hypokalemia, and fluid retention as a consequence of increased mineralocorticoid levels resulting from CYP17 inhibition. Caution should be exercised when treating subjects whose underlying medical conditions might be compromised by increases in blood pressure, hypokalemia, or fluid retention. Subjects should be monitored for hypertension, hypokalemia, and fluid retention at least once a month.

If hypokalemia persists despite optimal potassium supplementation and adequate oral intake, or if any of the other mineralocorticoid effects persist, the dose of prednisone may be increased from 5 to 10 mg/day and documented in the study medication eCRF.

7. TREATMENT COMPLIANCE

Accurate records of all drug shipments as well as tablets dispensed and returned will be maintained. This inventory must be available for inspection by designated sponsor or regulatory authority representatives at any time. Drug supplies are to be used only in accordance with this protocol and under the supervision of the investigator. Study drug administration and dosing compliance will be assessed at each study visit, starting with Cycle 2. A count of all study drug provided by the sponsor will be conducted during the Treatment Phase. Subjects will be required to maintain a medication diary. This medication diary will be provided by the sponsor and should be brought to study visits on Study Day 1 of each cycle for review of dosing compliance.

If dosing compliance is not 100% in the absence of toxicity, then subjects should be re-instructed regarding proper dosing procedures and may continue with the study. Subsequent dosing compliance procedure will be conducted at each study visit. If the number of study drug doses taken by the subject is ≤75% in the absence of toxicity or disease progression, then subjects should be re-instructed regarding proper dosing procedures. Subjects who have study drug dosing compliance of ≤75% for 2 consecutive cycles should be discontinued from the study.

The study site must maintain accurate records demonstrating dates and amount of study drug received, to whom dispensed (subject by subject accounting), and accounts of any study drug accidentally or deliberately destroyed. At the end of the study, reconciliation must be made between the amount of study drug supplied, dispensed, and subsequently destroyed or returned to sponsor or its representative.

8. PRESTUDY AND CONCOMITANT THERAPY

All pre-study systemic therapies administered up to 30 days before first dose of study drug must be recorded at screening. All concomitant systemic therapies must be recorded on the subject's eCRF throughout the study beginning when the first dose of study drug is administered.

Concurrent enrollment in another investigational drug or device study is prohibited. The sponsor must be notified in advance (or as soon as possible thereafter) of any instances in which prohibited therapies are administered.

All systemic therapies (prescriptions or over-the-counter medications, including vitamins and herbal supplements; non-pharmacologic therapies such as electrical stimulation, acupuncture, special diets, exercise regimens) different from study drug must be recorded in the concomitant therapy section of the eCRF.

8.1. Permitted Supportive Care and Interventions

- Surgical castration within 3 months prior to randomization is allowed.
- Use of opioid analgesics for cancer-related pain
- Bisphosphonates and denosumab for management of bone-related metastasis according to their market authorized approved label
- LHRH agonists to maintain testosterone <50 ng/dL (for those subjects who do not have orchiectomy)
- Conventional multivitamins, selenium and soy supplements
- Prednisone dose increase of up to 10 mg/day is permitted to manage refractory mineralocorticoid related toxicities
- Epelerenone can be used to manage mineralocorticoid-related toxicities
- Additional systemic glucocorticoid administration such as “stress dose” glucocorticoid is permitted when clinically indicated for a life threatening medical condition, and in such cases, the use of steroids will be documented as concomitant drug
- Transfusions and hematopoietic growth factors per institutional practice guidelines
- If the permissibility of a specific drug/treatment is in question, then please contact the study sponsor

8.2. Special Concomitant Therapy

The following concomitant therapies warrant special attention:

<u>CYP2D6 substrates:</u> ZYTIGA is an inhibitor of the hepatic drug-metabolizing enzyme CYP2D6. Avoid co-administration of ZYTIGA with CYP2D6 substrates that have a narrow therapeutic index. If an alternative treatment cannot be used, exercise caution and consider a dose reduction of the concomitant CYP2D6 substrate.

<u>CYP3A4</u>: ZYTIGA is a substrate of CYP3A4. Avoid, or use with caution, strong inhibitors and inducers of CYP3A4.

8.3. Prohibited Medications

- Investigational agents other than ZYTIGA
- Other antineoplastic agents
- Radiotherapy
- 5 α reductase inhibitors

- Chemotherapy
- Immunotherapy
- Anti-androgens (e.g., bicalutamide, nilutamide, flutamide, cyproterone acetate) except in situations as outlined for management of tumor flare anticipated with the initiation of continuous LHRH agonist (see Section 3.1)
- Systemic ketoconazole (or other azole drugs such as fluconazole or itraconazole)
- Diethylstilbestrol (DES) or similar and other preparations such as saw palmetto thought to have endocrine effects on prostate cancer
- Radiopharmaceuticals such as strontium (89Sr) or samarium (153Sm) or similar analogues such as radium-223
- Spironolactone
- Digoxin, digitoxin, and other digitalis drugs
- Fludrocortisone acetate (Florinef)

9. STUDY EVALUATIONS

9.1. Study Procedures

9.1.1. Overview

The Time and Events Schedule included in the Synopsis summarizes the frequency and timing of efficacy, MRU, and safety assessments. All visit-specific PRO assessments during a visit should be conducted before any tests, procedures, or other consultations for that visit to prevent influencing subject perceptions.

For each subject, approximately 20 to 25 mL of blood will be drawn at each study visit for hematology, clinical chemistry, liver function tests, and PSA. For microRNA analysis, 8.5 mL of blood will be obtained 3 times over the course of the study.

9.1.2. Screening Phase

A signed informed consent must be obtained before any study-specific assessments are performed. Informed consent may be obtained up to 28 days before the first dose of study drug. Screening procedures to evaluate subject eligibility for the study will be conducted within 28 days prior to Cycle 1 Day 1.

Imaging scans (CT/MRI or bone scans) taken within 3 months of randomization may be used in lieu of a new scan, if they are available and are deemed evaluable by a central reader.

Once eligibility is confirmed, subjects will be randomized to a treatment group according to the randomization schedule.

9.1.3. Double-blind Treatment Phase

The Treatment Phase begins on Cycle 1 Day 1 of treatment with study medication and continues until all study medication is discontinued. Subjects will be randomly assigned to either the active or control group, and must start taking study drug within 72 hours of randomization. Visits for each cycle will have a ± 2 day window. Study visits will be calculated from the Cycle 1 Day 1 date. Subjects may have additional imaging visits up to 8 days before cycles requiring images (Cycle 5 Day 1, Cycle 9 Day 1, and Cycle 13 Day 1 and every 4th cycle beyond Cycle 13) and at end-of-treatment visit.

During the Treatment Phase, investigators will review all clinical, laboratory and imaging data, and will use this information to make decisions about dose modification and study medication discontinuation. Results from PSA evaluations will not be available to the investigators.

9.1.4. End-of-Treatment Visit

The end-of-treatment visit will occur at the time of discontinuation of all study medication for any reason other than death of the subject. The end-of-treatment visit will be scheduled no more than 30 days after the final dose of study drug to collect post-treatment safety and efficacy data.

9.1.5. Follow-Up Phase

After discontinuation of all study medication, subject status will be monitored every 4 months for up to 60 months (5 years) or until subject death, lost to follow-up, withdrawal of consent, or study termination. Information regarding survival status and initiation of subsequent prostate cancer therapies will be collected. Visits to the clinic are not required; sites may collect the information by telephone interview, chart review, or other convenient methods. If the follow-up information is obtained via telephone contact, then written documentation of the communication must be available for review in the source documents. Completion of the EQ-5D-5L questionnaire every 4 months by the subject will continue for 12 months after the end of treatment. Deaths regardless of causality will be reported to the sponsor within 24 hours of discovery or notification of the event and documented (see Section 16.2.3).

9.1.6. Extension Phase

In the event of a positive study result at either of the interim analyses or at the final analysis, all subjects will have the opportunity to enroll in an open-label Extension Phase of this protocol. The Extension Phase will allow subjects to receive active drug (ZYTIGA plus prednisone) until ZYTIGA becomes commercially available in the subject's country. Subjects will be monitored for long-term survival and serious adverse events. Subjects who were previously receiving placebo in the Treatment Phase will have liver function and potassium monitoring completed per label recommendations for the first 3 months of ZYTIGA dosing in the Extension Phase. Those subjects who were previously receiving ZYTIGA will forego the additional liver function monitoring.

9.2. Efficacy

9.2.1. Evaluations

Unscheduled tumor assessment and appropriate imaging should be considered if signs or symptoms suggestive of disease progression, including escalating pain not referable to another cause, worsening ECOG performance status grade, or physical examination findings consistent with disease progression, are recorded.

A certain number of selected radiographic images obtained at screening will be reviewed by an independent central reader as outlined in the imaging charter.

9.2.2. Endpoints

Primary Endpoint

The primary efficacy endpoint is overall survival defined as the time from randomization to date of death from any cause.

Secondary Endpoints

- Radiographic progression-free survival (rPFS) based on Prostate Cancer Working Group 2 (PCWG2) and modified RECIST as the time from randomization to the occurrence of one of the following:
  - A subject is considered to have progressed by bone scan if the first bone scan with ≥2 new lesions compared to baseline is observed ≥16 weeks (Cycle 5 Day 1) from randomization. If the ≥2 new lesions are noted at or before Cycle 5 Day 1, a confirmatory bone scan will be performed ≥6 weeks later (to eliminate a false progression that could potentially be a flare phenomenon).
  - Progression of soft tissue lesions measured by CT or MRI as defined in modified RECIST criteria
  - Death from any cause
- Time to next skeletal-related event (one of the following):
  - Clinical fracture
  - Spinal cord compression
  - Palliative radiation to bone
  - Surgery to bone
- Time to PSA progression (by PCWG2 criteria)
- Time to next subsequent therapy for prostate cancer
- Time to initiation of chemotherapy

Exploratory Endpoints

- PSA response rate
- Patient Reported Outcome (PRO) measures: EQ-5D-5L, BPI - SF, FACT - P, BFI
- Pain measures (time to pain progression)
- Time to symptomatic local progression defined as occurrence of urethral obstruction or bladder outlet obstruction
- Prostate cancer specific survival

9.3. Biomarkers

FFPE tumor samples will be requested from the archival site at the Cycle 1 Day 1 visit. These samples will be collected from approximately 300 subjects. Serum samples will also be collected from approximately 300 subjects at the time points indicated in the Time and Events Schedule. Tumor and blood biomarker analyses are dependent upon the availability of appropriate biomarker assays and may be deferred or not performed, if during or at the end of the study, it becomes clear that the analysis will have no scientific value, or there are not enough samples or not enough responders to allow for adequate biomarker evaluation. In the event the study is terminated early or does not reach a positive primary endpoint, completion of biomarker assessments is based on justification and intended utility of the data.

Biomarker studies on the FFPE tumor samples may include immunohistochemistry analyses, global miRNA profiling, gene-expression profiling, and somatic mutational analyses.

MicroRNA (miRNAs) have been widely studied in prostate cancer and are small noncoding RNAs that regulate the expression of protein-coding genes by modulating both mRNA stability and translation.[10,20]Alteration of miRNA expression levels can alter cell function and induce cellular transformation[18] leading to cancer. A number of dysregulated miRNAs have been associated with different prostate cancer stages and some miRNAs are consistently dysregulated at early and advanced stages of disease or associated with more aggressive disease. Considering that some miRNAs are androgen controlled,[30,2,35,30,19,40,36,35,39] it is anticipated that dysregulated miRNAs patterns in baseline samples may predict response to drugs such as ZYTIGA. Changes in miRNA expression levels from baseline may allow elucidation of mechanisms leading to resistance. Dysregulation of steroid synthesis has been previously reported in xenograft models[24] and evaluation of steroid profiles may complement the miRNA studies. Herein, we plan to analyze dysregulated steroid transcript levels by gene expression profiling to confirm the predictive profiles found in these previously reported studies. Data collected from this study will be compared to data obtained from prior studies in asymptomatic or mildly symptomatic metastatic castration-resistant prostate cancer to identify miRNA and GEP profiles that correlate with response (or primary resistance) to abiraterone. The biomarker results from this study will then be used to inform future studies of anti-androgen therapies possibly leading to product differentiation by selection of responsive subjects.

miRNAs have also been identified that correlate with high risk prostate cancer.[40,39,2,29] Since it is difficult to histologically distinguish high risk from benign disease and because PSA velocity and doubling time may be an unreliable measure of disease aggression, we plan to investigate whether miRNA profiles may better define high risk prostate cancer in the early metastatic disease setting. This data may then be utilized in selection of high risk patients in future studies if these previously reported miRNA profiles are confirmed and found to be more sensitive than conventional clinical estimates of high risk disease.

9.4. Medical Resource Utilization

Medical resource usage data (MRU) associated with medical encounters will be collected in the eCRF by the investigator and staff for all subjects throughout the study. For each adverse event that occurs, MRU information associated with that adverse event should be recorded when possible. Protocol-mandated procedures, tests, and encounters are excluded. The following data may be used to conduct exploratory economic analyses:

- Number and duration of medical care encounters, including surgeries, and other selected procedures (inpatient and outpatient)
- Duration of hospitalization (total days length of stay, including duration by wards; eg, intensive care unit)
- Number and character of diagnostic and therapeutic tests and procedures
- Outpatient medical encounters and treatments (including physician or emergency room visits, tests and procedures, and medications)

9.5. Patient-Reported Outcomes

All patient reported outcomes will be administered within 2 days before randomization, and at the times specified in the Time and Events Schedule. PROs will be collected until radiographic or clinical progression, after which only the EQ-5D-5L will be collected for a period of 12 months.

9.5.1. Brief Pain Inventory-Short Form (BPI-SF)

Pain will be evaluated using the BPI-SF instrument. A sample of the BPI-SF questionnaire along with detailed instructions will be provided in the PRO Manual.

9.5.2. Functional Assessment of Cancer Therapy–Prostate (FACT-P)

The FACT-P questionnaire is a tool used for assessing the health-related quality of life in men with prostate cancer. A sample of the FACT-P questionnaire along with detailed instructions will be provided in the PRO Manual.

9.5.3. Brief Fatigue Inventory (BFI)

Fatigue will be evaluated using the BFI instrument. A sample of the BFI questionnaire along with detailed instructions will be provided in the PRO Manual.

9.5.4. EQ-5D-5L

The EQ-5D-5L (EuroQol Group, www.euroqol.org) is a validated tool that measures mobility, self-care, usual activities, pain, discomfort, and anxiety/depression. [8] The EQ-5D quality of life instrument consists of a 5-item questionnaire and a visual analogue scale ranging from 0 (worst imaginable health state) to 100 (best imaginable health state) that integrates many aspects of the subject's disease process into a single assessment. A sample of the EQ-5D questionnaire along with detailed instructions will be provided in the PRO Manual.

9.6. Safety Evaluations

The evaluation period for safety will start at the time a signed and dated informed consent form is obtained to at least 30 days after the last dose of study drug or recovery from all acute toxicities associated with the study drug administration. Adverse events will be reported by the subject for the duration of the study. Adverse events including laboratory AEs will be graded and summarized according to the NCI-CTCAE, Version 4.0.

Any clinically significant abnormalities persisting at the end of the study or during early withdrawal will be followed by the investigator until resolution or until a clinically stable endpoint is reached. The study will include the following evaluations of safety according to the time points provided in the Time and Events Schedule.

Adverse Events

Adverse events will be reported by the subject (or, when appropriate, by a caregiver, surrogate, or the subject's legally-acceptable representative). Adverse events will be followed by the investigator as specified in Section 12, Adverse Event Reporting.

Clinical Laboratory Tests

All laboratory tests will be performed by the central laboratory. Blood samples for serum chemistry and hematology will be collected. The following tests will be performed:

Hematology Panel	Serum Chemistry Panel	Liver Function Tests
• hemoglobin	• potassium	• alanine aminotransferase (ALT/SGPT)
• platelet count	• creatinine	• aspartate aminotransferase (AST/SGOT)
• white blood cells including neutrophil count	• fasting glucose	• total bilirubin
	• lactate dehydrogenase	

In addition, testosterone and PSA will be evaluated periodically throughout the study. PSA and testosterone results will not be available to the investigator.

The investigator must review the laboratory report, document this review, and record any clinically relevant changes occurring during the study in the adverse event section of the eCRF.

Electrocardiogram (ECG)

Electrocardiograms (ECGs) (12-lead) will be recorded at screening. Computer-generated interpretations of ECGs should be reviewed for data integrity and reasonableness by the investigator. During the collection of ECGs, subjects should be in a quiet setting without distractions (eg, television, cell phones). Subjects should rest in a supine position for at least 5 minutes before ECG collection and should refrain from talking or moving arms or legs. If blood sampling or vital sign measurement is scheduled at the same visit as ECG recording, then the procedures should be performed in the following order: ECG(s), vital signs, blood draw. ECGs should not be obtained when serum potassium is <3.5 mM/L. Hypokalemia should be corrected prior to ECG collection.

Vital Signs

Heart rate, blood pressure, respiratory rate, and oral or aural temperature will be measured at Screening. At all subsequent study visits, including the End-of-Treatment Visit, only blood pressure will be measured.

When measuring blood pressure, please ensure that the subject is comfortably seated in a chair with arm and back supported and legs uncrossed in a quiet area. Any restrictive clothing on the arm should be removed. At least 5 minutes should elapse before the reading is taken and the patient should be advised not to talk during the reading.

Physical Examination

Physical examinations will be directed at detecting AEs and should be completed as specified in the Time and Events Schedule. Physical examinations should be performed by the same evaluator throughout the study, whenever possible. If it is not possible to use the same evaluator to follow the subject, then it is recommended that evaluations overlap (ie, examine the subject together and discuss findings) for at least 1 visit. Subject height and weight (with the subject lightly clothed and without shoes) will be measured at screening only. Physical examination includes head, eyes, ears, nose and throat, chest, cardiac, abdominal, extremities, neurologic, and lymph node examinations. Any clinically significant change in physical findings noted during the study should be reported as an AE. Any clinically significant abnormalities persisting at the end of the study will be followed by the investigator until resolution or until reaching a clinically stable endpoint.

9.7. Sample Collection and Handling

The actual dates and times of sample collection must be recorded in the eCRF or laboratory requisition form. Refer to the Time and Events Schedule for the timing and frequency of all sample collections. Instructions for the collection, handling, and shipment of samples are found in the laboratory manual that will be provided for sample collection and handling.

10. SUBJECT COMPLETION/WITHDRAWAL

10.1. Completion

A subject will be considered to have completed the study if he has completed assessments through the end of the Follow-up Phase. Subjects who prematurely discontinue study treatment for any reason before completion of the Follow-up Phase will not be considered to have completed the study. A subject will be considered to have completed the Treatment Phase when a clinical endpoint of study is reached (death, radiographic progression per protocol definition, or unacceptable toxicity).

10.2. Discontinuation of Treatment

Discontinuation of a study drug will not result in automatic withdrawal from the study. A subject should ordinarily be maintained on study treatment until radiographic progression. However, a subject's study treatment may be discontinued for:

- Disease progression defined as radiographic evidence of new metastases (see Section 9.2.2)
- Administration of prohibited concomitant medications including need to initiate anticancer therapy or radiation therapy or surgical interventions for complications due to tumor progression even in the absence of radiographic evidence of disease progression
- Dosing noncompliance
- Withdrawal of consent for continued treatment
- The investigator believes that for safety reasons (eg, adverse event) it is in the best interest of the subject to stop treatment

Study treatment will be continued for subjects who have increasing PSA values in the absence of radiographic or clinical progression. Although serial PSA measurements will be performed in this study, progression or change in PSA values is not considered a reliable measure of disease progression and should not be used as the lone indicator for discontinuation.

Study data will continue to be collected after discontinuation of treatment, unless the subject withdraws consent for further study participation.

10.3. Withdrawal from the Study

A subject will be withdrawn from the study for any of the following reasons:

- Lost to follow-up

- Withdrawal of consent for subsequent data collection

If a patient is lost to follow up, all possible efforts must be made by the study site personnel to contact the subject and to determine endpoint status and the reason for the discontinuation/withdrawal. The measures taken for follow up must be documented. The informed consent will stipulate that even if a subject decides to discontinue double blind study drug, he will agree to be contacted periodically by the investigator to assess endpoint status. Furthermore, the subject will be asked to agree to grant permission for the investigator to consult family members or public records, including the use of locator agencies as permitted by local law, to determine the subject's endpoint status, in the event the subject is not reachable by conventional means (eg, office visit, telephone, e-mail, or certified mail.).

When a subject withdraws before completing the study, the reason for withdrawal is to be documented in the eCRF and in the source document. Study drug assigned to the withdrawn subject may not be re-assigned to another subject. Subjects who withdraw will not be replaced.

11. STATISTICAL METHODS

Statistical analysis will be performed by the sponsor or under the supervision of the sponsor. A general description of the statistical methods to be used to analyze the efficacy and safety data is outlined below. Complete details will be contained in the Statistical Analysis Plan.

11.1. Subject Information

ITT Population: The ITT population includes all randomized subjects and will be classified according to their assigned treatment group, regardless of the actual treatment received. Subject disposition and efficacy analyses will be performed on data from the ITT population.

Safety Population: The safety population includes all subjects who received at least 1 dose of study drug.

11.2. Sample Size Determination

It is assumed that failure will follow an exponential distribution with a constant hazard rate. Assuming a median overall survival (OS) of 33 months[21] for the control group (ADT alone), a planned sample size of approximately 1,270 subjects will provide 85% power to detect a hazard ratio (HR) of 0.81 (33 months versus 40.75 months, or >7 months of improvement) at a 2-tailed significance level of 0.05 and an enrollment duration of approximately 30 months over a total study duration of 74 months to obtain the required 842 events.

The OS endpoint will incorporate group sequential design by including 2 interim analyses and one final analysis with alpha spending function calculated as Wang-Tsiatis power boundaries of shape parameter 0.2 (East® software).[41] Subjects will be randomized in a 1:1 ratio to receive ZYTIGA plus low-dose prednisone plus ADT or ADT alone.

11.3. Efficacy Analyses

Efficacy endpoints are described in Section 9.2.2.

All continuous variables will be summarized using number of subjects (n), mean, standard deviation (SD), median, minimum, and maximum. Discrete variables will be summarized with n and percent. All efficacy endpoints will be analyzed using the ITT set of population. Kaplan-Meier product limit method and Cox model will be used to estimate the time to event variables and to obtain the HR. Non-stratified analysis will be conducted. Response rate variable will be evaluated using the Chi-square statistic, or the exact test if the cell counts are small.

11.3.1. Analysis of Primary Endpoint OS

The primary efficacy endpoint is overall survival defined as the time from randomization to date of death from any cause. Subjects without event will be censored on the last date known to be alive. The final analysis will occur when 842 events are observed. The primary analysis will be performed using the stratified log rank test, stratified based on ECOG performance (grade of 0, 1 vs. 2) and presence of visceral metastasis (Yes vs. No).

11.4. Biomarker Analyses

Biomarker studies are designed to confirm prior findings in asymptomatic or mildly symptomatic metastatic castration-resistant prostate cancer. Analyses will be conducted on tumor samples from approximately 300 subjects and serum samples collected at 3 timepoints from approximately 300 subjects. The following biomarkers may be analyzed (other DNA/RNA or IHC based-biomarkers may also be analyzed):

- Global miRNA expression levels in serial serum samples and in FFPE samples from baseline
- Global mRNA expression levels in FFPE samples
- AR abnormalities, such as AR mutations and splicing variants

The associations of the above biomarkers with clinical response or time-to-event endpoints will be assessed using the appropriate statistical methods (analysis of variance [ANOVA], categorical, or survival model will be used), depending on the endpoint. Analyses will be performed within and between each treatment group. Other clinical covariates (such as race, age, tumor burden) may also be included in the model. Correlation of baseline expression levels with response or time-to-event endpoints will identify responsive (or resistant) subgroups. Association of biomarkers with clinical response or time-to-event endpoints will also be evaluated in the overall population to identify "high risk" biomarker profiles that are correlated with poor outcome. Biomarker results will be reported separately from the clinical study report due to its exploratory nature.

11.5. Medical Resource Utilization

Medical resource utilization will be descriptively summarized by treatment group. This report will be prepared separately and will not be a part of the clinical study report.

11.6. Safety Analyses

Subjects who receive at least 1 dose of study drug will be analyzed for safety. The safety parameters to be evaluated are the incidence, intensity, and type of adverse events, clinically significant changes in the subject's physical examination findings, vital signs measurements, and clinical laboratory results. Exposure to study drug and reasons for discontinuation of study treatment will be tabulated.

Adverse Events

The verbatim terms used in the eCRF by investigators to identify adverse events will be coded using the Medical Dictionary for Regulatory Activities (MedDRA) and will be graded according to the NCI-CTCAE, Version 4.0. Treatment-emergent AEs are those events that occur or worsen on or after first dose of study drug through 30 days post last dose and will be included in the analysis. Adverse events will be summarized by system organ class and preferred term, and will be presented overall and by treatment group. Serious adverse events (SAEs) and deaths will be provided in a listing. All adverse events resulting in discontinuation, dose modification, dosing interruption, or treatment delay of study drug will also be listed and tabulated by preferred term.

Clinical Laboratory Tests

Clinical laboratory test results will be collected from screening and through 30 days post last dose of study drug. Laboratory data will be summarized by type of laboratory test. Parameters with predefined NCI-CTCAE, Version 4, toxicity grades will be summarized. Change from baseline to the worst adverse event grade experienced by the subject during the study will be provided as shift tables.

Vital Signs and Physical Examination

Descriptive statistics of blood pressure (systolic and diastolic) values and changes from baseline will be summarized at each scheduled time point. The percentage of subjects with values beyond clinically important limits will be summarized. Descriptive statistics of changes from baseline in physical examinations will be summarized at each scheduled time point.

11.7. Interim Analysis

Two interim analyses are planned for this study after observing 50% (421 events) and 65% (548 events) of the total number of required (842) events. The cumulative alpha spend will be

0.01 and 0.02 for each of the two interim analyses respectively. The exact significance levels will be determined according to the observed number of events at each interim analysis.

11.8. Independent Data Monitoring Committee (IDMC)

During the course of the study, an IDMC will review the safety and efficacy information. The committee will be composed of medical oncologists and/or urologists with experience in treating patients or medical experts in relevant therapeutic area, and a statistician, none of whom are involved in the study as investigators. The IDMC activities are described in detail in a separate charter.

12. ADVERSE EVENT REPORTING

Timely, accurate, and complete reporting and analysis of safety information from clinical studies are crucial for the protection of subjects, investigators, and the sponsor, and are mandated by regulatory agencies worldwide. The sponsor has established Standard Operating Procedures in conformity with regulatory requirements worldwide to ensure appropriate reporting of safety information; all clinical studies conducted by the sponsor or its affiliates will be conducted in accordance with those procedures.

12.1. Definitions

12.1.1. Adverse Event Definitions and Classifications

Adverse Event

An adverse event is any untoward medical occurrence in a clinical study subject administered a medicinal (investigational or non-investigational) product. An adverse event does not necessarily have a causal relationship with the treatment. An adverse event can therefore be any unfavorable and unintended sign (including an abnormal finding), symptom, or disease temporally associated with the use of a medicinal (investigational or non-investigational) product, whether or not related to that medicinal (investigational or non-investigational) product. (Definition per International Conference on Harmonisation [ICH])

This includes any occurrence that is new in onset or aggravated in severity or frequency from the baseline condition, or abnormal results of diagnostic procedures, including laboratory test abnormalities.

Note: The sponsor collects adverse events starting with the signing of the informed consent form (refer to Section 12.3.1, All Adverse Events, for time of last adverse event recording).

Serious Adverse Event

A serious adverse event based on ICH is any untoward medical occurrence that at any dose:

- Results in death
- Is life-threatening
  (The subject was at risk of death at the time of the event. It does not refer to an event that hypothetically might have caused death should it have been more severe.)
- Requires inpatient hospitalization or prolongation of existing hospitalization
- Results in persistent or significant disability/incapacity
- Is a congenital anomaly/birth defect
- Is a suspected transmission of any infectious agent via a medicinal product
- Is medically important*

*Medical and scientific judgment should be exercised in deciding whether expedited reporting is also appropriate in other situations, such as important medical events that may not be immediately life threatening or result in death or hospitalization but may jeopardize the subject or may require intervention to prevent one of the other outcomes listed in the definition above. These should usually be considered serious.

Unlisted (Unexpected) Adverse Event/Reference Safety Information

An adverse event is considered unlisted when the nature or severity is not consistent with the applicable product reference safety information. For an investigational product, the expectedness of an adverse event will be determined by whether or not it is listed in the Investigator's Brochure.

Adverse Event Associated With the Use of the Drug

An adverse event is considered associated with the use of the drug when the attribution is possible, probable, or very likely by the definitions listed in Section 12.1.2.

12.1.2. Attribution Definitions

Not Related

An adverse event that is not related to the use of the drug.

Doubtful

An adverse event for which an alternative explanation is more likely, eg, concomitant drug(s), concomitant disease(s), or the relationship in time suggests that a causal relationship is unlikely.

Possible

An adverse event that might be due to the use of the drug. An alternative explanation, eg, concomitant drug(s), concomitant disease(s), is inconclusive. The relationship in time is reasonable; therefore, the causal relationship cannot be excluded.

Probable

An adverse event that might be due to the use of the drug. The relationship in time is suggestive (eg, confirmed by dechallenge). An alternative explanation is less likely, eg, concomitant drug(s), concomitant disease(s).

Very Likely

An adverse event that is listed as a possible adverse reaction and cannot be reasonably explained by an alternative explanation, eg, concomitant drug(s), concomitant disease(s). The relationship in time is very suggestive (eg, it is confirmed by dechallenge and rechallenge).

12.1.3. Severity Criteria

The NCI-CTCAE v4.0 will be used to grade the severity of adverse events.

Any AE not listed in the CTCAE will be graded as follows:

Grade 1, Mild: Awareness of symptoms that are easily tolerated, causing minimal discomfort and not interfering with everyday activities.

Grade 2, Moderate: Sufficient discomfort is present to cause interference with normal activity.

Grade 3, Severe: Extreme distress, causing significant impairment of functioning or incapacitation. Prevents normal everyday activities.

Grade 4, Life-threatening: Urgent intervention indicated.

Grade 5, Death: Death.

The investigator should use clinical judgment in assessing the severity of events not directly experienced by the subject (eg, laboratory abnormalities).

12.2. Special Reporting Situations

Safety events of interest on a sponsor medicinal product that may require expedited reporting or safety evaluation include, but are not limited to:

- Overdose of a sponsor medicinal product
- Suspected abuse/misuse of a sponsor medicinal product
- Inadvertent or accidental exposure to a sponsor medicinal product
- Medication error involving a sponsor product (with or without subject/patient exposure to the sponsor medicinal product, eg, name confusion)

Special reporting situations should be recorded in the eCRF. Any special reporting situation that meets the criteria of a serious adverse event should be recorded on the serious adverse event page of the eCRF.

12.3. Procedures

12.3.1. All Adverse Events

All adverse events and special reporting situations, whether serious or non-serious, will be reported from the time a signed and dated informed consent form is obtained until 30 days after last dose of study drug. Serious adverse events, including those spontaneously reported to the investigator within 30 days after the last dose of study drug, must be reported using the Serious Adverse Event Form. The sponsor will evaluate any safety information that is spontaneously reported by an investigator beyond the time frame specified in the protocol.

All adverse events, regardless of seriousness, severity, or presumed relationship to study therapy, must be recorded using medical terminology in the source document and the eCRF. Whenever possible, diagnoses should be given when signs and symptoms are due to a common etiology (eg, cough, runny nose, sneezing, sore throat, and head congestion should be reported as "upper respiratory infection"). Investigators must record in the eCRF their opinion concerning the relationship of the adverse event to study therapy. All measures required for adverse event management must be recorded in the source document and reported according to sponsor instructions.

The sponsor assumes responsibility for appropriate reporting of adverse events to the regulatory authorities. The sponsor will also report to the investigator all serious adverse events that are unlisted (unexpected) and associated with the use of the drug. The investigator (or sponsor where required) must report these events to the appropriate Independent Ethics Committee/Institutional Review Board (IEC/IRB) that approved the protocol unless otherwise required and documented by the IEC/IRB.

Subjects (or their designees, when appropriate) must be provided with a "study card" indicating the following:

- Subject's name
- Subject number
- Subject's data of birth
- Study site number
- Investigator's name and 24-hour contact information
- Local sponsor's name and 24-hour contact information
- Statement that the subject is participating in a clinical trial

12.3.2. Serious Adverse Events

All serious adverse events occurring during clinical studies must be reported to the appropriate sponsor contact person by investigational staff within 24 hours of their knowledge of the event.

Information regarding serious adverse events will be transmitted to the sponsor using the Serious Adverse Event Form, which must be completed and signed by a member of the investigational staff, and transmitted to the sponsor within 24 hours. The initial and follow-up reports of a serious adverse event should be made by facsimile (fax).

All serious adverse events that have not resolved by the end of the study, or that have not resolved upon discontinuation of the subject's participation in the study, must be followed until any of the following occurs:

- The event resolves
- The event stabilizes
- The event returns to baseline, when a baseline value/status is available
- The event can be attributed to agents other than the study drug or to factors unrelated to study conduct
- It becomes unlikely that any additional information can be obtained (subject or health care practitioner refusal to provide additional information, lost to follow-up after demonstration of due diligence with follow-up efforts)

Suspected transmission of an infectious agent by a medicinal product will be reported as a serious adverse event. Any event requiring hospitalization (or prolongation of hospitalization) that occurs during the course of a subject's participation in a clinical study must be reported as a serious adverse event, except hospitalizations for the following:

- Social reasons in the absence of an adverse event
- Surgery or procedure planned before entry into the study (must be documented in the eCRF)

Disease progression should not be recorded as an adverse event or serious adverse event term; instead, signs and symptoms of clinical sequelae resulting from disease progression/lack of efficacy will be reported when they fulfill the serious adverse event definition (see Section 12.3.2).

During the follow-up phase of the study, deaths regardless of causality and serious adverse events thought to be related to blinded study drug will be collected and reported within 24 hours of discovery or notification of the event and documented.

12.3.3. Pregnancy

Because the effect of the blinded study drug on sperm is unknown, pregnancies in partners of male subjects included in the study will be reported by the investigational staff within 24 hours of their knowledge of the event using the appropriate pregnancy notification form.

Follow-up information regarding the outcome of the pregnancy and any postnatal sequelae in the infant will be required.

12.3.4. Contacting Sponsor Regarding Safety

The names (and corresponding telephone numbers) of the individuals who should be contacted regarding safety issues or questions regarding the study are listed on the Contact Information page(s), which will be provided as a separate document.

13. PRODUCT QUALITY COMPLAINT HANDLING

A product quality complaint (PQC) is defined as any suspicion of a product defect related to manufacturing, labeling, or packaging, ie, any dissatisfaction relative to the identity, quality, durability, or reliability of a product, including its labeling or package integrity. PQCs may have an impact on the safety and efficacy of the product. Timely, accurate, and complete reporting and analysis of PQC information from clinical studies are crucial for the protection of subjects, investigators, and the sponsor, and are mandated by regulatory agencies worldwide. The sponsor has established procedures in conformity with regulatory requirements worldwide to ensure appropriate reporting of PQC information; all clinical studies conducted by the sponsor or its affiliates will be conducted in accordance with those procedures.

13.1. Procedures

All initial PQCs must be reported to the sponsor by the investigational staff as soon as possible after being made aware of the event.

If the defect is combined with a serious adverse event, then the investigational staff must report the PQC to the sponsor according to the serious adverse event reporting timelines (refer to Section 12.3.2). A sample of the suspected product should be maintained for further investigation when requested by the sponsor.

13.2. Contacting Sponsor Regarding Product Quality

The names (and corresponding telephone numbers) of the individuals who should be contacted regarding product quality issues are listed on the Contact Information page(s), which will be provided as a separate document.

14. STUDY DRUG INFORMATION

14.1. Physical Description of Study Drug(s)

Abiraterone acetate 250 mg tablets are oval, white to off-white and contain ZYTIGA and compendial (USP/NF/EP) grade lactose monohydrate, microcrystalline cellulose, croscarmellose sodium, povidone, sodium lauryl sulfate, magnesium stearate, colloidal silicon dioxide, and purified water, in descending order of concentration (the water is removed during tabletting).

Placebo will be provided as a tablet formulation and will be matched in size, color (white to off-white), and shape (oval) to ZYTIGA tablets in order to maintain the study blind. Placebo prednisone tablets will also be provided.

14.2. Packaging

The study drugs will be packaged in individual subject kits. Subjects will be provided with a 30-day supply to allow for visits to occur every 28 days with a ± 2 day window and a 60-day supply to allow for visits to occur every 2 months with a ± 2 day window. Study drug will be dispensed in child-resistant packaging.

14.3. Labeling

Study drug labels will contain information to meet the applicable regulatory requirements.

14.4. Preparation, Handling, and Storage

The study drugs must be stored in a secure area and administered only to subjects entered into the clinical study in accordance with the conditions specified in this protocol. Bottles of study treatment should be stored at a room temperature between 15º to 30º C with the cap on tightly and should not be refrigerated. Additional information is provided in the ZYTIGA Investigator's Brochure.

14.5. Drug Accountability

The investigator is responsible for ensuring that all study drug received at the site is inventoried and accounted for throughout the study. The dispensing of study drug to the subject, and the return of study drug from the subject, must be documented on the drug accountability form. Subjects must be instructed to return all original containers, whether empty or containing study drug. These requirements also apply to prednisone/placebo tablets supplied by the Sponsor.

All study drugs must be handled in strict accordance with the protocol and the container label, and must be stored at the study site in a limited-access area or in a locked cabinet under appropriate environmental conditions. Unused study drug, and study drug returned by the subject, must be available for verification by the sponsor's site monitor during on-site monitoring visits. The return to the sponsor of unused study drug, or used returned study drug for destruction, will be documented on the drug return form. When the site is an authorized destruction unit and study drug supplies are destroyed on site, this must also be documented on the drug return form.

All study drugs should be dispensed under the supervision of the investigator or a qualified member of the investigational staff, or by a hospital/clinic pharmacist. Study drug will be supplied only to subjects participating in the study. Returned study drug must not be dispensed again, even to the same subject. Whenever a subject brings his or her study drug to the site for pill count, this is not seen as a return of supplies. Study drug may not be relabeled or reassigned for use by other subjects. The investigator agrees neither to dispense the study drug from, nor store it at, any site other than the study sites agreed upon with the sponsor.

15. STUDY-SPECIFIC MATERIALS

The investigator will be provided with the following supplies:

- Investigator Brochure
- Pharmacy manual/site investigational product procedures manual
- Laboratory manual
- NCI-CTCAE Version 4.0
- PRO Manual containing questionnaires and user instructions
- IVRS/IWRS Manual
- eDC Manual
- Medication diary for study drug

16. ETHICAL ASPECTS

16.1. Study-Specific Design Considerations

Potential subjects will be fully informed of the risks and requirements of the study and, during the study, subjects will be given any new information that may affect their decision to continue participation. They will be told that their consent to participate in the study is voluntary and may be withdrawn at any time with no reason given and without penalty or loss of benefits to which they would otherwise be entitled. Only subjects who are fully able to understand the risks, benefits, and potential adverse events of the study, and provide their consent voluntarily will be enrolled.

Ongoing medical review is completed by the sponsor. An interim analysis review will also be completed by the IDMC for safety and efficacy which will occur when 50% (421 events) and 65% (548 events) total number of required (842) have occurred.

The total blood volume to be collected is approximately 20 to 25 mL per study visit. As rPFS is a secondary endpoint, scheduled imaging is incorporated in the protocol. The timing of imaging is designed to capture progression events yet not to have patients be overexposed to radiation.

Subjects in the active group of this study will receive blinded low-dose prednisone 5mg/day. The prednisone dose used in this study is lower than the prednisone 10mg/day dose used in other ZYTIGA Phase 3 studies COU-AA-301 and COU-AA-302 that enrolled advanced symptomatic mCRPC patients. The higher prednisone dose in other phase 3 studies was selected because it is commonly used as the standard of care[11] in combination with approved chemotherapy agents or as monotherapy for palliation of symptoms in advanced prostate cancer patients. Prednisone is not considered standard of care for newly diagnosed mHNPC and therefore is not included in the control arm. The required use of prednisone in combination with ZYTIGA is to help mitigate the symptoms of mineralocorticoid excess caused by CYP17 inhibition, which is the mechanism of action of ZYTIGA. Data from early Phase 1/2 studies with ZYTIGA in patients with mCRPC demonstrate that dexamethasone 0.5mg once daily (equivalent to prednisone dose of 3.33 mg once daily) was effective in mitigating the mineralocorticoid effects of ZYTIGA.[3] The protocol allows for the dose of prednisone to be increased to 10 mg/day if required to manage mineralocorticoid toxicities.

16.2. Regulatory Ethics Compliance

16.2.1. Investigator Responsibilities

The investigator is responsible for ensuring that the clinical study is performed in accordance with the protocol, current ICH guidelines on Good Clinical Practice (GCP), and applicable regulatory and country-specific requirements.

Good Clinical Practice is an international ethical and scientific quality standard for designing, conducting, recording, and reporting studies that involve the participation of human subjects. Compliance with this standard provides public assurance that the rights, safety, and well-being of study subjects are protected, consistent with the principles that originated in the Declaration of Helsinki, and that the clinical study data are credible.

16.2.2. Independent Ethics Committee or Institutional Review Board

Before the start of the study, the investigator (or sponsor where required) will provide the IEC/IRB with current and complete copies of the following documents:

- Final protocol and, when applicable, amendments
- Sponsor-approved informed consent form (and any other written materials to be provided to the subjects)
- Investigator's Brochure (or equivalent information) and amendments/addenda
- Sponsor-approved subject recruiting materials
- Information on compensation for study-related injuries or payment to subjects for participation in the study, when applicable
- Investigator's curriculum vitae or equivalent information (unless not required, as documented by the IEC/IRB)
- Information regarding funding, name of the sponsor, institutional affiliations, other potential conflicts of interest, and incentives for subjects
- Any other documents that the IEC/IRB requests to fulfill its obligation

This study will be undertaken only after the IEC/IRB has given full approval of the final protocol, any amendments, the informed consent form, applicable recruiting materials, and subject compensation programs, and the sponsor has received a copy of this approval. This approval letter must be dated and must clearly identify the IEC/IRB and the documents being approved.

During the study the investigator (or sponsor where required) will send the following documents and updates to the IEC/IRB for their review and approval, where appropriate:

- Protocol amendments
- Revision(s) to informed consent form and any other written materials to be provided to subjects
- When applicable, new or revised subject recruiting materials approved by the sponsor
- Revisions to compensation for study-related injuries or payment to subjects for participation in the study, when applicable
- New edition(s) of the Investigator's Brochure and amendments/addenda
- Summaries of the status of the study at intervals stipulated in guidelines of the IEC/IRB (at least annually)
- Reports of adverse events that are serious, unlisted/unexpected, and associated with the investigational drug
- New information that may adversely affect the safety of the subjects or the conduct of the study
- Deviations from or changes to the protocol to eliminate immediate hazards to the subjects
- Report of deaths of subjects under the investigator's care
- Notification when a new investigator is responsible for the study at the site
- Development Safety Update Report and Line Listings, where applicable
- Any other requirements of the IEC/IRB

For all protocol amendments (excluding the ones that are purely administrative, with no consequences for subjects, data or trial conduct), the amendment and applicable informed consent form revisions must be submitted promptly to the IEC/IRB for review and approval before implementation of the change(s).

At least once a year, the IEC/IRB will be asked to review and reapprove this clinical study. The reapproval should be documented in writing (excluding the ones that are purely administrative, with no consequences for subjects, data, or study conduct).

At the end of the study, the investigator (or sponsor where required) will notify the IEC/IRB about the study completion.

16.2.3. Informed Consent

Each subject must give written consent according to local requirements after the nature of the study has been fully explained. The consent form must be signed before performance of any study-related activity. The consent form that is used must be approved by both the sponsor and by the reviewing IEC/IRB and be in a language that the subject can read and understand. The informed consent should be in accordance with principles that originated in the Declaration of Helsinki, current ICH and GCP guidelines, applicable regulatory requirements, and sponsor policy.

Before enrollment in the study, the investigator or an authorized member of the investigational staff must explain to potential subjects the aims, methods, reasonably anticipated benefits, and potential hazards of the study, and any discomfort participation in the study may entail. Subjects will be informed that their participation is voluntary and that they may withdraw consent to participate at any time. They will be informed that choosing not to participate will not affect the care the subject will receive for the treatment of his disease. Subjects will be told that alternative treatments are available should they refuse to take part and that such refusal will not prejudice future treatment. Finally, they will be told that the investigator will maintain a subject identification register for the purposes of long-term follow up when needed and that their records may be accessed by health authorities and authorized sponsor staff without violating the confidentiality of the subject, to the extent permitted by the applicable law(s) or regulations. By signing the informed consent form the subject is authorizing such access, and agrees to allow his or her study physician to recontact the subject for the purpose of obtaining consent for additional safety evaluations, and subsequent disease-related treatments or to obtain information about his survival status.

The subject will be given sufficient time to read the informed consent form and the opportunity to ask questions. After this explanation and before entry into the study, consent should be appropriately recorded by means of the subject's personally dated signature. After having obtained the consent, a copy of the informed consent form must be given to the subject.

If the subject is unable to read or write, then an impartial witness should be present for the entire informed consent process (which includes reading and explaining all written information) and should personally date and sign the informed consent form after the oral consent of the subject is obtained.

16.2.4. Privacy of Personal Data

The collection and processing of personal data from subjects enrolled in this study will be limited to those data that are necessary to fulfill the objectives of the study.

These data must be collected and processed with adequate precautions to ensure confidentiality and compliance with applicable data privacy protection laws and regulations. Appropriate technical and organizational measures to protect the personal data against unauthorized disclosures or access, accidental or unlawful destruction, or accidental loss or alteration must be put in place. Sponsor personnel whose responsibilities require access to personal data agree to keep the identity of study subjects confidential.

The informed consent obtained from the subject includes explicit consent for the processing of personal data and for the investigator to allow direct access to his or her original medical records for study-related monitoring, audit, IEC/IRB review, and regulatory inspection. This consent also addresses the transfer of the data to other entities and to other countries.

The subject has the right to request through the investigator access to his or her personal data and the right to request rectification of any data that are not correct or complete. Reasonable steps will be taken to respond to such a request, taking into consideration the nature of the request, the conditions of the study, and the applicable laws and regulations.

16.2.5. Country Selection

This study will only be conducted in those countries where the intent is to launch or otherwise help ensure access to the developed product, unless explicitly addressed as a specific ethical consideration in Section 16.1.

17. ADMINISTRATIVE REQUIREMENTS

17.1. Protocol Amendments

Neither the investigator nor the sponsor will modify this protocol without a formal amendment by the sponsor. All protocol amendments must be issued by the sponsor, and signed and dated by the investigator. Protocol amendments must not be implemented without prior IEC/IRB approval, or when the relevant competent authority has raised any grounds for non-acceptance, except when necessary to eliminate immediate hazards to the subjects, in which case the amendment must be promptly submitted to the IEC/IRB and relevant competent authority. Documentation of amendment approval by the investigator and IEC/IRB must be provided to the sponsor or its designee. When the change(s) involves only logistic or administrative aspects of the study, the IRB (and IEC where required) only needs to be notified.

During the course of the study, in situations where a departure from the protocol is unavoidable, the investigator or other physician in attendance will contact the appropriate sponsor representative (see Contact Information page(s) provided separately). Except in emergency situations, this contact should be made before implementing any departure from the protocol. In all cases, contact with the sponsor must be made as soon as possible to discuss the situation and agree on an appropriate course of action. The data recorded in the eCRF and source documents will reflect any departure from the protocol, and the source documents will describe this departure and the circumstances requiring it.

17.2. Regulatory Documentation

17.2.1. Regulatory Approval/Notification

This protocol and any amendment(s) must be submitted to the appropriate regulatory authorities in each respective country, when applicable. A study may not be initiated until all local regulatory requirements are met.

17.2.2. Required Prestudy Documentation

The following documents must be provided to the sponsor before shipment of blinded study drug to the investigational site:

- Protocol and any amendment(s), signed and dated by the principal investigator
- A copy of the dated and signed, written IEC/IRB approval of the protocol, amendments, informed consent form, any recruiting materials, and when applicable, subject compensation programs. This approval must clearly identify the specific protocol by title and number and must be signed by the chairman or authorized designee.
- Name and address of the IEC/IRB, including a current list of the IEC/IRB members and their function, with a statement that it is organized and operates according to GCP and the applicable laws and regulations. If accompanied by a letter of explanation, or equivalent, from the IEC/IRB, then a general statement may be substituted for this list. If an investigator or a member of the investigational staff is a member of the IEC/IRB, then documentation must be obtained to state that this person did not participate in the deliberations or in the vote/opinion of the study.
- Regulatory authority approval or notification, when applicable
- Signed and dated statement of investigator (eg, Form FDA 1572), when applicable
- Documentation of investigator qualifications (eg, curriculum vitae)
- Completed investigator financial disclosure form from the principal investigator, where required
- Signed and dated clinical trial agreement, which includes the financial agreement
- Any other documentation required by local regulations

The following documents must be provided to the sponsor before enrollment of the first subject:

- Completed investigator financial disclosure forms from all clinical subinvestigators
- Documentation of subinvestigator qualifications (eg, curriculum vitae)
- Name and address of any local laboratory conducting tests for the study, and a dated copy of current laboratory normal ranges for these tests, when applicable

- Local laboratory documentation demonstrating competence and test reliability (eg, accreditation/license), when applicable

17.3. Subject Identification, Enrollment, and Screening Logs

The investigator agrees to complete a subject identification and enrollment log to permit easy identification of each subject during and after the study. This document will be reviewed by the sponsor site contact for completeness.

The subject identification and enrollment log will be treated as confidential and will be filed by the investigator in the study file. To ensure subject confidentiality, no copy will be made. All reports and communications relating to the study will identify subjects by ID and date of birth. In cases where the subject is not randomized into the study, the date seen and the date of birth will be used.

The investigator must also complete a subject screening log, which reports on all subjects who were seen to determine eligibility for inclusion in the study.

17.4. Source Documentation

At a minimum, source documentation must be available for the following to confirm data collected in the eCRF: subject identification, eligibility, and study identification; study discussion and date of informed consent; dates of visits; results of safety and efficacy parameters as required by the protocol; record of all adverse events and follow-up of adverse events; concomitant medication; drug receipt/dispensing/return records; study drug administration information; and date of study completion and reason for early discontinuation of study drug or withdrawal from the study, when applicable.

In addition, the author of an entry in the source documents should be identifiable.

At a minimum, the type and level of detail of source data available for a study subject should be consistent with that commonly recorded at the site as a basis for standard medical care. Specific details required as source data for the study will be reviewed with the investigator before the study and will be described in the monitoring guidelines (or other equivalent document).

During the study, subjects will record PRO data on a tablet computer. After treatment discontinuation, PRO responses will be completed on paper and the site staff will enter data into the tablet computer. Both processes will be considered source documents.

17.5. Case Report Form Completion

Case report forms are provided for each subject in electronic format.

Electronic Data Capture (eDC) will be used for this study. The study data will be transcribed by study personnel from the source documents onto an eCRF, and transmitted in a secure manner to the sponsor within the timeframe agreed upon between the sponsor and the site. The electronic file will be considered to be the eCRF.

Worksheets may be used for the capture of some data to facilitate completion of the eCRF. Any such worksheets will become part of the subject's source documentation. All data relating to the study must be recorded in eCRFs prepared by the sponsor. Data must be entered into eCRFs in English. Designated site personnel must complete the eCRF as soon as possible after a subject visit, and the forms should be available for review at the next scheduled monitoring visit.

Every effort should be made to ensure that all subjective measurements (eg, pain scale information or other questionnaires) to be recorded in the eCRF are completed by the same individual who made the initial baseline determinations. The investigator must verify that all data entries in the eCRFs are accurate and correct.

All eCRF entries, corrections, and alterations must be made by the investigator or other authorized study-site personnel. When necessary, queries will be generated in the eDC tool. The investigator or an authorized member of the investigational staff must adjust the eCRF (when applicable) and complete the query.

When corrections to an eCRF are needed after the initial entry into the eCRF, this can be done in 3 different ways:

- Site personnel can make corrections in the eDC tool at their own initiative or as a response to an auto query (generated by the eDC tool)
- Site manager can generate a query for resolution by the investigational staff
- Clinical data manager can generate a query for resolution by the investigational staff

17.6. Data Quality Assurance/Quality Control

Steps to be taken to ensure the accuracy and reliability of data include the selection of qualified investigators and appropriate study sites, review of protocol procedures with the investigator and associated personnel before the study, and periodic monitoring visits by the sponsor. Written instructions will be provided for collection, preparation, and shipment of samples.

Guidelines for eCRF completion will be provided and reviewed with study personnel before the start of the study. The sponsor will review eCRFs for accuracy and completeness during on-site monitoring visits and after transmission to the sponsor; any discrepancies will be resolved with the investigator or designee, as appropriate. After upload of the data into the clinical study database they will be verified for accuracy and consistency with the data sources.

17.7. Record Retention

In compliance with the ICH/GCP guidelines, the investigator/institution will maintain all eCRFs and all source documents that support the data collected from each subject, as well as all study documents as specified in ICH/GCP Section 8, Essential Documents for the Conduct of a Clinical Trial, and all study documents as specified by the applicable regulatory requirement(s). The investigator/institution will take measures to prevent accidental or premature destruction of these documents.

Essential documents must be retained until at least 2 years after the last approval of a marketing application in an ICH region and until there are no pending or contemplated marketing applications in an ICH region or until at least 2 years have elapsed since the formal discontinuation of clinical development of the investigational product. These documents will be retained for a longer period when required by the applicable regulatory requirements or by an agreement with the sponsor. It is the responsibility of the sponsor to inform the investigator/institution as to when these documents no longer need to be retained.

If the responsible investigator retires, relocates, or for other reasons withdraws from the responsibility of keeping the study records, then custody must be transferred to a person who will accept the responsibility. The sponsor must be notified in writing of the name and address of the new custodian. Under no circumstance shall the investigator relocate or dispose of any study documents before having obtained written approval from the sponsor.

If it becomes necessary for the sponsor or the appropriate regulatory authority to review any documentation relating to this study, then the investigator must permit access to such reports.

17.8. Monitoring

The sponsor will perform on-site monitoring visits as frequently as necessary. The monitor will record dates of the visits in a study site visit log that will be kept at the site. The first post-initiation visit will be made as soon as possible after enrollment has begun. At these visits, the monitor will compare the data entered into the eCRFs with the hospital or clinic records (source documents). The nature and location of all source documents will be identified to ensure that all sources of original data required to complete the eCRF are known to the sponsor and investigational staff and are accessible for verification by the sponsor site contact. If electronic records are maintained at the investigational site, then the method of verification must be discussed with the investigational staff.

Direct access to source documentation (medical records) must be allowed for the purpose of verifying that the data recorded in the eCRF are consistent with the original source data. Findings from this review of eCRFs and source documents will be discussed with the investigational staff. The sponsor expects that, during monitoring visits, the relevant investigational staff will be available, the source documentation will be accessible, and a suitable environment will be provided for review of study-related documents. The monitor will meet with the investigator on a regular basis during the study to provide feedback on the study conduct.

17.9. Study Completion/Termination

17.9.1. Study Completion

The study is considered completed with the last study assessment for the last subject participating in the study. The final data from the investigational site will be sent to the sponsor (or designee) after completion of the final subject assessment at that site, in the time frame specified in the Clinical Trial Agreement.

17.9.2. Study Termination

The sponsor reserves the right to close the investigational site or terminate the study at any time for any reason at the sole discretion of the sponsor. Investigational sites will be closed upon study completion. An investigational site is considered closed when all required documents and study supplies have been collected and a site closure visit has been performed.

The investigator may initiate site closure at any time, provided there is reasonable cause and sufficient notice is given in advance of the intended termination.

Reasons for the early closure of an investigational site by the sponsor or investigator may include but are not limited to:

- Failure of the investigator to comply with the protocol, the requirements of the IEC/IRB or local health authorities, the sponsor's procedures, or GCP guidelines
- Inadequate recruitment of subjects by the investigator
- Discontinuation of further drug development

17.10. On-Site Audits

Representatives of the sponsor's clinical quality assurance department may visit the site at any time during or after completion of the study to conduct an audit of the study in compliance with regulatory guidelines and company policy. These audits will require access to all study records, including source documents, for inspection and comparison with the eCRFs. Subject privacy must, however, be respected. The investigator and staff are responsible for being present and available for consultation during routinely scheduled site audit visits conducted by the sponsor or its designees.

Similar auditing procedures may also be conducted by agents of any regulatory body, either as part of a national GCP compliance program or to review the results of this study in support of a regulatory submission. The investigator should immediately notify the sponsor when they have been contacted by a regulatory agency concerning an upcoming inspection.

17.11. Use of Information and Publication

All information, including but not limited to information regarding ZYTIGA or the sponsor's operations (eg, patent application, formulas, manufacturing processes, basic scientific data, prior clinical data, formulation information) supplied by the sponsor to the investigator and not previously published, and any data, including pharmacogenomic research data, generated as a result of this study, are considered confidential and remain the sole property of the sponsor. The investigator agrees to maintain this information in confidence and use this information only to accomplish this study, and will not use it for other purposes without the sponsor's prior written consent.

The investigator understands that the information developed in the clinical study will be used by the sponsor in connection with the continued development of ZYTIGA, and thus may be disclosed as required to other clinical investigators or regulatory agencies. To permit the information derived from the clinical studies to be used, the investigator is obligated to provide the sponsor with all data obtained in the study.

The results of the study will be reported in a Clinical Study Report generated by the sponsor and will contain eCRF data from all investigational sites that participated in the study. Recruitment performance or specific expertise related to the nature and the key assessment parameters of the study will be used to determine a coordinating investigator. Results of exploratory biomarker analyses performed after the Clinical Study Report has been issued will be reported in a separate report and will not require a revision of the Clinical Study Report. Study subject identifiers will not be used in publication of exploratory biomarker results. Any work created in connection with performance of the study and contained in the data that can benefit from copyright protection (except any publication by the investigator as provided for below) shall be the property of the sponsor as author and owner of copyright in such work.

The sponsor shall have the right to publish such data and information without approval from the investigator. If an investigator wishes to publish information from the study, then a copy of the manuscript must be provided to the sponsor for review at least 60 days before submission for publication or presentation. Expedited reviews will be arranged for abstracts, poster presentations, or other materials. If requested by the sponsor in writing, then the investigator will withhold such publication for up to an additional 60 days to allow for filing of a patent application. In the event that issues arise regarding scientific integrity or regulatory compliance, the sponsor will review these issues with the investigator. The sponsor will not mandate modifications to scientific content and does not have the right to suppress information. For multicenter study designs and substudy approaches, secondary results generally should not be published before the primary endpoints of a study have been published. Similarly, investigators will recognize the integrity of a multicenter study by not submitting for publication data derived from the individual site until the combined results from the completed study have been submitted for publication, within 12 months of the availability of the final data (tables, listings, graphs), or the sponsor confirms there will be no multicenter study publication. Authorship of publications resulting from this study will be based on the guidelines on authorship, such as those described in the Uniform Requirements for Manuscripts Submitted to Biomedical Journals, which state that the named authors must have made a significant contribution to the design of the study or analysis and interpretation of the data, provided critical review of the paper, and given final approval of the final version.

Registration of Clinical Studies and Disclosure of Results

The sponsor will register or disclose the existence of and the results of clinical studies as required by law.

REFERENCES

1. Adolfsson J, Steineck G, Hedlund PO. Deferred treatment of locally advanced non-metastatic prostate cancer: a long-term followup. J Urol 1999;161:505-508.
2. Ambs S, Prueitt R, Yi M, Hudson R, Howe T, Petrocca F, et al. Genomic profiling of MicroRNA and messenger RNA reveals deregulated microRNA expression in prostate cancer. Cancer Res 2008;68:15:6162-70.
3. Attard G, Reid AHM, Yap TA, Raynaud F, Dowsett M, Settatree S, et al. Phase I clinical trial of a selective inhibitor of CYP17, abiraterone acetate, confirms that castration-resistant prostate cancer commonly remains hormone driven. J Clin Oncol 2008;26(28);4563-4571.
4. Barrie SE, Potter GA, Goddard PM, et al. Pharmacology of novel steroidal inhibitors of cytochrome P45017α (17α hydroxylase/C17-20 lyase). J Steroid Biochem Mol Biol 1994;50:267–273
5. Cetin K, Beebe-Dimmer JL, Fryzek JP, Markus R, Carducci MA. Recent time trends in the epidemiology for stage IV prostate cancer in the United States: Analysis of data from the surveillance, epidemiology and end results program. Urology 2010;75:1396-1405.
6. deBono JS, Logothetis CJ, Molina A, Fizazi K, North S, Chu L. et al. Abiraterone and increased survival in metastatic prostate cancer. N Engl J Med 2011;364:1995-2005.
7. Efstathiou E, Davis JW, Troncoso P, Titus MA, Hoang A, Wen S, et al. Cytoreduction and androgen signalling modulation by abiraterone acetate (AA) plus leuprolide acetate (LHRHa) versus LHRHa in localized high-risk prostate cancer (PCa): Preliminary results of a randomized postoperative study. J Clin Oncol 30;2012 (suppl; abstr 4556).
8. EuroQol 1990. www.euroqol.org.
9. Flamand V et al. Observational survey on variations of prostate cancer incidence by stage in the Nord-Pas-de-Calais region between 1998 and 2004. Prog Urol. 2008;18:53-9.
10. He L, Hannon G. MicroRAs: small RNAs with a big role in gene regulation. Nat Rev Genet 2004;5:522-31.
11. Heidenreich A, Bolla M, Joniau S, Mason MD, Matveev V, Mottet N, et al. Guidelines on Prostate Cancer. European Association of Urology 2011. Update January 2011.
12. Howard GCW et al. Patterns of referral, management and survival of patients diagnosed with prostate cancer in Scotland during 1988 and 1993: results of a national retrospective population-based audit. BJUI. 2001;87:339-347.
13. Hussain M, Tangen CM, Higano C, Schelhammer PF, Faulkner J, Crawford ED, et al. Absolute Prostate-Specific Antigen Value After Androgen Deprivation Is a Strong Independent Predictor of Survival in New Metastatic Prostate Cancer: Data From Southwest Oncology Group Trial 9346 (INT-0162). J Clin Oncol 2006;24:3984-3990.
14. Hussain M, Tangen CM, Higano C, Crawford ED, Liu G, Wilding G, et al. Intermittent (IAD) versus continuouse androgen deprivation (CAD) in hormone sensitive metastatic prostate cancer (HSM1PC) patients: Results of S9346 (INT-0162), an international Phase III trial. J Clin Oncol 30;2012 (suppl; abstr 4).
15. International Agency for Research on Cancer (IARC). GLOBOCAN 2008 Database: Summary Table by Cancer. Available at http://www-dep.iarc.fr/. Accessed 27 August 2010.
16. Jack R et al. Prostate cancer incidence, stage at diagnosis, treatment and survival in ethnic groups in South-East England. BJUI. 2009;105:1226-1230.
17. Jonsson E et al. Adenocarcinoma of the prostate in Iceland: a population-based study of stage, Gleason grade, treatment and long-term survival in males diagnosed between 1983 and 1987. Scandinavian Journal of Urology and Nephrology. 2006;40:265-271.
18. Kumar MS, Lu J, Mercer KL, Golub TR, Jacks T. Impaired microRNA processing enhances cellular transformation and tumorigenesis. Nat Genet 2007;39:673–677.
19. Li B, Shi X, Xue L, Chen A, Valicenti R, deVere White R. Effect of down-regulation of microRNA 106b on p21-mediated cell cycle arrest in response to radiation in prostate cancer cells. ASCO Abstract 53 (2010 Genitourinary Cancers Symposium).

20. Lim L, Lau N, Garret-Engele P et al. Microarray analysis shows that some microRNAs downregulate large numbers of target mRNAs. Nature 2005;433:769-73.

21. Millikan RE, Wen S, Pagliaro LC, et al. Phase 3 trial of androgen ablation with or without three cycles of systemic chemotherapy for advanced prostate cancer. J Clin Oncol 2008;26:5936-5942.

22. Montgomery RB, Mostaghel EA, Vessella R, et al. Maintenance of intratumoral androgens in metastatic prostate cancer: a mechanism for castration-resistant tumor growth. Cancer Res 2008;68:4447-4454

23. Medical Research Council (MRC) Prostate Cancer Working Party Investigators Group. Immediate versus deferred treatment for advanced prostatic cancer: initial results of the Medical Research Council Trial. Br J Urol 1997;79:234-246.

24. Montgomery RB, Mostaghel EA, Vessella R, Hess DL, Kalhorn TF, Higano CS, et al. Maintenance of intratumoral androgens in metastatic prostate cancer: a mechanism for castration-resistant tumor growth. Cancer Res. 2008 68(11):4447-54.

25. Nair B, Wilt T, MacDonald R, Rutks I. Early vs. deferred androgen suppression in the treatment of advanced prostate cancer. Cochrane Database System Rev 2002;(1):CD 3506.

26. NCCN Clinical Practice Guidelines in Oncology (NCCN Guideline®) Prostate Cancer. Version 3.2012 (www.NCCN.org).

27. .Norgaard M, Jensen AO, Jacobsen JB, Cetin K, Fryzek JP, Sorensen HT. Skeletal related events, bone metastasis and survival of prostate cancer: a population based cohort study in Denmark. J Urol 2010;184:162-167.

28. O'Donnell A, Judson I, Dowsett M, et al. Hormonal impact of the 17α-hydroxylase/C17,20-lyase inhibitor abiraterone acetate (CB7630) in patients with prostate cancer. Br J Cancer 2004;90:2317 2325.

29. Ozen M, Creighton C, Ozdemir M, Ittmann M. Widespread deregulation of microRNA expression in human prostate cancer. Oncogene. 2008;27:1788-1793.

30. Porkka K, Pfeiffer M, Waltering K, Vessella R, Tammela T, Visakorpi T. MicroRNA expression profiling in Prostate cancer. Cancer Res 2007;67:6130-6135.

31. Quaglia A et al. Prostate cancer in Italy before and during the 'PSA era': survival trend and prognostic determinants. European Journal of Cancer Prevention. 2003;12:145-152.

32. Ryan CJ, Smith MR, De BonoJS, Molina A, Logothetis C, De Souza, PL, et al. Interim analysis (IA) results of COU-AA-302, a randomized, phase III study of abiraterone acetate (AA) in chemotherapy-naive patients (pts) with metastatic castration-resistant prostate cancer (mCRPC). J Clin Oncol 2012; 30(suppl; abstr LBA4518)

33. Schröder FH, Hugosson J, Carlsson S, Tammela T, Määttänen L, Auvinen A, et al. Screening for Prostate Cancer Decreases the Risk of Developing Metastatic Disease: Findings from the European Randomized Study of Screening for Prostate Cancer (ERSPC). Eur Urol. 2012 Jun 7 Abstract.

34. Shi X, Xue L, ang J, Ma A, Zhao J, Xu M, et al. An androgen regulated miRNA suppresses Bak1 expression and induces androgen-independent growth of prostate cancer cells. PNAS 2007:19983-19988.

35. Sun T, Yang M, Wang Q, Chen S, Balk S, Brown M, et al. Castration-Resistant Prostate cancer-related microRNAs. ASCO Abstract 150 (2010 Genitourinary Cancers Symposium).

36. Szczyrba J, Löprich E, Wach S, Jung V, Unteregger G, Barth S, et al. The microRNA profile of prostate carcinoma obtained by deep sequencing. Mol Cancer Res. 2010;8(4):529-38.

37. Tangen CM, Faulkner JR, Crawford Ed, Thompson IM, Hirano D, Eisenberger M, Hussain M. Ten-year survival in patients with metastatic prostate cancer. Clin Prostate Cancer 2003;2:41-45.

38. Taplin ME, Montgomery RB, Logothetis C, Bubley GJ, Richie JP, Dalkin BL, et al. Effect of neoadjuvant abiraterone acetate (AA) plus leuprolide acetate (LHRHa) on PSA, pathological complete response (pCR) and near pCR in localized high-risk prostate cancer (LHRPC): Results of a randomized phase II study. J Clin Oncol 30;2012 (suppl; abstr 4521).

39. Tong A, Fulgham P, Jay C, Chen P, Khalil I, Liu S, et al. MicroRNA profile analysis of human prostate cancers. Cancer Gene Therapy 2009: 16;206-216

40. Wang L, Tang H, Thayanithy V, Subramanian S, Oberg A, Cunningham J, Cerhan J, Steer C, Thibodeau S. Gene networks and microRNAs implicated in aggressive prostate cancer. Cancer Res. 2009: 69:9490-7.

41. Wang SK and Tsiatis AA. Approximately optimal one-parameter boundaries for group sequential trials. Biometrics 1987; 43:193-99.

42. Yamaoka M, Hara T, Kusaka M. Overcoming persistent dependency on androgen signaling after progression to castration-resistant prostate cancer. Clin Cancer Res 2010;16:4319-24.

Attachment 1: ECOG Performance Status

ECOG Grade	**Scale (with Karnofsky conversion)**
0	Fully active, able to carry on all predisease performance without restriction. (Karnofsky 90-100)
1	Restricted in physically strenuous activity but ambulatory and able to carry out work on a light or sedentary nature, eg, light housework, office work. (Karnofsky 70-80)
2	Ambulatory and capable of all self-care but unable to carry out any work activities. Up and about more than 50% of waking hours. (Karnofsky 50-60)
3	Capable of only limited self-care; confined to bed or chair more than 50% of waking hours. (Karnofsky 30-40)
4	Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair. (Karnofsky 10-20)
5	Dead. (Karnofsky 0)

Attachment 2: New York Heart Failure Criteria

The following table presents the New York Heart Association classification of cardiac disease:

Class	Functional Capacity	Objective Assessment
I	Patients with cardiac disease but without resulting limitations of physical activity. Ordinary physical activity does not cause undue fatigue, palpitation, dyspnea, or anginal pain.	No objective evidence of cardiovascular disease
II	Patients with cardiac disease resulting in slight limitation of physical activity. They are comfortable at rest. Ordinary physical activity results in fatigue, palpitation, dyspnea, or anginal pain.	Objective evidence of minimal cardiovascular disease
III	Patients with cardiac disease resulting in marked limitation of physical activity. They are comfortable at rest. Less than ordinary activity causes fatigue, palpitation, dyspnea, or anginal pain.	Objective evidence of moderately severe cardiovascular disease
IV	Patients with cardiac disease resulting in inability to carry on any physical activity without discomfort. Symptoms of heart failure or the anginal syndrome may be present even at rest. If any physical activity is undertaken, discomfort is increased.	Objective evidence of severe cardiovascular disease

LAST PAGE

Janssen Research & Development*

Clinical Protocol

A Randomized, Double-blind, Comparative Study of Abiraterone Acetate Plus Low-dose Prednisone Plus Androgen Deprivation Therapy (ADT) Versus ADT Alone in Newly Diagnosed Subjects With High-Risk, Metastatic Hormone-Naive Prostate Cancer (mHNPC)

Protocol 212082PCR3011; Phase 3

Amendment INT-3

JNJ-212082 (abiraterone acetate)

*Janssen Research & Development is a global organization that operates through different legal entities in various countries. Therefore, the legal entity acting as the sponsor for Janssen Research & Development studies may vary, such as, but not limited to Janssen Biotech, Inc; Janssen Products, LP; Janssen Biologics, B.V.; Janssen-Cilag International N.V.; Janssen, Inc; Janssen Infectious Diseases BVBA; Janssen R&D Ireland; or Janssen Research & Development, L.L.C. The term “sponsor” is used throughout the protocol to represent these various legal entities; the sponsor is identified on the Contact Information page that accompanies the protocol.

EudraCT NUMBER: 2012-002940-26

Status:	Approved
Date:	24 March 2016
Prepared by:	Janssen Research & Development, LLC
EDMS number:	EDMS-ERI-40268906; 4.0

Compliance: This study will be conducted in compliance with this protocol, Good Clinical Practice, and applicable regulatory requirements.

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

LIST OF ATTACHMENTS

PROTOCOL AMENDMENTS

Protocol Version	Issue Date
Original Protocol	31 July 2012
Amendment INT-1	27 November 2012
Amendment INT-2	18 April 2014
Amendment INT-3	24 March 2016

Amendments are listed beginning with the most recent amendment.

Amendment INT-3 (24 March 2016)

This amendment is considered to be non-substantial based on the criteria set forth in Article 10(a) of Directive 2001/20/EC of the European Parliament and the Council of the European Union.

The overall reason for the amendment: To revise protocol language to accommodate for the time gap between final radiographic progression-free survival (rPFS) and 1st interim analysis (IA) for overall survival (OS).

Applicable Section(s)	Description of Change(s)
Rationale: The death rate is slower than expected at protocol design; protocol language is revised to allow the 1st IA for OS to be performed with the final rPFS analysis.	
3.1 Overview of Study Design; 11.3.2 Analysis of Co-Primary Endpoint rPFS	Text was revised to clarify that the analysis of rPFS will occur at an estimated 565 rPFS events.
Synopsis Overview of Study Design; 3.1 Overview of Study Design; 11.3.1 Analysis of Co-Primary Endpoint OS 11.7 Interim Analysis; 16.1 Study-Specific Design Considerations	Text was revised to clarify timing of planned interim analyses for OS and final analysis for rPFS to occur at **approximately** 50% (~426 death events) and **approximately** 65% (~554 death events) of the total number of required (~852) death events for the final analysis.

Amendment INT-2 (18 April 2014)

This amendment is considered to be substantial based on the criteria set forth in Article 10(a) of Directive 2001/20/EC of the European Parliament and the Council of the European Union.

The overall reason for the amendment: To add radiographic progression-free survival (rPFS) as a co-primary endpoint with overall survival (OS)

Applicable Section(s)	Description of Change(s)
Rationale: Abiraterone acetate has already shown a statistically significant survival benefit in metastatic castration-resistant prostate cancer (mCRPC) patients who have received prior chemotherapy. In asymptomatic or mildly symptomatic mCRPC patients who were chemotherapy naïve, the demonstration of a highly significant and robust difference in radiographic Progression Free Survival (rPFS), significant improvements in the clinically important secondary endpoints of time to subsequent chemotherapy and time to opiate use for cancer pain and the strong trend in OS was considered clinically relevant and led to the approval of abiraterone acetate for use in this patient population. Recent data from studies in prostate cancer continue to support that the OS endpoint is challenging to show statistical significance in the metastatic prostate cancer patient population as well as in the more advanced mCRPC patient population. In this current study of subjects with newly diagnosed hormone-naïve high-risk metastatic prostate cancer, rPFS coupled with similar secondary end points is considered a measure of clinical benefit to patients. Radiographic PFS has been promoted from a secondary endpoint to a co-primary endpoint to provide an alternate measure of efficacy.	
Synopsis Primary Objective, Hypothesis, Efficacy Evaluations; Section 2.1. Objectives; Section 2.2. Hypothesis; Section 3.1. Overview of Study Design; Section 9.2.2. Efficacy Endpoints; Section 11.3.2. Analysis of Co-Primary Endpoint rPFS	Added rPFS as co-primary endpoint with OS. Moved the first Secondary Endpoint (rPFS) to Primary Endpoint along with OS. Updated definition of rPFS
Rationale: Change in the required number of events and change in sample size due to change in statistical parameters	
Statistical Methods; Section 3.1. Overview of Study Design; Section 11.2.1. Hypothesized OS Hazard Ratio of 0.81; Section 11.2.2. Hypothesized rPFS Hazard Ratio of 0.667	Changed the sample size from 1,270 to 1,200 subjects. Changed the anticipated duration of study to accumulate the new number of events required for OS analysis Added text regarding statistical parameters for rPFS
Rationale: Addition of Biomarker assessments on informed consent forms (ICFs) in the Time and Event Schedule to avoid confusion and delays in randomizing subjects,	
Time and Events Schedule	Added a row in the Biomarker subsection for informed consent to be completed.

Applicable Section(s)	Description of Change(s)
Rationale: Changed the 14-day window for Screening Phase assessments to a 28-day window to avoid having subjects make unnecessary visits to sites.	
Time and Events Schedule	In the Screening Phase, changed all 14-day windows to 28 days. Clarified that screening tests are to be done within the timeframe before randomization . Added the following footnote to PROs (Efficacy): “The subject is required to schedule the visit for PRO assessments either within 2 days prior to Randomization or on C1D1 (before any procedures are performed on C1D1).”
Rationale: Central review of bone scans can be performed on any scan not only those at baseline	
Time and Events Schedule; Section 9.2.1. Evaluations	Removed the word “baseline” so that central review is applicable to all scans; Added language to indicate that all scans will be collected and an audit may be performed using centralized independent review of scans from at least 160 subjects
Rationale: To allow End-of-Treatment bone scans to be eliminated if the subject had scans performed within 6 weeks prior to avoid the risk of additional diagnostic radiation exposure to the subject. To allow Screening Phase bone scans within 6 weeks prior to randomization	
Time and Events Schedule	Added the following footnote to CT/MRI and Bone Scan (Efficacy): “Bone scans or CT/MRI scans at End-Of-Treatment need not be done if scans have been done within 6 weeks prior.” Changed 28 days in the Screening Phase (time frame prior to randomization) to 6 weeks.
Rationale: To allow the end-of-treatment visit to occurr within 30 days of last dose	
Time and Events Schedule	Expanded the period for the end-of-treatment visit from 15 to 30 days to within 30 days of last dose
Rationale: To furher define Cycle 1 Day 1	
Time and Events Schedule	Added footnote to define Cycle 1 Day 1 as to occurr anytime within 72 hours after randomization
Rationale: Per Section 9.6 (Safety Evaluation) “ECGs should not be obtained when serum potassium is <3.5 mM/L. Hypokalemia should be corrected prior to ECG collection.” However, the trial protocol did not specifically state that the clinical laboratory results were to be available and evaluated before ECG recording.	
Time and Events Schedule; Section 9.6. Safety Evaluations	Added the following footnote to Electrocardiogram, 12-lead: “Note: Serum chemistry results must be available before ECG can be obtained.”
Rationale: To further define items required for subject monitoring in the Open-label Extension phase	
Time and Events Schedule	Removed the requirement to collect samples for hematology during the Open-label Extension Phase since long-term hematogical toxicity is not encountered with abiraterone acetate. Added dosing compliance evaluation during the open-label extension phase

Applicable Section(s)	Description of Change(s)
Rationale: Revised sections to allow the use of antiandrogens for 2 weeks only, after Cycle 1 Day 1 to adequately control tumor flare.	
Synopsis Overview of Study Design; Section 3.1. Overview of Study Design; Section 4.2. Exclusion Criteria; Section 8. Prestudy and Concomitant Therapy	Revised text such that subjects may also receive anti-androgens for 2 weeks after Cycle 1 Day 1 for subjects receiving an LHRH agonist. Modified exclusion criterion #5.1 to add "(for subjects receiving an LHRH agonist, antiandrogens are allowed for the first 2 weeks after Cycle 1 Day 1, any other use is prohibited)".
Rationale: Prednisone is used in this study; Prednisolone is not used	
Section 3.1 Overview of Study Design	Removed prednisolone from Figure 1.
Rationale: Clarified text to exclude subjects who have atrial fibrillation	
Section 4.2. Exclusion Criteria	Text was revised to exclude subjects who have existing atrial fibrillation or who have other cardiac arrhythmias that require pharmacotherapy.
Rationale: Defined ADT	
Synopsis Overview of Study Design; Section 3.1. Overview of Study Design; Section 4.2. Exclusion Criteria; Section 8. Prestudy or Concomitant Therapy	Text was revised to include LHRH antagonists as ADT allowed prior to study entry. During the study, subjects who did not have an orchiectomy will continue ADT with LHRH agonists only. Subjects may elect to undergo an orchiectomy during the study.
Rationale: Revised text to assure the integrity of the study is not compromised by unnecessary unblinding.	
Section 5. Treatment Allocation and Blinding	Text was revised to indicate that unblinding of any subject's data may be performed if the subject discontinues from the study because of disease progression and the investigator feels this information is necessary essential to determine the next course of therapy. Unblinding a subject for this situation can only take place after discussion with the sponsor's medical officer.
Rationale: Revised text to clarify enrollment in the study.	
Section 8. Prestudy and Concomitant Therapy	Text was revised to indicate that concurrent enrollment in another drug or device study during the Double-blind Treatment Phase or Open-Label Extension Phase is prohibited.
Rationale: To clarify administration of study drugs	
Section 6.1. Planned dose	Text was added to clarify that the prednisone/placebo dose need not be taken concurrently with the abiraterone acetate/placebo dose.
Rationale: Revised text to provide information and clarity to investigators with regard to the standard practice in monitoring LFT increase.	
Section 6.3. Guidelines for Abnormal Liver Function Test	Revised text to increase the frequency of liver function tests to ≥ 1 per week upon increases in ALT or AST to Grade 2 or 3.

Applicable Section(s)	Description of Change(s)
Rationale: Revised text to document increased doses of prednisone	
Section 6.4. Guidelines for Hypertension, Hypokalemia, and Fluid Retention/Edema due to Mineralocorticoid Excess	Added text to require the documentation of increased doses of prednisone as a concomitant medication.
Rationale: Revised text to clarify subjects' visits during the Double-blind Treatment Phase	
Time and Events Schedule; Section 9.1.3. Double Blind Treatment Phase.	Text was added to clarify clinic visits during the Double blind Treatment Phase as a footnote in the Time and Events Schedule and referred to the Schedule in Section 9.1.3.
Rationale: Revised text to emphasize the need for subjects to receive an LHRH agonist throughout the study or have had bilateral orchiectomy.	
Section 8. Prestudy and Concomitant Medications	Added text to require medical or surgical castration. Removed redundant text in Section 8.1 and modified Section 8.1 to only pertain to permitted supportive care and interventions during study.
Rationale: Text was updated to reflect current data on drug-drug interactions of abiraterone acetate and CYP2C8 and CYP3A4 substrates/inducers to align with the European Union (EU) Summary of Product Characteristics (SmPC) wording that was approved by the Committee for Human Medicinal Products on 25 July 2013 (labeling implementation 21 August 2013).	
Section 8.2. Special Concomitant Therapy	Precautionary text pertaining to the use of CYP3A4 inhibitors and abiraterone acetate was deleted. Text was added to describe results from interaction studies with CYP2C8 and CYP3A4 substrates/inducers.
Rationale: Revised blood volume table to account for protocol amendment changes in blood sampling	
Section 9.1.1. Overview, Table 1	Updated blood volume table to account for sampling changes; clarified treatment phases in footnotes to table
Rationale: Added text to define conditions to allow for retesting and rescreening of eligible subjects.	
Section 9.1.2. Screening Phase	Text was added to allow the rescreening of subjects under specific conditions as noted.
Rationale: Added time to pain progression as a secondary endpoint from an exploratory endpoint because of its appropriateness of measuring patients' wellness in this disease as a clinically relevant endpoint	
Synopsis Efficacy Endpoints; Section 9.2.2. Endpoints	Moved the pain measures: time to pain progression from an exploratory endpoint to a secondary endpoint
Rationale: Added text to clarify reasons for genetic studies and need for a separate informed consent form signature	
Section 9.3. Biomarkers	Text was added to provide reasoning for biomarker assessments and emphasis that the separate informed consent form needs to be signed by the subject

Applicable Section(s)	Description of Change(s)
Rationale: To define the conditions under which subjects are to be discontinued due to opioid use for cancer pain	
Section 10.2. Discontinuation of Treatment	Added text to define clinical progression as the initiation of new opioid analgesics or increased dose or frequency of existing opioid analgesics for ≥ 3 weeks orally or 7 days parenterally.
Rationale: Added text to provide instructions for storage of abiraterone acetate tablets by subjects and for handling of abiraterone acetate tablets by women.	
Section 14.2 Packaging; Section 14.4. Preparation, Handling, and Storage	Added text to describe storage conditions for abiraterone acetate and precautionary text regarding handling of abiraterone by women who may be pregnant or women of childbearing potential.
Rationale: Revised timing of interim analysis of OS based on new statistical parameters due to change in α from 0.05 to 0.049 for OS	
Synopsis; Section 3.1. Overview of Study Design; Section 11.7. Interim Analysis	The numbers of subjects to determine the point for interim and final analyses were modified based upon the change in α from 0.05 to 0.049 due to the addition of the rPFS co-primary endpoint
Rationale: Change in Statistical parameters due to addition of the rPFS co-primary endpoint	
Section 11.2. Sample Size Determination	Change the value of α from 0.05 for the OS determination to α = 0.049 for OS and α=0.001 for rPFS.
Rationale: To clarify the timing and conduct of PRO assessments	
Section 9.5. Patient-Reported Outcomes; Section 17.4. Source Documentation	Specified that PRO assessments be conducted before any other procedures and provided details on conducting PRO assessments by phone.
Rationale: To further define safety reporting during follow-up phase and later	
Section 12.3.2. Serious Adverse Events	Specified that deaths be reported via the CRF and serious adverse events via the CRF and SAE form if within 30 days of last dose and via the SAE form if > 30 days from last dose
Rationale: To define the time period to report Product Quality Complaints (PQC) to the Sponsor	
Section 13.1. Procedures	Changed text from PQCs must be reported to the sponsor as soon as possible to within 24 hours.
Rationale: Minor errors were noted and corrected	
Throughout the protocol	Minor grammatical, formatting, or spelling changes were made.

Amendment INT-1 (27 November 2012)

This amendment is considered to be substantial based on the criteria set forth in Article 10(a) of Directive 2001/20/EC of the European Parliament and the Council of the European Union.

The overall reason for the amendment: The overall reason for the amendment is to address requests and recommendations from health authorities (HA), Investigators, and Ethics Committees (ECs).

Applicable Section(s)	Description of Change(s)
Rationale: Request from HA to specify timeframe for duration of the Open-label Treatment Phase	
Synopsis, Section 3.1, Overview of Study Design	Removed text stating the abiraterone acetate would be provided until product is commercially available within a region to stating that abiraterone acetate would be provided for a maximum of 3 years. Added this information to the synopsis.
Rationale: Revised unblinding description to align with International Conference on Harmonisation (ICH) Good Clinical Practice (GCP) 5.13. 4. guidelines per request of HA	
Section 5, Treatment Allocation and Blinding	Replaced current text with protocol template text, which does follow the ICH guidelines
Rationale: Allowing sites to receive PSA and testosterone results per request of Investigators and ECs.	
Section 9.1.3, Double-blind Treatment Phase; Section 9.6, Safety Evaluations	Revised wording to state that PSA results will be made available to the investigators. Investigators will be notified if testosterone results do not fall below the definition of castration concentrations (<50 ng/dL or <1.7 nM).
Rationale: Per request of HA lists of strong inhibitors of CYP2D6, strong inhibitors and inducers of CYP3A4 should be included in the protocol. Additionally information regarding inhibitors of CYP2C8 was also included because abiraterone acetate was shown to be an inhibitor of CYP2C8 (in vitro).	
Section 8.2, Special Concomitant Therapy	Included additional information and links to appropriate websites in Attachment 4.
Rationale: The approach in the initial protocol was to capture the key components of the Child-Pugh classification in the Inclusion and Exclusion criteria. Based on the recommendation made by HA, the Inclusion and Exclusion criteria were updated to include all components of the Child-Pugh classification. These changes are meant to help minimize the complexity in assessing patients. Incorporating the components of Child-Pugh, as opposed to only stating the actual classification, makes it more prescriptive for the investigator. These proposed wording changes to the Inclusion and Exclusion criteria help ensure that those patients meeting the definition of moderate to severe classification of Child-Pugh will not be enrolled.	
Section 4.1, Inclusion Criteria; Section 4.2, Exclusion Criteria; Time and Events Schedule; Section 9.6, Safety Evaluations	Addition of serum albumin ≥ 3 g/dL to Inclusion Criterion #7 and change Exclusion Criterion #7 to read "Active or symptomatic viral hepatitis or chronic liver disease; ascites or bleeding disorders secondary to hepatic dysfunction" Serum albumin laboratory evaluation was added (Screening only)

Applicable Section(s)	Description of Change(s)
Rationale: The guidance in the protocol for dose modifications of abiraterone acetate and prednisone needed more detailed information.	
Section 6.2, Dose Modifications and Management of Toxicity; Section 6.4, Guidelines for Hypertension, Hypokalemia, and Fluid Retention/Edema Due to Mineralocorticoid Excess	Text was revised to include clearer guidance on dose modifications of abiraterone acetate or prednisone.
Rationale: Inclusion Criterion #5 was revised for greater clarity and consistency.	
Section 4.1, Inclusion Criteria	Inclusion Criterion #5 was revised to allow only CT or MRI for measurements of visceral metastases. Subjects may have more than 2 of the 3 required high-risk prognostic factors, therefore the criterion was revised to reflect this possibility.
Rationale: Correction to Exclusion Criterion #5 as curative intent is not applicable to metastatic disease and clarified allowed timing for prior therapy and resolution of adverse events for prior therapy	
Section 4.2, Exclusion Criteria	Removed “curative intent” wording from Exclusion Criterion #5. Revised timing for up to 3 months of ADT with LHRH agonists or orchiectomy with or without concurrent anti-androgens prior to Cycle 1 Day 1. Subjects may have one course of palliative radiation or surgical therapy to treat symptoms resulting from metastatic disease (eg, impending cord compression or obstructive symptoms) if it was administered at least 28 days prior to Cycle 1 Day 1. All adverse events associated with these procedures must be resolved at least to Grade 1 by Cycle 1 Day 1.
Rationale: Clearer definitions of treatment discontinuation were incorporated.	
Section 10.2 , Treatment Discontinuation	Included wording to better define clinical progression and allowing subjects to remain on study beyond radiographic progression if the Investigator thinks the subject will continue to receive benefit from treatment and no other treatment options are available.
Rationale: Safety information based on clinical studies was added for completeness	
Section 1.2, Clinical Experience with Abiraterone Acetate	Added updated safety information based on prescribing information.
Rationale: Added a column to the Time and Events Schedule to include Open-label Extension Phase for completeness.	
Time and Events Schedule	A column and information was added.
Rationale: Health Authorities and ECs requested to avoid the use of the brand name in the clinical protocol	
Throughout the protocol	ZYTIGA changed to abiraterone acetate
Rationale: To be consistent with the Imaging Charter, specify in the protocol that the number of scans read at baseline by the central reader will not exceed 100.	
Section 9.2.1, Efficacy Evaluations	Included a sentence stating that the number of scans read by the central reader at baseline will not exceed 100 as outlined in the Charter.

Applicable Section(s)	Description of Change(s)
Rationale: Confirmatory bone scan must show an additional 2 new lesions to be consistent with Prostate Cancer Working Group 2 (PCWG2) guidelines.	
Section 9.2.2	Added text for clarity
Rationale: Baseline echocardiograms are not required therefore Exclusion Criterion #9 was revised for clarity	
Section 4.2, Exclusion Criterion	Revised Exclusion Criteiron #9 to now read: Clinically significant heart disease as evidenced by myocardial infarction, or arterial thrombotic events or history of cardiac failure in the past 6 months, severe or unstable angina, or New York Heart Association (NYHA) Class II-IV heart disease ().
Rationale: Although this should rarely occur, there may be subjects whose testosterone does not fall below 50 ng/dL or 1.7 nM and therefore the product and dose of LHRH agonist may be changed	
Section 6.1, Planned Dose	Added sentence: Dosing (dose and frequency of administration) will be consistent with prescribing information and should only be adjusted if clinically indicated to achieve subcastrate concentrations of testosterone (<50 ng/dL or 1.7 nM)
Rationale: Axillary temperature readings are more commonly used in China therefore the wording “oral or aural” was removed for flexibility.	
Time and Events Schedule; Section 9.6, Safety Evaluations	Removed “oral or aural” from the Time and Events Schedule and body of protocol.
Rationale: To monitor for peripheral edema throughout treatment, removed statement that weight would only be obtained at screening	
Time and Events Schedule; Section 9.6, Safety Evaluations	Revised to clarify that weight will be measured at Screening and all visits during the Double-blind treatment Phase.
Rationale: Need uniform timing for baseline imaging to be obtained.	
Time and Events Schedule	Removed wording that scan taken with 3 months of randomization may be used if deemed evaluable by central reader.
Rationale: Inadvertently left out visit window for bone scans in original protocol	
Time and Events Schedule	Specified that imaging visits may occur up to 8 days before cycles requiring images
Rationale: Blood volumes were corrected	
Section 9.1.1, Overview	Blood volumes were corrected and a table was added for clarity.
Rationale: Room temperature definitions are not consistent across countries.	
Section 14.4, Preparation, Handling, and Storage	Revised wording to state that study drug will be stored at room temperature without specifying a specific temperature range.
Rationale: Inconsistencies noted during revisions of the amendment.	
Throughout the protocol	Inconsistencies were corrected.

Applicable Section(s)	Description of Change(s)
Rationale: References for the PCWG2 criteria and the Response Evaluation in Solid Tumors (RECIST) were missing from the original amendment.	
Reference section and in appropriate sections of the protocol	Added summary of RECIST criteria as Attachment 1 and Reference #8: Eisenhauer et al and reference #34: Scher et al. to the reference list and cited in the appropriate places in the body of the protocol.
Rationale: Minor errors were noted and updated sections based on new protocol template (Section 17.11)	
Throughout the protocol	Minor grammatical, formatting, or spelling changes were made. Updated wording in Section 17.11.

SYNOPSIS

A Randomized, Double-blind, Comparative Study of (Abiraterone Acetate) Plus Low-Dose Prednisone Plus Androgen Deprivation Therapy (ADT) Versus ADT Alone in Newly Diagnosed Subjects With High-Risk, Metastatic Hormone-naive Prostate Cancer (mHNPC)

ZYTIGA® (abiraterone acetate) is a prodrug of abiraterone, an irreversible inhibitor of 17α hydroxylase/C17, 20-lyase (cytochrome P450c17 [CYP17]), a key enzyme required for testosterone synthesis. Marketing authorization for ZYTIGA and prednisone/prednisolone was granted in April 2011 in the United States and in September 2011 in the European Union. Treatment with ZYTIGA improves survival in patients with metastatic castration-resistant prostate cancer. This study will test whether ZYTIGA improves the co-primary endpoints of radiographic progression-free survival (rPFS) and overall survival (OS) in patients with newly diagnosed metastatic, hormone-naive prostate cancer (mHNPC) who are at an increased risk for rapid disease progression.

OBJECTIVE AND HYPOTHESIS

Primary Objective

The primary objective is to determine whether abiraterone acetate in combination with low-dose prednisone and androgen deprivation therapy (ADT) is superior to ADT alone in improving rPFS and OS in subjects with mHNPC with high-risk prognostic factors.

Secondary Objective

The secondary objectives are to evaluate the clinically relevant improvements as well as the safety of abiraterone acetate plus low-dose prednisone and ADT compared with ADT alone and to identify microRNA (miRNA) and mRNA profiles predictive of abiraterone acetate response or resistance in this patient population.

Hypothesis

The hypothesis is that addition of abiraterone acetate to ADT as initial therapy (compared with ADT alone) will improve rPFS and OS in men with newly-diagnosed, high-risk mHNPC.

OVERVIEW OF STUDY DESIGN

This is a multinational, multicenter, randomized, double-blind, active-controlled study designed to determine if newly diagnosed subjects with mHNPC who have high-risk prognostic factors will benefit from the addition of abiraterone acetate and low-dose prednisone to ADT. The study population includes newly diagnosed (within 3 months prior to randomization) adult men with high-risk mHNPC (high-risk prognosis defined under Subject Selection). Subjects will be stratified by presence of visceral disease and Eastern Cooperative Oncology Group (ECOG) performance grade of 0, 1 vs. 2 prior to randomization. Subjects must have distant metastatic disease as documented by positive bone scan or metastatic lesions on CT or MRI to be eligible. Eligible subjects may have received ADT (LHRH agonist or antagonist) or had an orchiectomy within 3 months of randomization. Subjects may also receive anti-androgens for up to 3 months prior to randomization but the continued use of anti-androgens is allowed for only 2 weeks after Cycle 1 Day 1. Abiraterone acetate and low-dose prednisone or corresponding placebos will be considered as study drugs.

The study will consist of a Screening Phase of up to 28 days before randomization to establish study eligibility and document baseline measurements; a Double-blind Treatment Phase; and a Follow-up Phase of up to 60 months to monitor survival status and subsequent prostate cancer therapy. The Follow-up Phase will stop at the date of clinical cut-off for the final analysis of data for this study. Each cycle is 28 days. Treatment will continue until disease progression, withdrawal of consent, or the occurrence of unacceptable toxicity. Subjects who meet all of the inclusion criteria and none of the exclusion criteria will be centrally randomized in a 1:1 ratio to the active group (abiraterone acetate [1,000 mg once daily] plus low-dose prednisone [5 mg once daily] plus ADT [LHRH agonist or orchiectomy]) or the control group (abiraterone acetate and prednisone placebos plus ADT [LHRH agonist or orchiectomy]).

In the event of a positive study result at either of the interim analyses or at the time of the final analysis, all subjects will have the opportunity to enroll in an Open-label Extension Phase. The Open-label Extension Phase will allow subjects to receive active drug (abiraterone acetate plus prednisone) for up to 3 years. Two interim analyses in addition to the final analysis are planned for this study after observing approximately50% (~426 death events) and approximately 65% (~554 death events) of the total number of required (~852) death events for the final analysis.

Subjects will be monitored for safety throughout the study. Adverse events including laboratory adverse events will be graded and summarized using National Cancer Institute Common Terminology Criteria for Adverse Events (NCI-CTCAE), Version 4.0. Dose modification guidelines are provided. An Independent Data Monitoring Committee (IDMC) will be commissioned for the study to perform regular safety review and all data review at the planned interim analyses.

SUBJECT SELECTION

The study population includes newly diagnosed (within 3 months prior to randomization) men with mHNPC documented by positive bone scan or metastatic lesions at the time of diagnosis on CT or MRI, ≥18 years of age, with high-risk prognostic factors, and ECOG performance grade of 0, 1 or 2. High-risk prognosis is defined as having at least 2 of the following 3 risk factors: (1) Gleason score of ≥8; (2) presence of 3 or more lesions on bone scan; (3) presence of visceral metastasis.

DOSAGE AND ADMINISTRATION

Abiraterone acetate 1,000 mg (four 250 mg tablets) or placebo will be taken orally once daily continuously, concomitantly with oral low-dose prednisone 5 mg once daily. Abiraterone acetate/placebo must be taken on an empty stomach. No food should be consumed for at least 2 hours before the dose of abiraterone acetate/placebo is taken and for at least 1 hour after the dose of abiraterone acetate/placebo is taken. If an abiraterone acetate or placebo dose is missed, it should be omitted and will not be made up. Tablets should be swallowed whole with water.

All subjects will receive ADT (LHRH agonists) or have had surgical castration. The choice of the LHRH agonist will be at the Investigator's discretion. Dosage will be consistent with the prescribing information and should only be adjusted if clinically indicated to achieve or maintain subcastrate concentrations of testosterone (<50 ng/dL or <1.7 nM).

EFFICACY EVALUATIONS

Overall survival (OS) is defined as the time from randomization to date of death from any cause. Survival data will be collected throughout the Double-blind Treatment Phase and during the Follow-up Phase. Once a subject has completed the Double-blind Treatment Phase, OS follow-up will be performed every 4 months for up to 60 months (5 years) or until subject death, lost to follow-up, withdrawal of consent, or study termination.

Radiographic PFS (rPFS) is defined as the time from randomization to the occurrence of radiographic progression or death from any cause. Radiographic progression will be assessed by soft tissue lesion by CT/MRI per RECIST 1.1 or by bone lesion progression on bone scans per modified Prostate Cancer Working Group 2 (PCWG2) as defined in Section 9.2.2.

Patient-reported outcomes (PRO) data will be collected throughout the study as shown in the Time and Events Schedule.

Medical resource usage (MRU) data, associated with medical encounters, will be collected in the electronic case report form (eCRF) by the investigator and staff for all subjects throughout the study. For each adverse event that occurs, MRU information associated with that adverse event should be recorded whenever possible.

Efficacy Endpoints

Co-Primary: rPFS and OS

Secondary: Time to next skeletal related event, time to prostate specific antigen (PSA) progression, time to pain progression, and time to subsequent therapy for prostate cancer as well as time to initiation of chemotherapy

Exploratory: PSA response rate, PRO measures (EQ-5D-5L [Euro-QoL], Brief Pain Inventory-Short Form [BPI-SF], Functional Assessment Cancer Therapy-Prostate [FACT-P], Brief Fatigue Inventory [BFI]), time to symptomatic local progression defined as occurrence of urethral obstruction or bladder outlet obstruction, and prostate cancer specific survival.

BIOMARKER EVALUATIONS

Biomarker analyses are designed to confirm previously identified markers predictive of response (or resistance) to abiraterone acetate or correlated with poor prognosis. Tumor samples (formalin-fixed paraffin-embedded [FFPE] blocks or slides) obtained at the time of the original diagnosis will be collected from approximately 300 subjects to allow for molecular analyses. Serum samples will also be collected from approximately 300 subjects at multiple time points to allow for related analyses.

SAFETY EVALUATIONS

Safety evaluations include adverse events, vital signs measurements, physical examinations, and clinical laboratory tests.

STATISTICAL METHODS

The primary analysis population will use the intent-to-treat (ITT) population, which includes all randomized subjects. The ITT population will be used for the analysis of subject disposition and efficacy. The safety population includes all subjects who received at least 1 dose of study drug.

A sample size of approximately 1,200 subjects is planned.

Efficacy Analysis

Kaplan-Meier product limit method and Cox model will be used to estimate the time to event variables and to obtain the HR. Response rate variable will be evaluated using the Chi-square statistic.

Biomarkers Analysis

The associations of the above biomarkers with clinical response or survival endpoints will be assessed using appropriate statistical methods (analysis of variance [ANOVA], categorical, or survival model), depending on the endpoint. A detailed statistical analysis plan will be prepared for these exploratory studies.

Safety Analyses

The safety parameters to be evaluated are the incidence, intensity, and type of adverse events, clinically significant changes in the subject's physical examination findings, vital signs measurements, and clinical laboratory results. Exposure to study drug and reasons for discontinuation of study treatment will be tabulated.

TIME AND EVENTS SCHEDULE

PHASE	**Notes**	**Screening Phase**	**Double-blind Treatment Phase**[a]								**Follow-up Phase**	**Open-label Extension Phase**[b]
CYCLE (28 days):		Time frame prior to randomization	Cycle 1[c]		Cycle 2		Cycle 3		q1M from Cycle 4 to Cycle 13, then q2M until end of study treatment	End-of-Treatment Visit (within 30 days of last dose)	Every 4 months up to 60 months or until death, lost to follow up, withdrawal of consent or study termination	
CYCLE DAY:	Study visits/procedures have a ±2-day time window (except for CT/MRI and bone scans, see below).		1	15	1	15	1	15	1			
Screening												
Informed consent (must be performed before any study-specific procedures)		Within 28 days										X
Inclusion/exclusion criteria		Within 28 days										X
Prestudy anticancer therapy, palliative radiation, or surgical therapy for metastatic disease		Within 28 days										
Medical history and demographics[d]		Within 28 days										
Randomization			X									
Study Drug Administration												
Dispense study drug			X		X		X		X			X
Dosing compliance					X		X		X	X		X
Safety												
Electrocardiogram, 12-lead[e]	Serum potassium <3.5 mM should be corrected before ECG recording	Within 28 days										
Physical examination	Height recorded only at screening. Weight recorded at all visits[d]	Within 28 days			X		X		X	X		

PHASE	Notes	Screening Phase	Double-blind Treatment Phase[a]								Follow-up Phase	Open-label Extension Phase[b]
CYCLE (28 days):		Time frame prior to randomization	Cycle 1[c]		Cycle 2		Cycle 3		q1M from Cycle 4 to Cycle 13, then q2M until end of study treatment	End-of-Treatment Visit (within 30 days of last dose)	Every 4 months up to 60 months or until death, lost to follow up, withdrawal of consent or study termination	
CYCLE DAY:	Study visits/procedures have a ±2-day time window (except for CT/MRI and bone scans, see below).		1	15	1	15	1	15	1			
Vital signs	Blood pressure, heart rate, respiratory rate, and body temperature at screening; afterwards only blood pressure	Within 28 days	X		X		X		X	X		
Efficacy												
CT or MRI	The same modality (CT or MRI) should be used throughout the study. PET not acceptable. Imaging visits may occur up to 8 days before cycles requiring images. Unscheduled assessments if signs of disease progression are observed.	Within 28 days Central review of selected scans.							q4M starting with Cycle 5	X[f]		
Bone Scan	Scan visits may occur up to 8 days before cycles requiring images. Unscheduled assessments if signs of disease progression are observed.	Within 6 weeks Central review of selected scans							q4M starting with Cycle 5	X[f]		
ECOG		Within 28 days	X		X		X		X	X		
PROs[g]	EQ-5D-5L, BPI-SF, FACT-P, and BFI to be done before any other visit procedure.	Within 2 days	X		X		X		Until radiographic or clinical progression	X	EQ-5D-5L only; every 4 months for a total of 12 months.	
Medical Resource Utilization			X		X		X		X	X		

PHASE	**Notes**	**Screening Phase**	**Double-blind Treatment Phase[a]**								**Follow-up Phase**	**Open-label Extension Phase[b]**
CYCLE (28 days):		Time frame prior to randomization	Cycle 1[c]		Cycle 2		Cycle 3		q1M from Cycle 4 to Cycle 13, then q2M until end of study treatment	End-of-Treatment Visit (within 30 days of last dose)	Every 4 months up to 60 months or until death, lost to follow up, withdrawal of consent or study termination	
CYCLE DAY:	Study visits/procedures have a ±2-day time window (except for CT/MRI and bone scans, see below).		1	15	1	15	1	15	1			
Survival status and subsequent therapy	May be obtained by telephone or chart review.									X	X	X
Clinical Laboratory												
Hematology	Hemoglobin,WBCs including neutrophil count, platelet count	Within 28 days			X		X		X	X		
Serum chemistry	Potassium, creatinine, fasting glucose, lactate dehydrogenase, albumin (Screening only)	Within 28 days		X	X		X		X	X		X[b]
Liver function tests	ALT, AST, total bilirubin	Within 28 days		X	X	X	X	X	X	X		X[b]
PSA		Within 28 days	X		X		X		X			
Testosterone		Within 28 days	X				X		Cycle 6 only	X		
Ongoing Subject Review												
Concomitant therapy	*Continuous*											
Adverse events	*Continuous from time of Consent Form signature until 30 days after last study drug dose*											X
Biomarkers												
Biomarker Informed consent (must be performed before biomarker samples are obtained)		Within 28 days										
FFPE tumor blocks or slides for biomarker assessments	Samples obtained at time of original diagnosis from approximately 300 subjects		X									
Serum for microRNA profiling	Serum samples (8.5 mL) from approximately 300 subjects		X				X			X		

PHASE	Notes	Screening Phase	Double-blind Treatment Phase[a]								Follow-up Phase	Open-label Extension Phase[b]
CYCLE (28 days):		Time frame prior to randomization	Cycle 1[c]		Cycle 2		Cycle 3		q1M from Cycle 4 to Cycle 13, then q2M until end of study treatment	End-of-Treatment Visit (within 30 days of last dose)	Every 4 months up to 60 months or until death, lost to follow up, withdrawal of consent or study termination	
CYCLE DAY:	Study visits/procedures have a ±2-day time window (except for CT/MRI and bone scans, see below).		1	15	1	15	1	15	1			

ALT = alanine aminotransferase; AST = aspartate aminotransferase; BFI = Brief Fatigue Inventory; BPI-SF = Brief Pain Inventory - Short Form; CT = computed tomography; ECOG = Eastern Cooperative Oncology Group; EQ-5D-5L = Euro-QoL; FACT-P = Functional Assessment of Cancer Therapy-Prostate; FFPE = formalin-fixed paraffin-embedded; MRI = magnetic resonance imaging; PET = positron emission tomography; PROs = patient reported outcomes; PSA = prostate specific antigen; q1M = every month; q2M = every 2 months; q4M = every 4 months; WBCs=white blood cells

[a] Subjects should visit the clinic on Day 1 (+/- 2 days) of each cycle for assessments and on Day 15 (+/- 2 days) of each cycle for LFT assessments for the first 3 cycles. From Cycle 4 to Cycle 13, visits are monthly; After Cycle 13 visits are every other month until the end of double-blind treatment.

[b] During the Open-label Extension Phase subjects who were previously receiving placebos in the Double-blind Treatment Phase will be allowed to receive abiraterone acetate and prednisone treatment. Subjects formerly receiving placebos will have the same liver function and potassium monitoring for the first 3 months in the Open-label Extension Phase as was done in the Double-blind Treatment Phase. Those subjects who were receiving abiraterone acetate plus prednisone while in the Double-Blind Treatment Phase will forego the additional liver function monitoring. Subjects who were receiving placebos must meet entry eligibility requirements for hepatic function and cardiovascular disease before starting abiraterone acetate and prednisone treatment.

[c] Subjects must start study drug on Cycle1 Day 1; Cycle 1 Day 1 may occur anytime within 72 hours (3 calendar days) of randomization.

[d] Weight should be obtained with subjects wearing light clothing and no shoes.

[e] Note: Serum chemistry results must be available before ECG can be obtained. ECGs obtained within 28 days of randomization should be rechecked if prior ECG was obtained when potassium level was <3.5mm/L or unknown.

[f] Bone scans, CT, or MRI scans at End-Of-Treatment need not be done if scans have been done within 6 weeks prior.

[g] Only 1 baseline PRO assessment is required. Baseline PRO assessments can occur either within 2 days prior to randomization or on Cycle 1 Day 1 (before any procedures are performed).

ABBREVIATIONS

ACTH	adrenalcorticotrophic hormone
ADT	androgen deprivation therapy
ALT	alanine aminotransferase
AST	aspartate aminotransferase
BFI	Brief Fatigue Inventory
BPI-SF	Brief Pain Inventory-Short Form
CR	complete response
CRF	case report form (paper or electronic as appropriate for this study)
CTCAE	Common Terminology Criteria for Adverse Events
ECG	electrocardiogram
ECOG	Eastern Cooperative Oncology Group
eDC	electronic data capture
EQ-5D-5L	Euro-QoL
FACT-P	Functional Assessment of Cancer Therapy-Prostate
FFPE	formalin-fixed paraffin-embedded
GCP	Good Clinical Practice
HR	hazard ratio
IAC	Interim Analysis Committee
ICH	International Conference on Harmonisation
IDMC	Independent Data Monitoring Committee
IEC	Independent Ethics Committee
IRB	Institutional Review Board
IWRS	interactive web response system
LFT	liver function test
LHRH	lutenizing hormone releasing hormone
mCRPC	metastatic castration-resistant prostate cancer
MedDRA	Medical Dictionary for Regulatory Activities
mHNPC	metastatic hormone-naive prostate cancer
MRU	medical resource usage
NCI	National Cancer Institute
NE	not evaluable
PCWG2	Prostate Cancer Working Group 2
PQC	Product Quality Complaint
PR	partial response
PRO	patient-reported outcome(s)
PSA	prostate specific antigen
OS	overall survival
rPFS	radiographic progression-free survival
SWOG	Southwestern Oncology Group
ULN	upper limit of normal
USP	United States Pharmacopeia

1. INTRODUCTION

Prostate cancer is the most common noncutaneous cancer among men in Europe.[17] It is an androgen dependent disease and inhibition of testosterone is a key element in the control of prostate tumor growth. Of the 383,000 men in Europe diagnosed annually with prostate cancer,[17] approximately 15% to 30% present with metastatic disease.[18,19,10,14,29,33] On the contrary, patients presenting with newly diagnosed (M1) metastatic hormone-naive prostate cancer (mHNPC) account for approximately 4% of all prostate cancer diagnosis in the United States.[5] This disparity may result from differences in the use of prostate specific antigen (PSA) screening in different geographies.[36] Currently, the standard of care for patients with newly diagnosed mHNPC is androgen deprivation therapy (ADT), which consists of initiating a luteinizing hormone-releasing hormone (LHRH) agonist (medical castration) or less commonly, orchiectomy (surgical castration) with or without concurrent anti-androgens.[13] The Southwest Oncology Group (SWOG S8894) reported that 77% of men newly diagnosed with metastatic prostate cancer lived less than 5 years and only approximately 7% of men treated with hormonal therapy were alive at or after 10 years.[40] Several prognostic factors influence survival in M1 disease. Median overall survival (OS) has been reported to range from 13 months up to 75 months depending on the presence of high-risk prognostic features such as high PSA concentration at diagnosis, high Gleason score, increased volume of metastatic disease, as well as the presence of bony symptoms.[23]

In this study, high-risk patients with shortest median OS will be selected based on parameters described previously. The high-risk prognostic factor of Gleason score ≥8 was selected based on data from the SWOG 9346 study[15] which reported that it was a strong predictor for risk of death (hazard ratio[HR]=1.58 and $p<0.0001$). Baseline PSA alone was not considered a factor for selection of the high risk patient group because it was not as predictive of survival in univariate and multivariate models compared with Gleason score. In addition, baseline PSA alone did not show high association with post-baseline PSA decreases to below 4 ng/mL, a level that has been shown to have survival benefit.[15]

The second and third high-risk prognostic factors are both related to high-volume disease (defined as 3 or more lesions by bone scan or involvement of viscera). A single-center study of 286 patients has reported OS was 3.1 years in those with high-volume disease compared with 7.8 years in those with low-volume disease.[23]

Because the reduction of testosterone to castrate levels has been shown to improve survival of prostate cancer patients, initiation of ADT is standard of care for patients with M1 disease. A large randomized study in 938 men as well as a systematic review in men with locally advanced or asymptomatic metastatic disease demonstrated improved overall survival in those treated early.[25,27] The same study also demonstrated that the risk of deferring treatment increases the risk of developing debilitating symptoms such as pathological fractures and spinal cord compressions.[1,25] However, ADT is not curative and patients eventually progress to castration-resistance requiring further therapy including chemotherapy. Possible mechanisms of resistance to conventional ADT include the persistence of androgen production despite medical or surgical castration resulting from adrenal sources of testosterone or the up-regulation of intra-tumor testosterone production.[45]

This study evaluates the effectiveness of adding abiraterone acetate plus low-dose prednisone to ADT in men with newly diagnosed metastatic prostate cancer who have mHNPC and have a high-risk profile.

1.1. Abiraterone Acetate

ZYTIGA® (abiraterone acetate) is a prodrug of abiraterone, an irreversible inhibitor of 17α hydroxylase/C17, 20-lyase (cytochrome P450c17 [CYP17]), a key enzyme required for testosterone synthesis. This enzyme is found in the testes, adrenals, and prostate tumors.[4,24,30] The Marketing Authorisation for ZYTIGA and prednisone was granted in the United States in April 2011 and in the European Union in September 2011 for the treatment of metastatic castration-resistant prostate cancer (mCRPC) in adult men whose disease has progressed on or after a docetaxel-based chemotherapy regimen.

For the most comprehensive nonclinical and clinical information regarding the efficacy and safety of abiraterone acetate, refer to the latest version of the Investigator's Brochure for abiraterone acetate. The term sponsor used throughout this document refers to the entities listed in the Contact Information page(s), which will be provided as a separate document.

1.2. Clinical Experience with Abiraterone Acetate

Study COU-AA-301 was a Phase 3, multinational, randomized, double-blind, placebo-controlled study of oral abiraterone acetate and oral prednisone in 1,195 subjects with mCRPC whose disease had progressed on or after 1 or 2 chemotherapy regimens, at least one of which contained docetaxel. The study conclusively demonstrated that further lowering testosterone concentrations below those achieved with standard therapy to suppress androgen production (LHRH agonists or orchiectomy) using CYP17 inhibition with abiraterone acetate improves survival in patients with mCRPC.[6]

Study COU-AA-302 was a Phase 3, multinational, randomized, double-blind, placebo-controlled study of abiraterone acetate and oral prednisone in 1,088 asymptomatic or mildly symptomatic subjects with mCRPC who had not received chemotherapy. This study had co-primary endpoints of radiographic progression free survival (rPFS) and overall survival. Treatment with abiraterone acetate plus prednisone decreased the risk of radiographic progression or death by 57% compared with placebo plus prednisone (HR=0.425; $p<0.0001$). The study met its nominal significance level (0.01) for the co-primary endpoint of rPFS by independent review. There was a 25% decrease in the risk of death in the abiraterone acetate and prednisone group compared with the placebo plus prednisone group (HR=0.752; $p=0.0097$) when the Independent Data Monitoring Committee (IDMC) unanimously recommended unblinding the treatment and allowing subjects in the placebo group to receive abiraterone acetate. The median OS had not been reached for the abiraterone acetate group and was 27.2 months in the placebo group. The safety profile was similar, although the duration of treatment was longer, to that observed with abiraterone acetate plus prednisone in subjects in the post-docetaxel setting (COU-AA-301).

In healthy subjects after single dose administration of 1,000 mg abiraterone acetate, there is a substantial food effect and absorption of abiraterone acetate increases greatly with increasing fat content of a meal (Study COU-AA-009). Compared to administration after an overnight fast, geometric mean maximum concentration (C_{max}) and the area under the concentration-time curve (AUC) of abiraterone increased approximately 7-fold and 5-fold, respectively, when administered following a low-fat meal (estimated 2% of calories from fat) and increased by approximately 17-fold and 10-fold, respectively, when administered following a high-fat meal (estimated 56% of calories from fat). In the two large phase 3 randomized studies (COU-AA-301 and COU-AA-302), treatment with abiraterone acetate and prednisone had an acceptable safety profile and resulted in a favorable benefit/risk ratio. The safety profile of abiraterone acetate plus prednisone was distinct from that of cytotoxic agents. Adverse events usually did not interfere with administration of abiraterone acetate. In the combined dataset of safety for studies COU-AA-301 and COU-AA-302, the most frequently reported adverse events were fatigue (43.8%), back pain (32.6%), nausea (28.4%), arthralgia (29.5%), constipation (26.1%), bone pain (24.2%), peripheral edema (26.0%), hot flush (20.6%) and diarrhea (20.5%). Most events were Grade 1 or 2 in severity. Abiraterone acetate may cause hypertension, hypokalemia, and fluid retention as a consequence of increased mineralocorticoid levels resulting from CYP17 inhibition. Co-administration of a corticosteroid suppresses adrenocorticotropic hormone (ACTH) drive resulting in a reduction in incidence and severity of these adverse reactions. Caution is required in treating patients whose underlying medical conditions might be compromised by increases in blood pressure, hypokalemia (eg, those on cardiac glycosides), or fluid retention (eg, those with heart failure), severe or unstable angina pectoris, recent myocardial infarction or ventricular arrhythmia, and those with severe renal impairment.

1.3. Overall Rationale for the Study

Abiraterone acetate in combination with prednisone has been approved for the treatment of men with mCRPC who have received prior chemotherapy containing docetaxel. The efficacy and safety of abiraterone acetate (1,000 mg daily tablet dose) and prednisone (5 mg twice daily) therapy in patients with mCRPC is established by the results of Study COU-AA-301 and COU-AA-302, both Phase 3, multinational, randomized, double-blind, placebo-controlled studies. Study COU-AA-301 was the first Phase 3 study to demonstrate that further lowering testosterone concentrations below that achieved with ADT using CYP17 inhibition with abiraterone acetate improves survival in patients with mCRPC. The rationale for using abiraterone acetate in patients with high-risk prognostic factors in mHNPC is based on the positive results of Study COU-AA-301[6]and COU-AA-302[34] and the unmet medical need for alternative treatment options for these patients. Additionally, two studies using abiraterone acetate in addition to ADT in the neoadjuvant setting demonstrated higher rates of undetectable PSA and higher rates of complete pathologic response or near-complete pathologic response in patients undergoing prostatectomy for high risk localized prostate cancer, suggesting a potential therapeutic role for inhibiting extragonadal androgen synthesis prior to the emergence of castration-resistance.[41,7]

2. OBJECTIVES AND HYPOTHESIS

2.1. Objectives

Primary Objective

The primary objective is to determine whether abiraterone acetate in combination with low-dose prednisone and ADT is superior to ADT alone in improving rPFS and OS in subjects with mHNPC with high-risk prognostic factors.

Secondary Objective

The secondary objectives are

- To evaluate the clinically relevant improvements as well as the safety of abiraterone acetate plus low-dose prednisone and ADT compared to ADT alone.
- To identify microRNA (miRNA) and mRNA profiles predictive of abiraterone acetate response or resistance.

2.2. Hypothesis

The hypothesis is that addition of abiraterone acetate to ADT as initial therapy (compared with ADT alone) will improve rPFS and OS in men with newly-diagnosed, high-risk mHNPC.

3. STUDY DESIGN AND RATIONALE

3.1. Overview of Study Design

This is a multinational, randomized, double-blind, active-controlled study designed to determine if newly diagnosed subjects with mHNPC who have high-risk prognostic factors will benefit from the addition of abiraterone acetate and low-dose prednisone to ADT. Approximately 1,200 subjects will be enrolled. The study population includes newly diagnosed (within 3 months prior to randomization) adult men with high-risk mHNPC (high-risk prognosis defined in Section 4.1). Subjects will be stratified by presence of visceral disease (yes/no) and Eastern Cooperative Oncology Group (ECOG) performance grade (0, 1, versus 2) prior to randomization. Subjects must have distant metastatic disease as documented by positive bone scan or metastatic lesions on computed tomography (CT) or magnetic resonance imaging (MRI) to be eligible. Eligible subjects may have received ADT (LHRH agonist or antagonist) or had an orchiectomy within 3 months of randomization. Subjects may also receive an anti-androgen for up to 3 months prior to randomization. Upon entering the study, medical ADT should be continued with the use of an LHRH agonist only. For subjects receiving an LHRH agonist, the use of an anti-androgen is allowed for a maximum of 2 weeks after Cycle 1 Day 1. Subjects may elect to undergo an orchiectomy instead of medical ADT with an LHRH agonist during the study.

If the LHRH agonist is not started until subjects are randomized into the study, then investigators should initiate an anti-androgen shortly before or at the start of an LHRH agonist and continue its use for at least 7 days and up to 2 weeks after the start of an LHRH agonist.[28,13] This is to address the tumor flare that may be associated with the initiation of LHRH agonist. The continued use of anti-androgens beyond the first 2 weeks after Cycle 1 Day 1 is prohibited.

The study will consist of a Screening Phase of up to 28 days before randomization to establish study eligibility and document baseline measurements; a Double-blind Treatment Phase; and a Follow-up Phase of up to 60 months to monitor survival status and subsequent prostate cancer therapy. Each cycle is 28 days. Treatment will continue until disease progression, withdrawal of consent or the occurrence of unacceptable toxicity (see Section 10). In the event of a positive study result (efficacy boundary is crossed) at either of the interim analyses or at the final analysis, all subjects will have the opportunity to enroll in an Open-label Extension Phase of this protocol. The Open-label Extension Phase will allow subjects to receive active drug (abiraterone acetate plus prednisone) for up to 3 years (See Section 9.1.6).

Subjects who meet all of the inclusion criteria and none of the exclusion criteria will be centrally randomized in a 1:1 ratio to the active group (abiraterone acetate [1,000 mg once daily] plus low-dose prednisone [5 mg once daily]) plus ADT (LHRH agonist or orchiectomy) or control group (abiraterone acetate and prednisone placebos) plus ADT (LHRH agonist or orchiectomy). Selection of the LHRH agonist is by Investigator's choice provided that the dosing (dose and frequency of administration) is consistent with prescribing information. Each subject will be reviewed by the sponsor before randomization to ensure that select eligibility criteria have been met.

Two interim analyses for OS in addition to the final analysis are planned for this study after observing approximately 50% (~426 death events) and approximately 65% (~554 death events) of the total number of required (~852) death events for the final analysis.

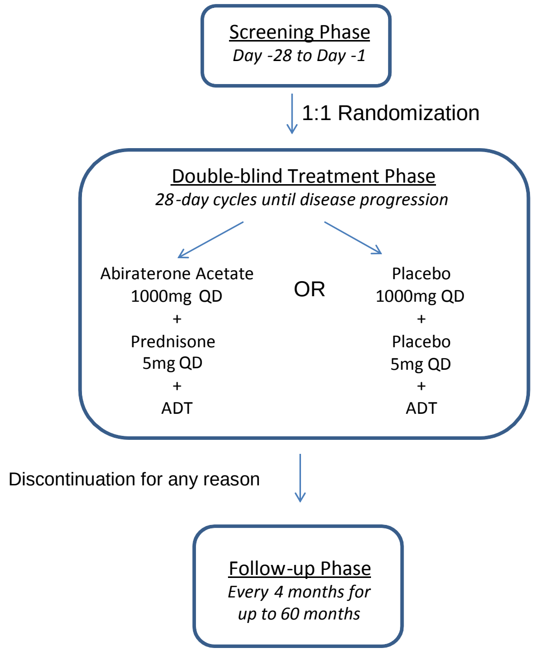
Only one analysis for rPFS is planned. The analysis of rPFS will be carried out at the two-tailed 0.001 level of significance at the estimated 565 rPFS events. It is expected that the rPFS analysis will likely occur in conjunction with the first interim OS analysis. The timing of the first interim analysis will be determined according to both rPFS and OS events required, so that the analysis will take place when the required number of events in both measures has been reached.

Tumor samples from FFPE blocks or slides collected at the time of diagnosis, and serum samples collected at multiple time points during the study, will be obtained from approximately 300 subjects for biomarker evaluations.

Subjects will be monitored for safety throughout the study. Adverse events including laboratory adverse events will be graded and summarized using National Cancer Institute Common Terminology Criteria for Adverse Events (NCI-CTCAE), Version 4.0. Dose modifications will be made as required according to dose modification rules. An Independent Data Monitoring Committee (IDMC) will be commissioned for the study to perform regular review of the safety data and all data review at the planned interim analyses.

A study flowchart is provided in Figure 1.

Figure 1: Study Flowchart



ADT=androgen deprivation therapy (may include surgical castration); QD=once a day

3.2. Study Design Rationale

Abiraterone acetate in combination with prednisone has been approved for the treatment of men with mCRPC who have received prior chemotherapy containing docetaxel. The efficacy and safety of abiraterone acetate (1,000 mg daily tablet dose) and prednisone (10 mg daily) therapy in patients with mCRPC is established by the results of Study COU-AA-301, a Phase 3, multinational, randomized, double-blind, placebo-controlled study that demonstrated that further lowering testosterone concentrations below that achieved with ADT using CYP17 inhibition with abiraterone acetate improves survival in patients with mCRPC. In addition, results have been reported from two randomized Phase 2 studies conducted in the neoadjuvant setting before a prostatectomy was performed in patients with high-risk localized disease. In these studies, the addition of abiraterone acetate to ADT compared with ADT alone demonstrated an improved PSA response, a higher rate of undetectable PSA concentration, and a higher rate of complete or near complete pathologic responses.[7,41] These results support the addition of abiraterone acetate to ADT in patients with hormone-naive disease, prior to the emergence of castration-resistance. Both Phase 2 studies used once-daily prednisone 5 mg instead of twice-daily as in Study COU-AA-301 and the lower prednisone dose sufficiently managed mineralocorticoid side effects. Inhibition of androgen biosynthesis with abiraterone acetate and prednisone is predicted to improve OS among patients with high-risk prognostic factors in mHNPC. Finally, the use of continuous ADT is supported by recent data from men with newly diagnosed hormone-sensitive metastatic prostate cancer presented by Hussain et al. In this large study of 1,535 eligible patients who were randomized to receive intermittent ADT or continuous ADT, intermittent ADT was not proven to be noninferior to continuous ADT.[16]

Blinding, Control, Study Phase/Periods, Treatment Groups

This is a randomized, double-blind, active-controlled study. Randomization will be used to minimize bias in the assignment of subjects to treatment groups, to increase the likelihood that known and unknown subject attributes (eg, demographic and baseline characteristics) are evenly balanced across treatment groups, and to enhance the validity of statistical comparisons across treatment groups. Blinding will also enhance the validity of PRO data.

Biomarker Collection

The goal of the biomarker study is to confirm biomarkers from other studies that are predictive of response to abiraterone acetate. The markers will be used in the development of future studies of anti-androgen therapies. The second goal is to confirm prognostic biomarkers from previous reports that were correlated with poor clinical outcome. These latter biomarkers are important for selection of “high risk” patients with earlier disease stage who could be treated in future clinical studies.

Patient Reported Outcomes (PROs)

The goal of collecting PRO data is to explore patient benefits in pain, fatigue, and functional status. In both Studies COU-AA-301 and COU-AA-302, significant delay in pain progression and improvements in functional status were found for abiraterone acetate plus prednisone compared with prednisone plus placebo. Patient reported outcomes will be measured utilizing the EuroQol EQ-5D-5L, Brief Pain Inventory-Short Form (BPI-SF), Functional Assessment of Cancer Therapy–Prostate (FACT-P), and Brief Fatigue Inventory (BFI). Data will be collected throughout the study as well as during the Follow-up Period (up to 12 months) as specified in the Time and Events Schedule.

Medical Resource Usage Data and Health Economics Data Collection

Medical resource usage data (MRU), associated with medical encounters, will be collected in the eCRF by the investigator and staff for all subjects throughout the study. For each adverse event that occurs, MRU information associated with that adverse event should be recorded when possible. Protocol-mandated procedures, tests, and encounters are excluded. Data may be used to conduct exploratory economic analyses (see Section 9.4).

4. SUBJECT SELECTION

The inclusion and exclusion criteria for enrolling subjects in this study are described in the following 2 subsections.

4.1. Inclusion Criteria

Each potential subject must satisfy all of the following criteria to be enrolled in the study:

1. Willing and able to provide written informed consent

2. Men age 18 years and older

3 Newly diagnosed prostate cancer with metastases within 3 months prior to randomization with histologically or cytologically confirmed adenocarcinoma of the prostate without neuroendocrine differentiation or small cell histology.

4. Distant metastatic disease documented by positive bone scan or metastatic lesions on CT or MRI.

5. Criterion modified per amendment

5.1 At least 2 of the 3 following high-risk prognostic factors:

- Gleason score of ≥8
- Presence of 3 or more lesions on bone scan
- Presence of measurable visceral (excluding lymph node disease) metastasis on CT or MRI (RECIST 1.1) (Attachment 1)

6. Eastern Cooperative Oncology Group (ECOG) performance status grade of 0, 1 or 2 (Attachment 2)

7. Criterion modified per amendment

7.1 Adequate hematologic, hepatic, and renal function:

- hemoglobin ≥9.0 g/dL independent of transfusions
- neutrophils ≥1.5 x 10^9/L
- platelets ≥100 x 10^9/L
- total bilirubin ≤1.5X upper limit of normal (ULN) [except for subjects with documented Gilbert's disease in which case total bilirubin not to exceed 10X ULN]
- alanine (ALT) and aspartate (AST) aminotransferase ≤2.5X ULN
- serum creatinine <1.5X ULN or calculated creatinine clearance ≥50 mL/min
- serum potassium ≥3.5 mM
- serum albumin ≥ 3.0 g/dL

8. Ability to swallow study medication tablets

9. Agrees to use a condom and another effective method of birth control if he is having sex with a woman of childbearing potential or agrees to use a condom if he is having sex with a woman who is pregnant

4.2. Exclusion Criteria

Subjects who meet any of the following criteria will be excluded from participating in the study:

1. Active infection or other medical condition that would make prednisone use contraindicated

2. Any chronic medical condition requiring a higher systemic dose of corticosteroid than 5 mg prednisone per day

3. Pathological finding consistent with small cell carcinoma of the prostate

4. Known brain metastasis

5. Criterion modified per amendment

5.1 Any prior pharmacotherapy, radiation therapy, or surgery for metastatic prostate cancer. The following exceptions are permitted:

   – Up to 3 months of ADT with LHRH agonists or antagonists or orchiectomy with or without concurrent anti-androgens prior to Cycle 1 Day 1 (for subjects receiving an LHRH agonist, antiandrogens are allowed for the first 2 weeks after Cycle 1 Day 1, any other use is prohibited).

   – Subjects may have one course of palliative radiation or surgical therapy to treat symptoms resulting from metastatic disease (eg, impending cord compression or obstructive symptoms) if it was administered at least 28 days prior to Cycle 1 Day 1. All adverse events associated with these procedures must be resolved at least to Grade 1 by Cycle 1 Day 1.

6. Uncontrolled hypertension (systolic blood pressure ≥160 mmHg or diastolic BP ≥95 mmHg). Subjects with a history of hypertension are allowed provided blood pressure is controlled by anti-hypertensive treatment

7. Criterion modified per amendment

7.1 Active or symptomatic viral hepatitis or chronic liver disease; ascites or bleeding disorders secondary to hepatic dysfunction

8. History of adrenal dysfunction

9. Criterion modified per amendment

9.1 Clinically significant heart disease as evidenced by myocardial infarction, or arterial thrombotic events or history of cardiac failure in the past 6 months, severe or unstable angina, or New York Heart Association (NYHA) Class II-IV heart disease (Attachment 3)

10. Criterion modified by amendment

10.1 Existing atrial fibrillation with or without pharmacotherapy. Other cardiac arrhythmia requiring pharmacotherapy

11 Other malignancy (within 5 years), except non-melanoma skin cancer

12 Administration of an investigational therapeutic or invasive surgical procedure (not including surgical castration) within 28 days of Cycle 1 Day 1 or currently enrolled in an investigational study.

13 Any condition or situation which, in the opinion of the investigator, would put the subject at risk, may confound study results, or interfere with the subject's participation in this study.

NOTE: Each subject will be reviewed by sponsor before randomization to ensure that select eligibility criteria have been met.

5. TREATMENT ALLOCATION AND BLINDING

Treatment Allocation

Subjects will be randomly assigned to the active or control group in a 1:1 ratio after the investigator has verified that all eligibility criteria have been met. Randomization will take place across all study sites using a centralized Interactive Web Response System (IWRS). A country by country randomization scheme will be performed using permuted block randomization. Subjects will be stratified by presence of visceral disease and ECOG performance grade.

At randomization, the IWRS will assign a unique subject identification number to each subject. The subject's identification number will be used on all study-related documents including electronic case report forms (eCRFs). A treatment number will also be assigned to each subject. This treatment number is the link between a subject's eCRF and blinded treatment group assignment. Subject identification numbers will not be reused. Subjects withdrawn from the study will not be replaced. All subjects must commence treatment within 72 hours (3 calendar days) of randomization.

Blinding

The investigator will not be provided with randomization codes. The codes will be maintained within the IWRS, which has the functionality to allow the investigator to break the blind for an individual subject.

- Under normal circumstances, the blind should not be broken until completion of the study or the IDMC recommendation for unblinding is accepted by the sponsor.
- The blind should be broken only if specific emergency treatment/course of action would be dictated by knowing the treatment status of the subject. In such cases, the investigator may in an emergency determine the identity of the treatment by contacting the IWRS. It is recommended that the investigator contact the sponsor or its designee if possible to discuss the particular situation before breaking the blind. Telephone contact with the sponsor or its designee will be available 24 hours per day, 7 days per week. In the event the blind is broken, the sponsor must be informed as soon as possible. The date, time, and reason for unblinding must be documented by the IWRS, in the appropriate section of the eCRF, and in the source document. The documentation received from the IWRS indicating the code break must be retained with the subject's source documents in a secure manner. Subjects who have had their treatment assignment unblinded should be discontinued from the Double-blind Treatment Phase and entered in the Follow-up Phase.
- Unblinding will be performed in the event that the IDMC recommends unblinding and subjects are to be crossed-over to receive abiraterone acetate.

- Unblinding may be performed if the subject discontinues from the study because of disease progression and the investigator feels this information is essential to determine the next course of therapy. Unblinding a subject for this situation can only take place after discussion with the sponsor's medical officer.
- In general, randomization codes will be disclosed fully only when the study is completed and the clinical database is closed.

6. DOSAGE AND ADMINISTRATION

6.1. Planned Dose

Abiraterone acetate 1,000 mg (four 250 mg tablets) or matching placebo should be taken orally once daily, in combination with oral low-dose prednisone 5mg or prednisone placebo. Placebos will be provided as tablets (abiraterone acetate placebo) or capsules (prednisone placebo) and will be matched in size, color, and shape to abiraterone acetate and prednisone to maintain the study blind. Abiraterone acetate/placebo must be taken on an empty stomach. No food should be consumed for at least 2 hours before the dose of abiraterone acetate/placebo is taken and for at least 1 hour after the dose of abiraterone acetate/placebo is taken. Tablets should be swallowed whole with water. Prednisone/placebo dose need not be taken at the same time as the abiraterone acetate/placebo dose. If an abiraterone acetate or placebo dose is missed, it should be omitted and will not be made up. Similarly if the prednisone/placebo dose is missed, it should be omitted and not made up.

All subjects will receive and remain on a stable regimen of ADT (LHRH agonists) or have had surgical castration. The choice of the LHRH agonist will be at the Investigator's discretion and is to be selected prior to randomization. Dosing (dose and frequency of administration) will be consistent with the prescribing information and should only be adjusted if clinically indicated to achieve and maintain subcastrate concentrations of testosterone (50 ng/dL or 1.7 nM).

6.2. Dose Modification and Management of Toxicity

In clinical studies in subjects with mCRPC, the most common adverse events related to abiraterone acetate have included fatigue most likely attributable to the underlying disease; and hypertension, hypokalemia, fluid retention/edema, and due to mineralocorticoid excess caused by compensatory ACTH drive. In this study, low-dose prednisone is expected to mitigate these effects through abrogation of the ACTH drive (see Section 6.4).

Following prolonged therapy with corticosteroids, subjects may develop Cushings syndrome characterized by central adiposity, thin skin, easy bruising, and proximal myopathy. Withdrawal of the corticosteroid may result in symptoms that include fever, myalgia, fatigue, arthralgia, and malaise. This may occur even without evidence of adrenal insufficiency.

In the event where dose-reduction is used for adverse event management, 2 dose reductions are allowed. At each dose reduction, 1 tablet of abiraterone acetate/placebo will be removed, eg, 4→3 tablets, and 3→2 tablets. Any return to protocol dose level (4 tablets) after dose reduction must follow documentation of adverse event resolution and a discussion with the sponsor.

6.3. Guidelines for Abnormal Liver Function Test

For subjects who develop liver function test abnormalities (ALT and/or AST greater than 5 X ULN but not exceeding 20 X ULN or total bilirubin greater than 3 X ULN but not exceeding 10X ULN), study drug must be interrupted. If increases in ALT and/or AST of Grade 2 (increase in ALT and/or AST to >2.5 to 5X ULN) or Grade 3 (increase in ALT and/or AST to > 5X ULN) occur, the frequency of liver function test monitoring must be increased to at least once a week. Treatment may be restarted at a reduced dose of 750 mg (3 tablets) once daily following return of liver function tests to the subject's baseline or to AST and ALT less than or equal to 2.5 X ULN and total bilirubin less than or equal to 1.5 X ULN.

For subjects with ALT and/or AST greater than 20 X ULN or total bilirubin greater than 10 X ULN, study drug must be interrupted. The decision to restart treatment at a reduced dose will be made in consultation with the sponsor medical monitor on an individual basis.

For subjects who resume treatment, serum transaminases and bilirubin should be monitored at a minimum of every 2 weeks for 3 months and monthly thereafter.

If liver function test abnormality recurs at the dose of 750 mg (3 tablets) once daily, re-treatment may be restarted at a reduced dose of 500 mg (2 tablets) once daily following return of liver function tests to the subject's baseline or to AST and ALT less than or equal to 2.5X ULN and total bilirubin less than or equal to 1.5X ULN. If liver function test abnormality recurs at the reduced dose of 500 mg (2 tablets) once daily, study drug must be discontinued.

6.4. Guidelines for Hypertension, Hypokalemia and Fluid Retention/Edema Due to Mineralocorticoid Excess

The study drug should be used with caution in subjects with a history of cardiovascular disease. The study drug may cause hypertension, hypokalemia, and fluid retention as a consequence of increased mineralocorticoid levels resulting from CYP17 inhibition. Caution should be exercised when treating subjects whose underlying medical conditions might be compromised by increases in blood pressure, hypokalemia, or fluid retention.

In patients with pre-existing hypokalemia or those who develop hypokalemia on study treatment, consider maintaining the subject’s potassium level at 4.0 mM or higher. If hypokalemia persists despite optimal potassium supplementation and adequate oral intake, or if any of the other mineralocorticoid effects persist, the dose of prednisone may be increased by 5 mg/day and documented in the study medication eCRF. The increased dose of prednisone (or prednisolone) must be obtained locally of an open-label source (eg, a prescription provided by investigator) and documented as a concomitant medication.

For subjects who develop drug-related Grade 3 or higher toxicities, including hypertension, hypokalemia, edema, and other non-mineralocorticoid toxicities, treatment must be withheld and appropriate medical management should be instituted. Treatment with abiraterone acetate must not be reinitiated until symptoms of the toxicity have resolved to Grade 1 or baseline. In the event where dose reduction is used for adverse event management, please refer to Section 6.2.

7. TREATMENT COMPLIANCE

Accurate records of all drug shipments as well as tablets dispensed and returned will be maintained. This inventory must be available for inspection by designated sponsor or regulatory authority representatives at any time. Drug supplies are to be used only in accordance with this protocol and under the supervision of the investigator. Study drug administration and dosing compliance will be assessed at each study visit, starting with Cycle 2. A count of all study drug provided by the sponsor will be conducted during the Double-blind Treatment Phase. Subjects will be required to maintain a medication diary. This medication diary will be provided by the sponsor and should be brought to study visits on Study Day 1 of each cycle for review of dosing compliance.

If dosing compliance is not 100% in the absence of toxicity, then subjects should be re-instructed regarding proper dosing procedures and may continue with the study. Subsequent dosing compliance procedure will be conducted at each study visit. If the number of study drug doses taken by the subject is ≤75% in the absence of toxicity or disease progression, then subjects should be re-instructed regarding proper dosing procedures. Subjects who have study drug dosing compliance of ≤75% for 2 consecutive cycles should be discontinued from the study.

The study site must maintain accurate records demonstrating dates and amount of study drug received, to whom dispensed (subject by subject accounting), and accounts of any study drug accidentally or deliberately destroyed. At the end of the study, reconciliation must be made between the amount of study drug supplied, dispensed, and subsequently destroyed or returned to sponsor or its representative.

8. PRESTUDY AND CONCOMITANT THERAPY

All pre-study therapies administered up to 28 days before first dose of study drug must be recorded at screening. All concomitant therapies must be recorded on the subject’s eCRF throughout the study beginning when the first dose of study drug is administered.

Subjects without bilateral orchiectomy within 3 months prior to randomization are required to receive a LHRH agonist throughout the study to maintain castrate testosterone levels (<50 ng/dL or <1.7 nM). Subjects who receive an LHRH agonist may receive up to 2 weeks of anti-androgen treatment following Cycle 1 Day 1 to prevent LHRH agonist mediated tumor flare (the continued use of antiandrogens beyond the first 2 weeks after Cycle 1 Day 1 is prohibited). Subjects who are initiated on LHRH agonist on Cycle 1 Day 1 must receive anti-androgens for a minimum of 7 days and a maximum of 2 weeks. Subjects are allowed to undergo a bilateral orchiectomy instead of ADT with an LHRH agonist during the study.

Concurrent enrollment in another investigational drug or device study during the Treatment Phase and the Open-Label Extension Phase is prohibited. The sponsor must be notified in advance (or as soon as possible thereafter) of any instances in which prohibited therapies are administered.

All therapies (prescriptions or over-the-counter medications, including vitamins and herbal supplements; non-pharmacologic therapies such as electrical stimulation, acupuncture, special diets, exercise regimens) different from study drug must be recorded in the concomitant therapy section of the eCRF.

8.1. Permitted Supportive Care and Interventions During Study

- Use of opioid analgesics as needed for cancer-related pain (Section 10.2)
- Bisphosphonates and denosumab for management of bone-related metastasis according to their market authorized approved label
- Conventional multivitamins, selenium and soy supplements
- Prednisone dose increase by 5 mg/day is permitted to manage refractory mineralocorticoid related toxicities
- Epelerenone can be used to manage mineralocorticoid-related toxicities
- Additional systemic glucocorticoid administration such as "stress dose" glucocorticoid is permitted when clinically indicated for a life threatening medical condition, and in such cases, the use of steroids will be documented as concomitant drug
- Transfusions and hematopoietic growth factors per institutional practice guidelines
- If the permissibility of a specific drug/treatment is in question, then please contact the study sponsor

8.2. Special Concomitant Therapy

The following concomitant therapies warrant special attention:

Caution is advised when abiraterone acetate is administered with medicinal products activated by or metabolized by CYP2D6, particularly with medicinal products that have a narrow therapeutic index. Dose reduction of medicinal products with a narrow therapeutic index that are metabolized by CYP2D6 should be considered. Examples of medicinal products metabolized by CYP2D6 include metoprolol, propranolol, desipramine, venlafaxine, haloperidol, risperidone, propafenone, flecainide, codeine, oxycodone and tramadol (the latter 3 products requiring CYP2D6 to form their active analgesic metabolites).

Based on *in vitro* data, abiraterone acetate is an inhibitor of the hepatic drug-metabolizing enzyme, CYP2C8. Examples of medicinal products metabolized by CYP2C8 include paclitaxel and repaglinide. In a CYP2C8 drug-drug interaction trial in healthy subjects, the AUC of pioglitazone was increased by 46% and the AUCs for M III and M IV, the active metabolites of pioglitazone, each decreased by 10% when pioglitazone was given together with a single dose of 1000 mg abiraterone acetate. Although these results indicate that no clinically meaningful increases in exposure are expected when ZYTIGA is combined with drugs that are predominantly eliminated by CYP2C8, patients should be monitored for signs of toxicity related to a CYP2C8 substrate with a narrow therapeutic index (e.g. paclitaxel) if used concomitantly with ZYTIGA.

Based on *in vitro* data, abiraterone is a substrate of CYP3A4. In a clinical pharmacokinetic interaction study of healthy subjects pretreated with a strong CYP3A4 inducer, rifampicin, 600 mg daily for 6 days followed by a single dose of abiraterone acetate 1,000 mg, the mean plasma AUC_∞ of abiraterone was decreased by 55%. Strong inducers of CYP3A4 (e.g., phenytoin, carbamazepine, rifampicin, rifabutin, rifapentine, phenobarbital, St John's wort [Hypericum perforatum]) during treatment are to be avoided, unless there is no therapeutic alternative. In a separate clinical pharmacokinetic interaction study of healthy subjects, co administration of ketoconazole, a strong inhibitor of CYP3A4, had no clinically meaningful effect on the pharmacokinetics of abiraterone.

For the most current information regarding potential drug-drug interactions with abiraterone acetate, refer to the latest version of the reference safety information (Investigator's Brochure or SmPC) for abiraterone acetate. For more information see Attachment 4.

8.3. Prohibited Medications

- Investigational agents other than abiraterone acetate
- Other antineoplastic agents
- Radiotherapy
- 5-α-reductase inhibitors
- Chemotherapy
- Immunotherapy
- Anti-androgens (e.g., bicalutamide, nilutamide, flutamide, cyproterone acetate) except in situations as outlined for management of tumor flare anticipated with the initiation of continuous LHRH agonist (see Section 3.1)
- Systemic ketoconazole (or other azole drugs such as fluconazole or itraconazole)
- Diethylstilbestrol (DES) or similar
- Other preparations such as saw palmetto thought to have endocrine effects on prostate cancer
- Radiopharmaceuticals such as strontium (89Sr) or samarium (153Sm) or similar analogues such as radium-223 (^{223}Ra)
- Spironolactone
- Digoxin, digitoxin, and other digitalis drugs
- Fludrocortisone acetate (Florinef)

9. STUDY EVALUATIONS

9.1. Study Procedures

9.1.1. Overview

The Time and Events Schedule included in the Synopsis summarizes the frequency and timing of efficacy, MRU, and safety assessments.

For subjects not participating in the biomarker study, blood volumes drawn will range from 2.5 to 7 mL per visit. Subjects participating in the biomarker study will have blood volumes drawn that range from 2.5 to 15.5 mL per visit. See table below for details.

Table 1: Summary of Blood Volumes per Visit

	SCR	C1D1	C1D15	C2D1	C2D15	C3D1/ C6D1	C3D15	C(n)D1	EOT	OLE
Serum	5.0	2.5	2.5	5.0	2.5	5.0	2.5	5.0	5.0	2.5
Hematology	2.0			2.0		2.0		2.0	2.0	
Total per visit	**7.0**	**2.5**	**2.5**	**7.0**	**2.5**	**7.0**	**2.5**	**7.0**	**7.0**	**2.5**
miRNA		8.5				8.5			8.5	
Total W/Biomarkers per visit	**7.0**	**11.0**	**2.5**	**7.0**	**2.5**	**15.5**	**2.5**	**7.0**	**15.5**	**2.5**
C=cycle; D=day; EOT=end-of-treatment; OLE= open-label extension; SCR=screening; C(n) D1 refers to sampling every cycle from Cycle 4 to Cycle 13 and every other cycle from Cycle 14 to EOT.										

9.1.2. Screening Phase

A signed informed consent must be obtained before any study-specific assessments are performed. Informed consent may be obtained up to 28 days before the first dose of study drug. Screening procedures to evaluate subject eligibility for the study will be conducted within 28 days prior to Cycle 1 Day 1.

Subjects who do not meet all inclusion criteria or who meet an exclusion criterion may, at the discretion of the investigator, be rescreened once, upon the sponsor’s approval and agreement, if the non-eligibility is related to duration of an event that led to ineligibility or those events that can be managed by appropriate clinical measures leading to study eligibility (eg, resolution of a medical condition which required treatment with a contraindicated medication, treatment of cause of abnormal laboratory values, vital signs).

Subjects who are to be rescreened must sign a new informed consent before rescreening. Subjects rescreened within 4 weeks of planned randomization may use the initial screening laboratory results, CT/MRI and bone scans to determine eligibility. Re-screening and subsequent randomization activities (with the exception of use of original screening labs and CT/MRI and Bone scans) must be conducted in accordance with all protocol defined windows and timelines (eg, subject must still meet inclusion criteria of newly diagnosed metastatic prostate cancer within 3 months prior to randomization).

Once select eligibility criteria are confirmed by the sponsor, subjects will be randomized to a treatment group according to the randomization schedule.

Retesting of any laboratory assessments must be approved by the sponsor.

9.1.3. Double-Blind Treatment Phase

The Double-blind Treatment Phase begins on Cycle 1 Day 1 of treatment with study medication and continues until all study medication is discontinued. Subjects will be randomly assigned to either the active or control group, and must start taking study drug within 72 hours (3 calendar days) of randomization. Visits for each cycle will have a ± 2 day window. Study visits will be calculated from the Cycle 1 Day 1 date. Subjects may have imaging visits up to 8 days before cycles requiring images (Cycle 5 Day 1, Cycle 9 Day 1, and Cycle 13 Day 1 and every 4th cycle beyond Cycle 13) and at End-of-Treatment Visit. Please refer to T&E Schedule for treatment visits and assessments during the double-blind treatment phase.

During the Double-blind Treatment Phase, investigators will review all clinical, laboratory and imaging data, and will use this information to make decisions about dose modification and study medication discontinuation. Prostate-specific antigen results from every other cycle starting with Cycle 5 will be available to the investigators. Investigators will be notified if testosterone results do not fall below the definition of castration concentrations (<50 ng/dL or <1.7 nM).

9.1.4. End-of-Treatment Visit

The End-of-Treatment Visit should be scheduled within 30 days after the final dose of study drug to collect posttreatment safety and efficacy data.

9.1.5. Follow-Up Phase

After discontinuation of all study medication, subject status will be monitored every 4 months for up to 60 months (5 years) or until subject death, lost to follow-up, withdrawal of consent, or study termination. Information regarding survival status and initiation of subsequent prostate cancer therapies will be collected. Visits to the clinic are not required; sites may collect the information by telephone interview, chart review, or other convenient methods. If the follow-up information is obtained via telephone contact, then written documentation of the communication must be available for review in the source documents. Completion of the EQ-5D-5L questionnaire every 4 months by the subject will continue for 12 months after the end of treatment. Deaths regardless of causality will be reported to the sponsor within 24 hours of discovery or notification of the event and documented (see Section 12.3.2).

9.1.6. Open-Label Extension Phase

In the event of a positive study result at either of the interim analyses or at the final analysis, all subjects in the Double-Blind Treatment Phase will have the opportunity to enroll in an Open-label Extension Phase of this protocol. The Open-label Extension Phase will allow subjects to receive active drug (abiraterone acetate plus prednisone) for up to 3 years. Subjects will be monitored for long-term survival and adverse events. During the Open-label Extension Phase subjects who were previously receiving placebos in the Double-Blind Treatment Phase will be allowed to receive abiraterone acetate plus prednisone treatment. Subjects formerly receiving placebos will have the same liver function and potassium monitoring for the first 3 months in the Open-label Extension Phase as was done in the Double-Blind Treatment Phase. Those subjects who were previously receiving abiraterone acetate plus prednisone while in the

Double-Blind Treatment Phase will forego the additional liver function monitoring. Subjects who were receiving placebos must meet entry eligibility requirements for hepatic function and cardiovascular disease before starting abiraterone acetate plus prednisone treatment as described in the T&E schedule.

9.2. Efficacy

9.2.1. Evaluations

Unscheduled tumor assessment and appropriate imaging should be considered if signs or symptoms suggestive of disease progression, including escalating pain not referable to another cause, worsening ECOG performance status grade, or physical examination findings consistent with disease progression, are recorded.

The Company intends to collect all scans in a central location so that a valid audit can be performed using independent centralized review of blinded radiographic data for a random sample of at least 160 subjects.

9.2.2. Endpoints

Primary Endpoints

- The co-primary efficacy endpoint of OS is defined as the time from randomization to date of death from any cause.

- The co-primary endpoint of rPFS, based on Prostate Cancer Working Group 2 (PCWG2)[35] and RECIST 1.1 (Attachment 1), is defined as the time from randomization to the occurrence of radiographic progression or death from any cause.

Radiographic progression in this protocol is defined as one of the following:

I. Progression of soft tissue lesions measured by CT or MRI as defined in RECIST 1.1 (Attachment 1)

II. Progression by bone lesions observed on bone scan defining as one of the following:

Each subject on study would fall into one of the following scenarios, depending on what is observed at Cycle 5 Day 1 scan and what is observed on the confirmatory scan. Note that Cycle 5 Day 1 scan need to be at 16 weeks or later from Cycle 1 Day 1.

1. Subjects who are observed to have ≥2 new bone lesions on Cycle 5 Day 1 scans compared with baseline scans, will need to have a confirmatory scan performed 6 or more weeks later and would fall into one of the 2 categories below:

    a. Subjects with confirmatory scans that show at least 2 new lesions compared to the Cycle 5 Day 1 scans (so a total of at least 4 new lesions compared with baseline scan) will be considered to have disease progression by bone scan.

b. Subjects who on confirmatory scan did not have at least 2 new lesions compared with the Cycle 5 Day 1 scan will not be considered to have disease progression at that time. For these subjects all subsequent scans would serve as "confirmatory scans" for the Cycle 5 Day 1 scan findings. In order to be considered to have disease progression, these subjects will need to have subsequent scans with at least 2 new lesions compared with Cycle 5 Day 1 scans (ie, at least 4 total new lesions compared to baseline scan).

2. Subjects who at Cycle 5 Day 1 do not have ≥2 new bone lesions compared with baseline would not need to have confirmatory scan. For any subsequent scan beyond the Cycle 5 scan time point, the FIRST scan that would show at least 2 new lesions compared with baseline scans, would be considered to demonstrate disease progression by bone scan.

Secondary Endpoints

- Time to next skeletal-related event (one of the following):
  - Clinical fracture
  - Spinal cord compression
  - Palliative radiation to bone
  - Surgery to bone
- Time to PSA progression (by PCWG2 criteria)[35]
- Time to next subsequent therapy for prostate cancer
- Time to initiation of chemotherapy
- Pain Measures (Time to Pain Progression)

Exploratory Endpoints

- PSA response rate
- Patient Reported Outcome (PRO) measures: EQ-5D-5L, BPI-SF, FACT - P, BFI
- Time to symptomatic local progression defined as occurrence of urethral obstruction or bladder outlet obstruction
- Prostate cancer specific survival

9.3. Biomarkers

A separate informed consent form needs to be signed by subjects for biomarker assessments. Tumor samples from FFPE blocks or slides will be requested from the archival site at the Cycle 1 Day 1 visit. These samples will be collected from approximately 300 subjects. Serum samples will also be collected from approximately 300 subjects at the time points indicated in the Time and Events Schedule. Tumor and blood biomarker analyses are dependent upon the availability of appropriate biomarker assays and may be deferred or not performed, if during or at the end of the study, it becomes clear that the analysis will have no scientific value, or there are not enough samples or not enough responders to allow for adequate biomarker evaluation. In the event the study is terminated early or does not reach a positive primary endpoint, completion of biomarker assessments is based on justification and intended utility of the data.

Biomarker studies on the FFPE blocks or slides of tumor samples may include immunohistochemistry analyses, global miRNA profiling, gene-expression profiling, and somatic mutational analyses.

MicroRNAs (miRNAs) have been widely studied in prostate cancer and are small noncoding RNAs that regulate the expression of protein-coding genes by modulating both mRNA stability and translation.[12,22]Alteration of miRNA expression levels can alter cell function and induce cellular transformation[20] leading to cancer. A number of dysregulated miRNAs have been associated with different prostate cancer stages and some miRNAs are consistently dysregulated at early and advanced stages of disease or associated with more aggressive disease. Considering that some miRNAs are androgen controlled,[32,2,38,32,21,43,39,38,42] it is anticipated that dysregulated miRNAs patterns in baseline samples may predict response to drugs such as abiraterone acetate. Changes in miRNA expression levels from baseline may allow elucidation of mechanisms leading to resistance. Dysregulation of steroid synthesis has been previously reported in xenograft models[26] and evaluation of steroid profiles may complement the miRNA studies. Herein, we plan to analyze dysregulated steroid transcript levels by gene expression profiling to confirm the predictive profiles found in these previously reported studies. Data collected from this study will be compared to data obtained from prior studies in asymptomatic or mildly symptomatic metastatic castration-resistant prostate cancer to identify miRNA and gene expression profiling (GEP) profiles that correlate with response (or primary resistance) to abiraterone. The biomarker results from this study will then be used to inform future studies of anti-androgen therapies possibly leading to product differentiation by selection of responsive subjects.

miRNAs have also been identified that correlate with high risk prostate cancer.[43,42,2,31] Since it is difficult to histologically distinguish high risk from benign disease and because PSA velocity and doubling time may be an unreliable measure of disease aggression, we plan to investigate whether miRNA profiles may better define high risk prostate cancer in the early metastatic disease setting. This data may then be utilized in selection of high risk patients in future studies if these previously reported miRNA profiles are confirmed and found to be more sensitive than conventional clinical estimates of high risk disease.

9.4. Medical Resource Utilization

Medical resource usage (MRU) data associated with medical encounters will be collected in the eCRF by the investigator and staff for all subjects throughout the study. For each adverse event that occurs, MRU information associated with that adverse event should be recorded when possible. Protocol-mandated procedures, tests, and encounters are excluded. The following data may be used to conduct exploratory economic analyses:

- Number and duration of medical care encounters, including surgeries, and other selected procedures (inpatient and outpatient)
- Duration of hospitalization (total days length of stay, including duration by wards; eg, intensive care unit)
- Number and character of diagnostic and therapeutic tests and procedures
- Outpatient medical encounters and treatments (including physician or emergency room visits, tests and procedures, and medications)

9.5. Patient-Reported Outcomes

All visit-specific PRO assessments should be conducted before any tests, procedures, or other consultations for that visit to prevent influencing subject perceptions. All PRO measures will be collected at the times specified in the Time and Events Schedule. Only 1 baseline PRO assessment is required. Baseline PRO assessments can occur either within 2 days prior to randomization or on Cycle 1 Day 1 (before any procedures are performed). Patient-reported outcome measures will be collected until radiographic or clinical progression, after which only the EQ-5D-5L will be collected for a period of 12 months.

For subjects that discontinued from study for reasons other than death, radiographic or clinical progression, all 4 PRO measures should be collected until death, or radiographic/clinical progression. After such progression only the EQ-5D-5L will be collected for a period of 12 months.

9.5.1. Brief Pain Inventory-Short Form (BPI-SF)

Pain will be evaluated using the BPI-SF instrument. A sample of the BPI-SF questionnaire along with detailed instructions will be provided in the PRO Manual.

9.5.2. Functional Assessment of Cancer Therapy–Prostate (FACT-P)

The FACT-P questionnaire is a tool used for assessing the health-related quality of life in men with prostate cancer. A sample of the FACT-P questionnaire along with detailed instructions will be provided in the PRO Manual.

9.5.3. Brief Fatigue Inventory (BFI)

Fatigue will be evaluated using the BFI instrument. A sample of the BFI questionnaire along with detailed instructions will be provided in the PRO Manual.

9.5.4. EQ-5D-5L

The EQ-5D-5L (EuroQol Group, www.euroqol.org) is a validated tool that measures mobility, self-care, usual activities, pain, discomfort, and anxiety/depression.[9] The EQ-5D quality of life instrument consists of a 5-item questionnaire and a visual analogue scale ranging from 0 (worst imaginable health state) to 100 (best imaginable health state) that integrates many aspects of the subject's disease process into a single assessment. A sample of the EQ-5D questionnaire along with detailed instructions will be provided in the PRO Manual.

9.6. Safety Evaluations

The evaluation period for safety will start at the time a signed and dated informed consent form is obtained to at least 30 days after the last dose of study drug or recovery from all acute toxicities associated with the study drug administration. Adverse events will be reported by the subject for the duration of the study. Adverse events including laboratory adverse events will be graded and summarized according to the NCI-CTCAE, Version 4.0.

Any clinically significant abnormalities persisting at the end of the study or during early withdrawal will be followed by the investigator until resolution or until a clinically stable endpoint is reached. The study will include the following evaluations of safety according to the time points provided in the Time and Events Schedule.

Adverse Events

Adverse events will be reported by the subject (or, when appropriate, by a caregiver, surrogate, or the subject's legally-acceptable representative). Adverse events will be followed by the investigator as specified in Section 12, Adverse Event Reporting.

Clinical Laboratory Tests

All laboratory tests will be performed by the central laboratory. Blood samples for serum chemistry and hematology will be collected. The following tests will be performed:

Hematology Panel	Serum Chemistry Panel	Liver Function Tests
• hemoglobin	• potassium	• alanine aminotransferase (ALT/SGPT)
• platelet count	• creatinine	• aspartate aminotransferase (AST/SGOT)
• white blood cells including neutrophil count	• fasting glucose • lactate dehydrogenase • albumin (Screening only)	• total bilirubin

In addition, testosterone and PSA will be evaluated periodically throughout the study. Prostate-specific antigen results from every other cycle starting with Cycle 5 will be available to the investigators. Investigators will be notified if testosterone results do not fall below the definition of castration concentrations (<50 ng/dL or <1.7 nM).

The investigator must review the laboratory report, document this review, and record any clinically relevant changes occurring during the study in the adverse event section of the eCRF.

Electrocardiogram (ECG)

Electrocardiograms (ECGs) (12-lead) will be recorded at screening (ie, within 28 days of randomization). Computer-generated interpretations of ECGs should be reviewed for data integrity and reasonableness by the investigator. During the collection of ECGs, subjects should be in a quiet setting without distractions (eg, television, cell phones). Subjects should rest in a supine position for at least 5 minutes before ECG collection and should refrain from talking or moving arms or legs. ECGs should not be obtained when serum potassium is <3.5 mmol/L; thus potassium results must be available before ECG can be obtained. An ECG obtained within 28 days of randomization when the potassium level was <3.5mmol/L or unknown should be rechecked prior to randomization. Hypokalemia should be corrected prior to ECG collection.

Vital Signs

Vital sign measurements should be obtained before any blood draws.

Heart rate, blood pressure, respiratory rate, and body temperature will be measured at Screening. At all subsequent study visits, including the End-of-Treatment Visit, only blood pressure will be measured.

When measuring blood pressure, please ensure that the subject is comfortably seated in a chair with arm and back supported and legs uncrossed in a quiet area. Any restrictive clothing on the arm should be removed. At least 5 minutes should elapse before the reading is taken and the patient should be advised not to talk during the reading.

Physical Examination

Physical examinations will be directed at detecting adverse events and should be completed as specified in the Time and Events Schedule. Physical examinations should be performed by the same evaluator throughout the study, whenever possible. If it is not possible to use the same evaluator to follow the subject, then it is recommended that evaluations overlap (ie, examine the subject together and discuss findings) for at least 1 visit. Subject height will be measured at screening only. Weight will be measured at Screening and during the Double-blind Treatment Phase. Physical examination includes head, eyes, ears, nose and throat, chest, cardiac, abdominal, extremities, neurologic, and lymph node examinations. Any clinically significant change in physical findings noted during the study should be reported as an adverse event. Any clinically significant abnormalities persisting at the end of the study will be followed by the investigator until resolution or until reaching a clinically stable endpoint.

9.7. Sample Collection and Handling

The actual dates and times of sample collection must be recorded in the eCRF or laboratory requisition form. Refer to the Time and Events Schedule for the timing and frequency of all sample collections. Instructions for the collection, handling, and shipment of samples are found in the laboratory manual that will be provided for sample collection and handling.

10. SUBJECT COMPLETION/WITHDRAWAL

10.1. Completion

A subject will be considered to have completed the study if assessments have been completed through the end of the Follow-up Phase. Subjects who prematurely discontinue study treatment for any reason before completion of the Follow-up Phase will not be considered to have completed the study. A subject will be considered to have completed the Double-blind Treatment

Phase when a clinical endpoint of study is reached (death, radiographic progression or clinical progression per protocol definition, or unacceptable toxicity).

10.2. Discontinuation of Treatment

Discontinuation of a study drug will not result in automatic withdrawal from the study.

A subject should ordinarily be maintained on study treatment until radiographic progression as defined Section 9.2.2.

If the subject has radiographic progression without clinical progression and alternate therapy is not initiated, treatment may continue at the discretion of the investigator.

However, a subject's study treatment must be discontinued for:

- Clinical Progression defined as:
  - Deterioration in ECOG performance status to grade 3 or higher
  - Need to initiate any of the following because of <u>tumor progression</u> (even in the absence of radiographic evidence of disease)
    - Anticancer therapy for prostate cancer
    - Radiation therapy
    - Surgical interventions for complications due to tumor progression
    - Chronic opioid analgesics for cancer pain (new opioid analgesics, or increased dose or frequency of the existing opioid analgesics from Cycle 1 Day 1 for ≥ 3 weeks orally; 7 days parenterally, ie, non-oral formulation)
- Dosing noncompliance
- Withdrawal of consent for continued treatment
- The investigator believes that for safety reasons (eg, adverse event) it is in the best interest of the subject to stop treatment

All attempts to obtain imaging studies at the time of treatment discontinuation or End-of-Treatment Visit should be made to assess for radiographic progression.

Study treatment will be continued for subjects who have increasing PSA values in the absence of radiographic or clinical progression. Although serial PSA measurements will be performed in this study, progression or change in PSA values is not considered a reliable measure of disease progression and should not be used as the lone indicator for discontinuation.

Study data will continue to be collected after discontinuation of treatment, unless the subject withdraws consent for further study participation.

10.3. Withdrawal from the Study

A subject will be withdrawn from the study for any of the following reasons:

- Lost to follow-up
- Withdrawal of consent for subsequent data collection

If a patient is lost to follow up, all possible efforts must be made by the study site personnel to contact the subject and to determine endpoint status and the reason for the discontinuation/withdrawal. The measures taken for follow up must be documented. The informed consent will stipulate that even if a subject decides to discontinue double-blind study drug, he will agree to be contacted periodically by the investigator to assess endpoint status. Furthermore, the subject will be asked to agree to grant permission for the investigator to consult family members or public records, including the use of locator agencies as permitted by local law, to determine the subject's endpoint status, in the event the subject is not reachable by conventional means (eg, office visit, telephone, e-mail, or certified mail.).

When a subject withdraws before completing the study, the reason for withdrawal is to be documented in the eCRF and in the source document. Study drug assigned to the withdrawn subject may not be re-assigned to another subject. Subjects who withdraw will not be replaced.

11. STATISTICAL METHODS

Statistical analysis will be performed by the sponsor or under the supervision of the sponsor. A general description of the statistical methods to be used to analyze the efficacy and safety data is outlined below. Complete details will be contained in the Statistical Analysis Plan.

11.1. Subject Information

ITT Population: The ITT population includes all randomized subjects and will be classified according to their assigned treatment group, regardless of the actual treatment received. Subject disposition and efficacy analyses will be performed on data from the ITT population.

Safety Population: The safety population includes all subjects who received at least 1 dose of study drug.

11.2. Sample Size Determination

The overall level of significance for the study will be 0.05, with 0.049 allocated in the testing of OS, and 0.001 allocated in the testing of rPFS. Subjects will be randomized in a 1:1 ratio to receive abiraterone acetate plus low-dose prednisone plus ADT or ADT alone.

11.2.1. Hypothesized OS Hazard Ratio of 0.81

It is assumed that failure will follow an exponential distribution with a constant hazard rate. Assuming a median overall survival (OS) of 33 months[22] for the control group (ADT alone), a planned sample size of approximately 1,200 subjects will provide 85% power to detect a hazard ratio (HR) of 0.81 (33 months versus 40.75 months, or >7 months of improvement) at a 2-tailed significance level of 0.049 and an enrollment duration of approximately 24 months over a total study duration of 78 months to obtain the required 852 events.

The OS endpoint will incorporate group sequential design by including 2 interim analyses and 1 final analysis with the Wang-Tsiatis power boundaries of shape parameter 0.2 as implemented by the Lan-DeMets alpha spending function (East® software).[43]

11.2.2. Hypothesized rPFS Hazard Ratio of 0.667

Only one analysis will be conducted for rPFS. It is assumed that failure will follow an exponential distribution with a constant hazard rate. It is estimated that 565 rPFS events would be required to provide 94% power in detecting a HR of 0.667 (median rPFS of 20 months for the ADT alone group versus 30 months for the abiraterone acetate plus low-dose prednisone plus ADT group) at a 2-tailed level of significance of 0.001; the same enrollment assumptions as described for OS are assumed.

11.3. Efficacy Analyses

Efficacy endpoints are described in Section 9.2.2.

All continuous variables will be summarized using number of subjects (n), mean, standard deviation (SD), median, minimum, and maximum. Discrete variables will be summarized with n and percent. All efficacy endpoints will be analyzed using the ITT set of population. Kaplan-Meier product limit method and Cox model will be used to estimate the time to event variables and to obtain the hazard ratio. Non-stratified analysis will be conducted. Response rate variable will be evaluated using the Chi-square statistic or the exact test if the cell counts are small.

11.3.1. Analysis of Co-Primary Endpoint OS

The co-primary efficacy endpoint is OS defined as the time from randomization to date of death from any cause. Subjects without event will be censored on the last date known to be alive. The final analysis will occur when ~852 events are observed. The primary analysis will be performed using the stratified log rank test, stratified based on ECOG performance (grade of 0, 1 vs. 2) and presence of visceral metastasis (Yes vs. No).

11.3.2. Analysis of Co-Primary Endpoint rPFS

The co-primary efficacy endpoint of rPFS is defined as the time from randomization to the occurrence of radiographic progression or death. The analysis of the rPFS endpoint will use the investigator-reviewed radiographic assessment of progression. It is expected that the analysis with the estimated 565 rPFS events will coincide with the first OS interim analysis. The timing of the first interim analysis will be determined according to both rPFS and OS events required, so that the analysis will take place when the required number of events in both measures has been reached.

11.4. Biomarker Analyses

Biomarker studies are designed to confirm prior findings in asymptomatic or mildly symptomatic metastatic castration-resistant prostate cancer. Analyses will be conducted on tumor samples from approximately 300 subjects and serum samples collected at 3 timepoints from approximately 300 subjects. The following biomarkers may be analyzed (other DNA/RNA or immunohistochemistry (IHC) based-biomarkers may also be analyzed):

- Global miRNA expression levels in serial serum samples and in FFPE samples or slides from baseline
- Global mRNA expression levels in FFPE samples or slides
- AR abnormalities, such as AR mutations and splicing variants

The associations of the above biomarkers with clinical response or time-to-event endpoints will be assessed using the appropriate statistical methods (analysis of variance [ANOVA], categorical, or survival model will be used), depending on the endpoint. Analyses will be performed within and between each treatment group. Other clinical covariates (such as race, age, tumor burden) may also be included in the model. Correlation of baseline expression levels with response or time-to-event endpoints will identify responsive (or resistant) subgroups. Association of biomarkers with clinical response or time-to-event endpoints will also be evaluated in the overall population to identify “high risk” biomarker profiles that are correlated with poor outcome. Biomarker results will be reported separately from the clinical study report due to its exploratory nature.

11.5. Medical Resource Utilization

Medical resource utilization will be descriptively summarized by treatment group. This report will be prepared separately and will not be a part of the clinical study report.

11.6. Safety Analyses

Subjects who receive at least 1 dose of study drug will be analyzed for safety. The safety parameters to be evaluated are the incidence, intensity, and type of adverse events, clinically significant changes in the subject’s physical examination findings, vital signs measurements, and clinical laboratory results. Exposure to study drug and reasons for discontinuation of study treatment will be tabulated.

Adverse Events

The verbatim terms used in the eCRF by investigators to identify adverse events will be coded using the Medical Dictionary for Regulatory Activities (MedDRA) and will be graded according to the NCI-CTCAE, Version 4.0. Treatment-emergent adverse events are those events that occur or worsen on or after first dose of study drug through 30 days post last dose and will be included in the analysis. Adverse events will be summarized by system organ class and preferred term, and will be presented overall and by treatment group. Serious adverse events and deaths will be provided in a listing. All adverse events resulting in discontinuation, dose modification, dosing interruption, or treatment delay of study drug will also be listed and tabulated by preferred term.

Clinical Laboratory Tests

Clinical laboratory test results will be collected from screening and through 30 days post last dose of study drug. Laboratory data will be summarized by type of laboratory test. Parameters with predefined NCI-CTCAE, Version 4, toxicity grades will be summarized. Change from baseline to the worst adverse event grade experienced by the subject during the study will be provided as shift tables.

Vital Signs and Physical Examination

Descriptive statistics of blood pressure (systolic and diastolic) values and changes from baseline will be summarized at each scheduled time point. The percentage of subjects with values beyond clinically important limits will be summarized. Descriptive statistics of changes from baseline in physical examinations will be summarized at each scheduled time point.

11.7. Interim Analysis

Two interim analyses for OS are planned for this study after observing approximately 50% (~426 events) and approximately 65% (~554 events) of the total number of required (~852) events. The cumulative alpha spend will be 0.011 and 0.022 for each of the two interim analyses respectively. The exact significance levels will be determined according to the observed number of events at each interim analysis. It is expected that the first interim OS analysis will likely occur in conjunction with the rPFS analysis. The timing of the first interim analysis will be determined according to both rPFS and OS events required, so that the analysis will take place when the required number of events in both measures has been reached.

11.8. Independent Data Monitoring Committee (IDMC)

During the course of the study, an IDMC will review the safety and efficacy information. The committee will be composed of medical oncologists and/or urologists with experience in treating patients or medical experts in relevant therapeutic area, and a statistician, none of whom are involved in the study as investigators. The IDMC activities are described in detail in a separate charter.

12. ADVERSE EVENT REPORTING

Timely, accurate, and complete reporting and analysis of safety information from clinical studies are crucial for the protection of subjects, investigators, and the sponsor, and are mandated by regulatory agencies worldwide. The sponsor has established Standard Operating Procedures in conformity with regulatory requirements worldwide to ensure appropriate reporting of safety information; all clinical studies conducted by the sponsor or its affiliates will be conducted in accordance with those procedures.

12.1. Definitions

12.1.1. Adverse Event Definitions and Classifications

Adverse Event

An adverse event is any untoward medical occurrence in a clinical study subject administered a medicinal (investigational or non-investigational) product. An adverse event does not necessarily have a causal relationship with the treatment. An adverse event can therefore be any unfavorable and unintended sign (including an abnormal finding), symptom, or disease temporally associated with the use of a medicinal (investigational or non-investigational) product, whether or not related to that medicinal (investigational or non-investigational) product. (Definition per International Committee on Harmonisation [ICH])

This includes any occurrence that is new in onset or aggravated in severity or frequency from the baseline condition, or abnormal results of diagnostic procedures, including laboratory test abnormalities.

Note: The sponsor collects adverse events starting with the signing of the informed consent form (refer to Section 12.3.1, All Adverse Events, for time of last adverse event recording).

Serious Adverse Event

A serious adverse event based on ICH and EU Guidelines on Pharmacovigilance for Medicinal Products for Human Use is any untoward medical occurrence that at any dose:

- Results in death
- Is life-threatening
  (The subject was at risk of death at the time of the event. It does not refer to an event that hypothetically might have caused death should it have been more severe.)
- Requires inpatient hospitalization or prolongation of existing hospitalization
- Results in persistent or significant disability/incapacity
- Is a congenital anomaly/birth defect
- Is a suspected transmission of any infectious agent via a medicinal product
- Is medically important*

*Medical and scientific judgment should be exercised in deciding whether expedited reporting is also appropriate in other situations, such as important medical events that may not be immediately life threatening or result in death or hospitalization but may jeopardize the subject or may require intervention to prevent one of the other outcomes listed in the definition above. These should usually be considered serious.

Unlisted (Unexpected) Adverse Event/Reference Safety Information

An adverse event is considered unlisted when the nature or severity is not consistent with the applicable product reference safety information. For an investigational product, the expectedness of an adverse event will be determined by whether or not it is listed in the Investigator's Brochure.

Adverse Event Associated With the Use of the Drug

An adverse event is considered associated with the use of the drug when the attribution is possible, probable, or very likely by the definitions listed in Section 12.1.2.

12.1.2. Attribution Definitions

Not Related

An adverse event that is not related to the use of the drug.

Doubtful

An adverse event for which an alternative explanation is more likely, eg, concomitant drug(s), concomitant disease(s), or the relationship in time suggests that a causal relationship is unlikely.

Possible

An adverse event that might be due to the use of the drug. An alternative explanation, eg, concomitant drug(s), concomitant disease(s), is inconclusive. The relationship in time is reasonable; therefore, the causal relationship cannot be excluded.

Probable

An adverse event that might be due to the use of the drug. The relationship in time is suggestive (eg, confirmed by dechallenge). An alternative explanation is less likely, eg, concomitant drug(s), concomitant disease(s).

Very Likely

An adverse event that is listed as a possible adverse reaction and cannot be reasonably explained by an alternative explanation, eg, concomitant drug(s), concomitant disease(s). The relationship in time is very suggestive (eg, it is confirmed by dechallenge and rechallenge).

12.1.3. Severity Criteria

The NCI-CTCAE v4.0 will be used to grade the severity of adverse events.

Any adverse event not listed in the CTCAE will be graded as follows:

Grade 1, Mild: Awareness of symptoms that are easily tolerated, causing minimal discomfort and not interfering with everyday activities.

Grade 2, Moderate: Sufficient discomfort is present to cause interference with normal activity.

Grade 3, Severe: Extreme distress, causing significant impairment of functioning or incapacitation. Prevents normal everyday activities.

Grade 4, Life-threatening: Urgent intervention indicated.

Grade 5, Death: Death.

The investigator should use clinical judgment in assessing the severity of events not directly experienced by the subject (eg, laboratory abnormalities).

12.2. Special Reporting Situations

Safety events of interest on a sponsor medicinal product that may require expedited reporting or safety evaluation include, but are not limited to:

- Overdose of a sponsor medicinal product
- Suspected abuse/misuse of a sponsor medicinal product
- Inadvertent or accidental exposure to a sponsor medicinal product

- Medication error involving a sponsor product (with or without subject/patient exposure to the sponsor medicinal product, eg, name confusion)

Special reporting situations should be recorded in the eCRF. Any special reporting situation that meets the criteria of a serious adverse event should be recorded on the serious adverse event page of the eCRF.

12.3. Procedures

12.3.1. All Adverse Events

All adverse events and special reporting situations, whether serious or non-serious, will be reported from the time a signed and dated informed consent form is obtained until 30 days after last dose of study drug. Serious adverse events, including those spontaneously reported to the investigator within 30 days after the last dose of study drug, must be reported using the Serious Adverse Event Form. The sponsor will evaluate any safety information that is spontaneously reported by an investigator beyond the time frame specified in the protocol.

All adverse events, regardless of seriousness, severity, or presumed relationship to study therapy, must be recorded using medical terminology in the source document and the eCRF. Whenever possible, diagnoses should be given when signs and symptoms are due to a common etiology (eg, cough, runny nose, sneezing, sore throat, and head congestion should be reported as "upper respiratory infection"). Investigators must record in the eCRF their opinion concerning the relationship of the adverse event to study therapy. All measures required for adverse event management must be recorded in the source document and reported according to sponsor instructions.

The sponsor assumes responsibility for appropriate reporting of adverse events to the regulatory authorities. The sponsor will also report to the investigator all serious adverse events that are unlisted (unexpected) and associated with the use of the drug. The investigator (or sponsor where required) must report these events to the appropriate Independent Ethics Committee/Institutional Review Board (IEC/IRB) that approved the protocol unless otherwise required and documented by the IEC/IRB.

Subjects (or their designees, when appropriate) must be provided with a "study card" indicating the following:

- Subject's name
- Subject number
- Subject's data of birth
- Study site number
- Investigator's name and 24-hour contact information
- Local sponsor's name and 24-hour contact information
- Statement that the subject is participating in a clinical study

12.3.2. Serious Adverse Events

All serious adverse events occurring during clinical studies must be reported to the appropriate sponsor contact person by investigational staff within 24 hours of their knowledge of the event.

Information regarding serious adverse events will be transmitted to the sponsor using the Serious Adverse Event Form, which must be completed and signed by a member of the investigational staff, and transmitted to the sponsor within 24 hours. The initial and follow-up reports of a serious adverse event should be made by facsimile (fax).

All serious adverse events that have not resolved by the end of the study, or that have not resolved upon discontinuation of the subject's participation in the study, must be followed until any of the following occurs:

- The event resolves
- The event stabilizes
- The event returns to baseline, when a baseline value/status is available
- The event can be attributed to agents other than the study drug or to factors unrelated to study conduct
- It becomes unlikely that any additional information can be obtained (subject or health care practitioner refusal to provide additional information, lost to follow-up after demonstration of due diligence with follow-up efforts)

Suspected transmission of an infectious agent by a medicinal product will be reported as a serious adverse event. Any event requiring hospitalization (or prolongation of hospitalization) that occurs during the course of a subject's participation in a clinical study must be reported as a serious adverse event, except hospitalizations for the following:

- Social reasons in the absence of an adverse event
- Surgery or procedure planned before entry into the study (must be documented in the eCRF)

Disease progression should not be recorded as an adverse event or serious adverse event term; instead, signs and symptoms of clinical sequelae resulting from disease progression/lack of efficacy will be reported when they fulfill the serious adverse event definition (see Section 12.1.1).

During the follow-up phase of the study, deaths regardless of causality will be reported via the CRF. Serious adverse events thought to be related to blinded study drug will be collected and reported via the CRF and the SAE form if within 30 days of last study drug administration. Serious adverse events that occur after 30 days following the last drug administration thought to be related to study drug will be collected and reported via the SAE form within 24 hours of discovery or notification of the event and documented.

12.3.3. Pregnancy

Because the effect of the blinded study drug on sperm is unknown, pregnancies in partners of male subjects included in the study will be reported by the investigational staff within 24 hours of their knowledge of the event using the appropriate pregnancy notification form.

Follow-up information regarding the outcome of the pregnancy and any postnatal sequelae in the infant will be required.

12.3.4. Contacting Sponsor Regarding Safety

The names (and corresponding telephone numbers) of the individuals who should be contacted regarding safety issues or questions regarding the study are listed on the Contact Information page(s), which will be provided as a separate document.

13. PRODUCT QUALITY COMPLAINT HANDLING

A product quality complaint (PQC) is defined as any suspicion of a product defect related to manufacturing, labeling, or packaging, ie, any dissatisfaction relative to the identity, quality, durability, or reliability of a product, including its labeling or package integrity. PQCs may have an impact on the safety and efficacy of the product. Timely, accurate, and complete reporting and analysis of PQC information from clinical studies are crucial for the protection of subjects, investigators, and the sponsor, and are mandated by regulatory agencies worldwide. The sponsor has established procedures in conformity with regulatory requirements worldwide to ensure appropriate reporting of PQC information; all clinical studies conducted by the sponsor or its affiliates will be conducted in accordance with those procedures.

13.1. Procedures

All initial PQCs must be reported to the sponsor by the investigational staff within 24 hours after being made aware of the event.

If the defect is combined with a serious adverse event, then the investigational staff must report the PQC to the sponsor according to the serious adverse event reporting timelines (refer to Section 12.3.2). A sample of the suspected product should be maintained for further investigation when requested by the sponsor.

13.2. Contacting Sponsor Regarding Product Quality

The names (and corresponding telephone numbers) of the individuals who should be contacted regarding product quality issues are listed on the Contact Information page(s), which will be provided as a separate document.

14. STUDY DRUG INFORMATION

14.1. Physical Description of Study Drug(s)

Abiraterone acetate 250 mg tablets are oval, white to off-white and contain abiraterone acetate and compendial (USP/NF/EP) grade lactose monohydrate, microcrystalline cellulose, croscarmellose sodium, povidone, sodium lauryl sulfate, magnesium stearate, colloidal silicon dioxide, and purified water, in descending order of concentration (the water is removed during tabletting).

Placebo will be provided as a tablet formulation and will be matched in size, color (white to off-white), and shape (oval) to abiraterone acetate tablets in order to maintain the study blind. Placebo prednisone capsules will also be provided.

14.2. Packaging

The study drugs will be packaged and supplied by the sponsor. Abiraterone acetate / placebo are packaged in high-density polyethylene bottles with child-resistant closures. There are 120 tablets per bottle. Prednisone/placebo is packaged in blister packs. There are 30 capsules per pack.

14.3. Labeling

Study drug labels will contain information to meet the applicable regulatory requirements.

14.4. Preparation, Handling, and Storage

The study drugs must be stored in a secure area and administered only to subjects entered into the clinical study in accordance with the conditions specified in this protocol. Study drugs should be stored at room temperature. Additional information is provided in the abiraterone acetate Investigator's Brochure. Subjects should be advised to keep all medications out of reach and sight of children.

Abiraterone Acetate Handling Requirements: Abiraterone acetate is contraindicated in women who are or may potentially be pregnant. There are no human data on the use of abiraterone acetate in pregnancy. Maternal use of CYP17 inhibitor is expected to produce changes in hormonal levels that may affect the development of the fetus. Women who are pregnant or may be pregnant should not handle abiraterone acetate without protection, eg gloves.

In an oral developmental toxicity study in the rat, abiraterone acetate affected pregnancy.

It is not known whether abiraterone or its metabolites are present in semen. A condom is required if the subject is engaged in sexual activity with a pregnant woman. If the subject is engaged in sex with a woman of childbearing potential, a condom is required along with another effective contraceptive method.

14.5. Drug Accountability

The investigator is responsible for ensuring that all study drug received at the site is inventoried and accounted for throughout the study. The dispensing of study drug to the subject, and the return of study drug from the subject, must be documented on the drug accountability form. Subjects must be instructed to return all original containers, whether empty or containing study drug. These requirements also apply to prednisone/placebo capsules supplied by the sponsor.

All study drugs must be handled in strict accordance with the protocol and the container label, and must be stored at the study site in a limited-access area or in a locked cabinet under appropriate environmental conditions. Unused study drug, and study drug returned by the subject, must be available for verification by the sponsor's site monitor during on-site monitoring visits. The return to the sponsor of unused study drug, or used returned study drug for destruction, will be documented on the drug return form. When the site is an authorized destruction unit and study drug supplies are destroyed on site, this must also be documented on the drug return form.

All study drugs should be dispensed under the supervision of the investigator or a qualified member of the investigational staff, or by a hospital/clinic pharmacist. Study drug will be supplied only to subjects participating in the study. Returned study drug must not be dispensed again, even to the same subject. Whenever a subject brings his or her study drug to the site for pill count, this is not seen as a return of supplies. Study drug may not be relabeled or reassigned for use by other subjects. The investigator agrees neither to dispense the study drug from, nor store it at, any site other than the study sites agreed upon with the sponsor.

15. STUDY-SPECIFIC MATERIALS

The investigator will be provided with the following supplies:

- Investigator Brochure
- Pharmacy manual/site investigational product procedures manual
- Laboratory manual
- PRO Manual containing questionnaires and user instructions
- Trial Slate
- IWRS Manual
- eDC Manual
- Medication diary for study drug

16. ETHICAL ASPECTS

16.1. Study-Specific Design Considerations

Potential subjects will be fully informed of the risks and requirements of the study and, during the study, subjects will be given any new information that may affect their decision to continue participation. They will be told that their consent to participate in the study is voluntary and may be withdrawn at any time with no reason given and without penalty or loss of benefits to which they would otherwise be entitled. Only subjects who are fully able to understand the risks, benefits, and potential adverse events of the study, and provide their consent voluntarily will be enrolled.

Ongoing medical review is completed by the sponsor. An interim analysis review will also be completed by the IDMC for safety and efficacy which will occur when approximately 50% (~426 events) and approximately 65% (~554 events) total number of required (~852) have occurred.

The blood volume to be collected is approximately 2.5 to 7.0 mL per study visit. For subjects participating in the biomarker study, the blood volumes will be 2.5 to 15.5 mL per study visit. As rPFS is a secondary endpoint, scheduled imaging is incorporated in the protocol. The timing of imaging is designed to capture progression events yet not to have patients be overexposed to radiation.

Subjects in the active group of this study will receive blinded low-dose prednisone 5mg/day. The prednisone dose used in this study is lower than the prednisone 10mg/day dose used in other abiraterone acetate Phase 3 studies COU-AA-301 and COU-AA-302 that enrolled advanced symptomatic mCRPC patients. The higher prednisone dose in other Phase 3 studies was selected because it is commonly used as the standard of care[13] in combination with approved chemotherapy agents or as monotherapy for palliation of symptoms in advanced prostate cancer patients. Prednisone is not considered standard of care for newly diagnosed mHNPC and therefore is not included in the control arm. The required use of prednisone in combination with abiraterone acetate is to help mitigate the symptoms of mineralocorticoid excess caused by CYP17 inhibition, which is the mechanism of action of abiraterone acetate. Data from early Phase 1/2 studies with abiraterone acetate in patients with mCRPC demonstrate that dexamethasone 0.5mg once daily (equivalent to prednisone dose of 3.33 mg once daily) was effective in mitigating the mineralocorticoid effects of abiraterone acetate.3 The protocol allows for the dose of prednisone to be increased by 5 mg/day if required to manage mineralocorticoid toxicities.

16.2. Regulatory Ethics Compliance

16.2.1. Investigator Responsibilities

The investigator is responsible for ensuring that the clinical study is performed in accordance with the protocol, current ICH guidelines on Good Clinical Practice (GCP), and applicable regulatory and country-specific requirements.

Good Clinical Practice is an international ethical and scientific quality standard for designing, conducting, recording, and reporting studies that involve the participation of human subjects. Compliance with this standard provides public assurance that the rights, safety, and well-being of study subjects are protected, consistent with the principles that originated in the Declaration of Helsinki, and that the clinical study data are credible.

16.2.2. Independent Ethics Committee or Institutional Review Board

Before the start of the study, the investigator (or sponsor where required) will provide the IEC/IRB with current and complete copies of the following documents:

- Final protocol and, when applicable, amendments
- Sponsor-approved informed consent form (and any other written materials to be provided to the subjects)
- Investigator's Brochure (or equivalent information) and amendments/addenda
- Sponsor-approved subject recruiting materials

- Information on compensation for study-related injuries or payment to subjects for participation in the study, when applicable
- Investigator's curriculum vitae or equivalent information (unless not required, as documented by the IEC/IRB)
- Information regarding funding, name of the sponsor, institutional affiliations, other potential conflicts of interest, and incentives for subjects
- Any other documents that the IEC/IRB requests to fulfill its obligation

This study will be undertaken only after the IEC/IRB has given full approval of the final protocol, any amendments, the informed consent form, applicable recruiting materials, and subject compensation programs, and the sponsor has received a copy of this approval. This approval letter must be dated and must clearly identify the IEC/IRB and the documents being approved.

During the study the investigator (or sponsor where required) will send the following documents and updates to the IEC/IRB for their review and approval, where appropriate:

- Protocol amendments
- Revision(s) to informed consent form and any other written materials to be provided to subjects
- When applicable, new or revised subject recruiting materials approved by the sponsor
- Revisions to compensation for study-related injuries or payment to subjects for participation in the study, when applicable
- New edition(s) of the Investigator's Brochure and amendments/addenda
- Summaries of the status of the study at intervals stipulated in guidelines of the IEC/IRB (at least annually)
- Reports of adverse events that are serious, unlisted/unexpected, and associated with the investigational drug
- New information that may adversely affect the safety of the subjects or the conduct of the study
- Deviations from or changes to the protocol to eliminate immediate hazards to the subjects
- Report of deaths of subjects under the investigator's care
- Notification when a new investigator is responsible for the study at the site
- Development Safety Update Report and Line Listings, where applicable
- Any other requirements of the IEC/IRB

For all protocol amendments (excluding the ones that are purely administrative, with no consequences for subjects, data or study conduct), the amendment and applicable informed consent form revisions must be submitted promptly to the IEC/IRB for review and approval before implementation of the change(s).

At least once a year, the IEC/IRB will be asked to review and reapprove this clinical study. The reapproval should be documented in writing (excluding the ones that are purely administrative, with no consequences for subjects, data, or study conduct).

At the end of the study, the investigator (or sponsor where required) will notify the IEC/IRB about the study completion.

16.2.3. Informed Consent

Each subject must give written consent according to local requirements after the nature of the study has been fully explained. The consent form must be signed before performance of any study-related activity. The consent form that is used must be approved by both the sponsor and by the reviewing IEC/IRB and be in a language that the subject can read and understand. The informed consent should be in accordance with principles that originated in the Declaration of Helsinki, current ICH and GCP guidelines, applicable regulatory requirements, and sponsor policy.

Before enrollment in the study, the investigator or an authorized member of the investigational staff must explain to potential subjects the aims, methods, reasonably anticipated benefits, and potential hazards of the study, and any discomfort participation in the study may entail. Subjects will be informed that their participation is voluntary and that they may withdraw consent to participate at any time. They will be informed that choosing not to participate will not affect the care the subject will receive for the treatment of his disease. Subjects will be told that alternative treatments are available should they refuse to take part and that such refusal will not prejudice future treatment. Finally, they will be told that the investigator will maintain a subject identification register for the purposes of long-term follow up when needed and that their records may be accessed by health authorities and authorized sponsor staff without violating the confidentiality of the subject, to the extent permitted by the applicable law(s) or regulations. By signing the informed consent form the subject is authorizing such access, and agrees to allow his or her study physician to recontact the subject for the purpose of obtaining consent for additional safety evaluations, and subsequent disease-related treatments or to obtain information about his survival status.

The subject will be given sufficient time to read the informed consent form and the opportunity to ask questions. After this explanation and before entry into the study, consent should be appropriately recorded by means of the subject's personally dated signature. After having obtained the consent, a copy of the informed consent form must be given to the subject.

If the subject is unable to read or write, then an impartial witness should be present for the entire informed consent process (which includes reading and explaining all written information) and should personally date and sign the informed consent form after the oral consent of the subject is obtained.

16.2.4. Privacy of Personal Data

The collection and processing of personal data from subjects enrolled in this study will be limited to those data that are necessary to fulfill the objectives of the study.

These data must be collected and processed with adequate precautions to ensure confidentiality and compliance with applicable data privacy protection laws and regulations. Appropriate technical and organizational measures to protect the personal data against unauthorized disclosures or access, accidental or unlawful destruction, or accidental loss or alteration must be put in place. Sponsor personnel whose responsibilities require access to personal data agree to keep the identity of study subjects confidential.

The informed consent obtained from the subject includes explicit consent for the processing of personal data and for the investigator to allow direct access to his or her original medical records for study-related monitoring, audit, IEC/IRB review, and regulatory inspection. This consent also addresses the transfer of the data to other entities and to other countries.

The subject has the right to request through the investigator access to his or her personal data and the right to request rectification of any data that are not correct or complete. Reasonable steps will be taken to respond to such a request, taking into consideration the nature of the request, the conditions of the study, and the applicable laws and regulations.

16.2.5. Country Selection

This study will only be conducted in those countries where the intent is to launch or otherwise help ensure access to the developed product, unless explicitly addressed as a specific ethical consideration in Section 16.1.

17. ADMINISTRATIVE REQUIREMENTS

17.1. Protocol Amendments

Neither the investigator nor the sponsor will modify this protocol without a formal amendment by the sponsor. All protocol amendments must be issued by the sponsor, and signed and dated by the investigator. Protocol amendments must not be implemented without prior IEC/IRB approval, or when the relevant competent authority has raised any grounds for non-acceptance, except when necessary to eliminate immediate hazards to the subjects, in which case the amendment must be promptly submitted to the IEC/IRB and relevant competent authority. Documentation of amendment approval by the investigator and IEC/IRB must be provided to the sponsor or its designee. When the change(s) involves only logistic or administrative aspects of the study, the IRB (and IEC where required) only needs to be notified.

During the course of the study, in situations where a departure from the protocol is unavoidable, the investigator or other physician in attendance will contact the appropriate sponsor representative (see Contact Information page(s) provided separately). Except in emergency situations, this contact should be made before implementing any departure from the protocol. In all cases, contact with the sponsor must be made as soon as possible to discuss the situation and agree on an appropriate course of action. The data recorded in the eCRF and source documents will reflect any departure from the protocol, and the source documents will describe this departure and the circumstances requiring it.

17.2. Regulatory Documentation

17.2.1. Regulatory Approval/Notification

This protocol and any amendment(s) must be submitted to the appropriate regulatory authorities in each respective country, when applicable. A study may not be initiated until all local regulatory requirements are met.

17.2.2. Required Prestudy Documentation

The following documents must be provided to the sponsor before shipment of blinded study drug to the investigational site:

- Protocol and any amendment(s), signed and dated by the principal investigator
- A copy of the dated and signed, written IEC/IRB approval of the protocol, amendments, informed consent form, any recruiting materials, and when applicable, subject compensation programs. This approval must clearly identify the specific protocol by title and number and must be signed by the chairman or authorized designee.
- Name and address of the IEC/IRB, including a current list of the IEC/IRB members and their function, with a statement that it is organized and operates according to GCP and the applicable laws and regulations. If accompanied by a letter of explanation, or equivalent, from the IEC/IRB, then a general statement may be substituted for this list. If an investigator or a member of the investigational staff is a member of the IEC/IRB, then documentation must be obtained to state that this person did not participate in the deliberations or in the vote/opinion of the study.
- Regulatory authority approval or notification, when applicable
- Signed and dated statement of investigator (eg, Form FDA 1572), when applicable
- Documentation of investigator qualifications (eg, curriculum vitae)
- Completed investigator financial disclosure form from the principal investigator, where required
- Signed and dated clinical trial agreement, which includes the financial agreement
- Any other documentation required by local regulations

The following documents must be provided to the sponsor before enrollment of the first subject:

- Completed investigator financial disclosure forms from all clinical subinvestigators
- Documentation of subinvestigator qualifications (eg, curriculum vitae)
- Name and address of any local laboratory conducting tests for the study, and a dated copy of current laboratory normal ranges for these tests, when applicable
- Local laboratory documentation demonstrating competence and test reliability (eg, accreditation/license), when applicable

17.3. Subject Identification, Enrollment, and Screening Logs

The investigator agrees to complete a subject identification and enrollment log to permit easy identification of each subject during and after the study. This document will be reviewed by the sponsor site contact for completeness.

The subject identification and enrollment log will be treated as confidential and will be filed by the investigator in the study file. To ensure subject confidentiality, no copy will be made. All reports and communications relating to the study will identify subjects by ID and date of birth. In cases where the subject is not randomized into the study, the date seen and the date of birth will be used.

The investigator must also complete a subject screening log, which reports on all subjects who were seen to determine eligibility for inclusion in the study.

17.4. Source Documentation

At a minimum, source documentation must be available for the following to confirm data collected in the eCRF: subject identification, eligibility, and study identification; study discussion and date of informed consent; dates of visits; results of safety and efficacy parameters as required by the protocol; record of all adverse events and follow-up of adverse events; concomitant medication; drug receipt/dispensing/return records; study drug administration information; and date of study completion and reason for early discontinuation of study drug or withdrawal from the study, when applicable.

In addition, the author of an entry in the source documents should be identifiable.

At a minimum, the type and level of detail of source data available for a study subject should be consistent with that commonly recorded at the site as a basis for standard medical care. Specific details required as source data for the study will be reviewed with the investigator before the study and will be described in the monitoring guidelines (or other equivalent document).

During the study, subjects will record PRO data on a tablet computer. After treatment discontinuation, subjects have the option to conduct follow-up visits via phone and in these cases, the site staff will enter data into the tablet computer on the subject’s behalf. Both processes will be considered source documents.

17.5. Case Report Form Completion

Case report forms are provided for each subject in electronic format.

Electronic Data Capture (eDC) will be used for this study. The study data will be transcribed by study personnel from the source documents onto an eCRF, and transmitted in a secure manner to the sponsor within the timeframe agreed upon between the sponsor and the site. The electronic file will be considered to be the eCRF.

Worksheets may be used for the capture of some data to facilitate completion of the eCRF. Any such worksheets will become part of the subject's source documentation. All data relating to the study must be recorded in eCRFs prepared by the sponsor. Data must be entered into eCRFs in English. Designated site personnel must complete the eCRF as soon as possible after a subject visit, and the forms should be available for review at the next scheduled monitoring visit.

Every effort should be made to ensure that all subjective measurements to be recorded in the eCRF are completed by the same individual who made the initial baseline determinations. The investigator must verify that all data entries in the eCRFs are accurate and correct.

All eCRF entries, corrections, and alterations must be made by the investigator or other authorized study-site personnel. When necessary, queries will be generated in the eDC tool. The investigator or an authorized member of the investigational staff must adjust the eCRF (when applicable) and complete the query.

When corrections to an eCRF are needed after the initial entry into the eCRF, this can be done in 3 different ways:

- Site personnel can make corrections in the eDC tool at their own initiative or as a response to an auto query (generated by the eDC tool)
- Site manager can generate a query for resolution by the investigational staff
- Clinical data manager can generate a query for resolution by the investigational staff

17.6. Data Quality Assurance/Quality Control

Steps to be taken to ensure the accuracy and reliability of data include the selection of qualified investigators and appropriate study sites, review of protocol procedures with the investigator and associated personnel before the study, and periodic monitoring visits by the sponsor. Written instructions will be provided for collection, preparation, and shipment of samples.

Guidelines for eCRF completion will be provided and reviewed with study personnel before the start of the study. The sponsor will review eCRFs for accuracy and completeness during on-site monitoring visits and after transmission to the sponsor; any discrepancies will be resolved with the investigator or designee, as appropriate. After upload of the data into the clinical study database they will be verified for accuracy and consistency with the data sources.

17.7. Record Retention

In compliance with the ICH/GCP guidelines, the investigator/institution will maintain all eCRFs and all source documents that support the data collected from each subject, as well as all study documents, including imaging scans as specified in ICH/GCP Section 8, Essential Documents for the Conduct of a Clinical Trial, and all study documents as specified by the applicable regulatory requirement(s). The investigator/institution will take measures to prevent accidental or premature destruction of these documents.

Essential documents must be retained until at least 2 years after the last approval of a marketing application in an ICH region and until there are no pending or contemplated marketing applications in an ICH region or until at least 2 years have elapsed since the formal discontinuation of clinical development of the investigational product. These documents will be retained for a longer period when required by the applicable regulatory requirements or by an agreement with the sponsor. It is the responsibility of the sponsor to inform the investigator/institution as to when these documents no longer need to be retained.

If the responsible investigator retires, relocates, or for other reasons withdraws from the responsibility of keeping the study records, then custody must be transferred to a person who will accept the responsibility. The sponsor must be notified in writing of the name and address of the new custodian. Under no circumstance shall the investigator relocate or dispose of any study documents before having obtained written approval from the sponsor.

If it becomes necessary for the sponsor or the appropriate regulatory authority to review any documentation relating to this study, then the investigator must permit access to such reports.

17.8. Monitoring

The sponsor will perform on-site monitoring visits as frequently as necessary. The monitor will record dates of the visits in a study site visit log that will be kept at the site. The first post-initiation visit will be made as soon as possible after enrollment has begun. At these visits, the monitor will compare the data entered into the eCRFs with the hospital or clinic records (source documents). The nature and location of all source documents will be identified to ensure that all sources of original data required to complete the eCRF are known to the sponsor and investigational staff and are accessible for verification by the sponsor site contact. If electronic records are maintained at the investigational site, then the method of verification must be discussed with the investigational staff.

Direct access to source documentation (medical records) must be allowed for the purpose of verifying that the data recorded in the eCRF are consistent with the original source data. Findings from this review of eCRFs and source documents will be discussed with the investigational staff. The sponsor expects that, during monitoring visits, the relevant investigational staff will be available, the source documentation will be accessible, and a suitable environment will be provided for review of study-related documents. The monitor will meet with the investigator on a regular basis during the study to provide feedback on the study conduct.

17.9. Study Completion/Termination

17.9.1. Study Completion

The study is considered completed with the last study assessment for the last subject participating in the study. The final data from the investigational site will be sent to the sponsor (or designee) after completion of the final subject assessment at that site, in the time frame specified in the Clinical Trial Agreement.

17.9.2. Study Termination

The sponsor reserves the right to close the investigational site or terminate the study at any time for any reason at the sole discretion of the sponsor. Investigational sites will be closed upon study completion. An investigational site is considered closed when all required documents and study supplies have been collected and a site closure visit has been performed.

The investigator may initiate site closure at any time, provided there is reasonable cause and sufficient notice is given in advance of the intended termination.

Reasons for the early closure of an investigational site by the sponsor or investigator may include but are not limited to:

- Failure of the investigator to comply with the protocol, the requirements of the IEC/IRB or local health authorities, the sponsor's procedures, or GCP guidelines
- Inadequate recruitment of subjects by the investigator
- Discontinuation of further drug development

17.10. On-Site Audits

Representatives of the sponsor's clinical quality assurance department may visit the site at any time during or after completion of the study to conduct an audit of the study in compliance with regulatory guidelines and company policy. These audits will require access to all study records, including source documents, for inspection and comparison with the eCRFs. Subject privacy must, however, be respected. The investigator and staff are responsible for being present and available for consultation during routinely scheduled site audit visits conducted by the sponsor or its designees.

Similar auditing procedures may also be conducted by agents of any regulatory body, either as part of a national GCP compliance program or to review the results of this study in support of a regulatory submission. The investigator should immediately notify the sponsor when they have been contacted by a regulatory agency concerning an upcoming inspection.

17.11. Use of Information and Publication

All information, including but not limited to information regarding abiraterone acetate or the sponsor's operations (eg, patent application, formulas, manufacturing processes, basic scientific data, prior clinical data, formulation information) supplied by the sponsor to the investigator and not previously published, and any data, including exploratory biomarker data, generated as a result of this study, are considered confidential and remain the sole property of the sponsor. The investigator agrees to maintain this information in confidence and use this information only to accomplish this study, and will not use it for other purposes without the sponsor's prior written consent.

The investigator understands that the information developed in the clinical study will be used by the sponsor in connection with the continued development of abiraterone acetate, and thus may be disclosed as required to other clinical investigators or regulatory agencies. To permit the information derived from the clinical studies to be used, the investigator is obligated to provide the sponsor with all data obtained in the study.

The results of the study will be reported in a Clinical Study Report generated by the sponsor and will contain eCRF data from all investigational sites that participated in the study, and direct transmission of clinical laboratory data from a central laboratory into the sponsor's database. Recruitment performance or specific expertise related to the nature and the key assessment parameters of the study will be used to determine a coordinating investigator. Results of exploratory biomarker analyses performed after the Clinical Study Report has been issued will be reported in a separate report and will not require a revision of the Clinical Study Report. Study subject identifiers will not be used in publication of exploratory biomarker results. Any work created in connection with performance of the study and contained in the data that can benefit from copyright protection (except any publication by the investigator as provided for below) shall be the property of the sponsor as author and owner of copyright in such work.

Consistent with Good Publication Practices and International Committee of Medical Journal Editors guidelines, the sponsor shall have the right to publish such primary (multicenter) data and information without approval from the investigator. The investigator has the right to publish site-specific data after the primary data are published. If an investigator wishes to publish information from the study, then a copy of the manuscript must be provided to the sponsor for review at least 60 days before submission for publication or presentation. Expedited reviews will be arranged for abstracts, poster presentations, or other materials. If requested by the sponsor in writing, then the investigator will withhold such publication for up to an additional 60 days to allow for filing of a patent application. In the event that issues arise regarding scientific integrity or regulatory compliance, the sponsor will review these issues with the investigator. The sponsor will not mandate modifications to scientific content and does not have the right to suppress information. For multicenter study designs and substudy approaches, secondary results generally should not be published before the primary endpoints of a study have been published. Similarly, investigators will recognize the integrity of a multicenter study by not submitting for publication data derived from the individual site until the combined results from the completed study have been submitted for publication, within 12 months of the availability of the final data (tables, listings, graphs), or the sponsor confirms there will be no multicenter study publication. Authorship of publications resulting from this study will be based on the guidelines on authorship, such as those described in the Uniform Requirements for Manuscripts Submitted to Biomedical Journals, which state that the named authors must have made a significant contribution to the design of the study or analysis and interpretation of the data, provided critical review of the paper, and given final approval of the final version.

Registration of Clinical Studies and Disclosure of Results

The sponsor will register or disclose the existence of and the results of clinical studies as required by law.

REFERENCES

1. Adolfsson J, Steineck G, Hedlund PO. Deferred treatment of locally advanced non-metastatic prostate cancer: a long-term followup. J Urol 1999;161:505-508.
2. Ambs S, Prueitt R, Yi M, Hudson R, Howe T, Petrocca F, et al. Genomic profiling of MicroRNA and messenger RNA reveals deregulated microRNA expression in prostate cancer. Cancer Res 2008;68:6162-70.
3. Attard G, Reid AHM, Yap TA, Raynaud F, Dowsett M, Settatree S, et al. Phase I clinical trial of a selective inhibitor of CYP17, abiraterone acetate, confirms that castration-resistant prostate cancer commonly remains hormone driven. J Clin Oncol 2008;26(28);4563-4571.
4. Barrie SE, Potter GA, Goddard PM, et al. Pharmacology of novel steroidal inhibitors of cytochrome P45017α (17α hydroxylase/C17-20 lyase). J Steroid Biochem Mol Biol 1994;50:267–273
5. Cetin K, Beebe-Dimmer JL, Fryzek JP, Markus R, Carducci MA. Recent time trends in the epidemiology for stage IV prostate cancer in the United States: Analysis of data from the surveillance, epidemiology and end results program. Urology 2010;75:1396-1405.
6. deBono JS, Logothetis CJ, Molina A, Fizazi K, North S, Chu L. et al. Abiraterone and increased survival in metastatic prostate cancer. N Engl J Med 2011;364:1995-2005.
7. Efstathiou E, Davis JW, Troncoso P, Titus MA, Hoang A, Wen S, et al. Cytoreduction and androgen signalling modulation by abiraterone acetate (AA) plus leuprolide acetate (LHRHa) versus LHRHa in localized high-risk prostate cancer (PCa): Preliminary results of a randomized postoperative study. J Clin Oncol 30;2012 (suppl; abstr 4556).
8. Eisenhauer EA, Therasse P, Bogaerts et al., New response evaluation criteria in solid tumours: Revised RECIST guideline (version 1.1). Eur J Cancer 2009;45:228-247.
9. EuroQol 1990. www.euroqol.org.
10. Flamand V et al. Observational survey on variations of prostate cancer incidence by stage in the Nord-Pas-de-Calais region between 1998 and 2004. Prog Urol. 2008;18:53-9.
11. Gravis G, Fiazzi, F, Joly F, et al. Androgen-deprivation therapy alone or with docetaxel in non-castrate metastatic prostate cancer (GETUG-AFU 15): a randomised, open-label, phase 3 trial. Lancet Oncol 2013;14:149-158.
12. He L, Hannon G. MicroRAs: small RNAs with a big role in gene regulation. Nat Rev Genet 2004;5:522-31.
13. Heidenreich A, Bolla M, Joniau S, Mason MD, Matveev V, Mottet N, et al. Guidelines on Prostate Cancer. European Association of Urology 2011. Update January 2011.
14. Howard GCW et al. Patterns of referral, management and survival of patients diagnosed with prostate cancer in Scotland during 1988 and 1993: results of a national retrospective population-based audit. BJUI. 2001;87:339-347.
15. Hussain M, Tangen CM, Higano C, Schelhammer PF, Faulkner J, Crawford ED, et al. Absolute Prostate-Specific Antigen Value After Androgen Deprivation Is a Strong Independent Predictor of Survival in New Metastatic Prostate Cancer: Data From Southwest Oncology Group Trial 9346 (INT-0162). J Clin Oncol 2006;24:3984-3990.
16. Hussain M, Tangen CM, Higano C, Crawford ED, Liu G, Wilding G, et al. Intermittent (IAD) versus continuouse androgen deprivation (CAD) in hormone sensitive metastatic prostate cancer (HSM1PC) patients: Results of S9346 (INT-0162), an international Phase III trial. J Clin Oncol 30;2012 (suppl; abstr 4).
17. International Agency for Research on Cancer (IARC). GLOBOCAN 2008 Database: Summary Table by Cancer. Available at http://www-dep.iarc.fr/. Accessed 27 August 2010.
18. Jack R et al. Prostate cancer incidence, stage at diagnosis, treatment and survival in ethnic groups in South-East England. BJUI. 2009;105:1226-1230.
19. Jonsson E et al. Adenocarcinoma of the prostate in Iceland: a population-based study of stage, Gleason grade, treatment and long-term survival in males diagnosed between 1983 and 1987. Scandinavian Journal of Urology and Nephrology. 2006;40:265-271.

20. Kumar MS, Lu J, Mercer KL, Golub TR, Jacks T. Impaired microRNA processing enhances cellular transformation and tumorigenesis. Nat Genet 2007;39:673–677.

21. Li B, Shi X, Xue L, Chen A, Valicenti R, deVere White R. Effect of down-regulation of microRNA 106b on p21-mediated cell cycle arrest in response to radiation in prostate cancer cells. ASCO Abstract 53 (2010 Genitourinary Cancers Symposium).

22. Lim L, Lau N, Garret-Engele P et al. Microarray analysis shows that some microRNAs downregulate large numbers of target mRNAs. Nature 2005;433:769-73.

23. Millikan RE, Wen S, Pagliaro LC, et al. Phase 3 trial of androgen ablation with or without three cycles of systemic chemotherapy for advanced prostate cancer. J Clin Oncol 2008;26:5936-5942.

24. Montgomery RB, Mostaghel EA, Vessella R, et al. Maintenance of intratumoral androgens in metastatic prostate cancer: a mechanism for castration-resistant tumor growth. Cancer Res 2008;68:4447-4454

25. Medical Research Council (MRC) Prostate Cancer Working Party Investigators Group. Immediate versus deferred treatment for advanced prostatic cancer: initial results of the Medical Research Council Trial. Br J Urol 1997;79:234-246.

26. Montgomery RB, Mostaghel EA, Vessella R, Hess DL, Kalhorn TF, Higano CS, et al. Maintenance of intratumoral androgens in metastatic prostate cancer: a mechanism for castration-resistant tumor growth. Cancer Res. 2008 68(11):4447-54.

27. Nair B, Wilt T, MacDonald R, Rutks I. Early vs. deferred androgen suppression in the treatment of advanced prostate cancer. Cochrane Database System Rev 2002;(1):CD 3506.

28. NCCN Clinical Practice Guidelines in Oncology (NCCN Guideline®) Prostate Cancer. Version 3.2012 (www.NCCN.org).

29. .Norgaard M, Jensen AO, Jacobsen JB, Cetin K, Fryzek JP, Sorensen HT. Skeletal related events, bone metastasis and survival of prostate cancer: a population based cohort study in Denmark. J Urol 2010;184:162-167.

30. O'Donnell A, Judson I, Dowsett M, et al. Hormonal impact of the 17α-hydroxylase/C17,20-lyase inhibitor abiraterone acetate (CB7630) in patients with prostate cancer. Br J Cancer 2004;90:2317 2325.

31. Ozen M, Creighton C, Ozdemir M, Ittmann M. Widespread deregulation of microRNA expression in human prostate cancer. Oncogene. 2008;27:1788-1793.

32. Porkka K, Pfeiffer M, Waltering K, Vessella R, Tammela T, Visakorpi T. MicroRNA expression profiling in Prostate cancer. Cancer Res 2007;67:6130-6135.

33. Quaglia A et al. Prostate cancer in Italy before and during the ‘PSA era’: survival trend and prognostic determinants. European Journal of Cancer Prevention. 2003;12:145-152.

34. Ryan CJ, Smith MR, De BonoJS, Molina A, Logothetis C, De Souza, PL, et al. Interim analysis (IA) results of COU-AA-302, a randomized, phase III study of abiraterone acetate (AA) in chemotherapy-naive patients (pts) with metastatic castration-resistant prostate cancer (mCRPC). J Clin Oncol 2012; 30(suppl; abstr LBA4518)

35. Scher HI, Halabi S, Tannock I et al. Design and end points of clinical trials for patients with progressive prostate cancer and castrate levels of testosterone: Recommendations of the prostate cancer clinical trials working group. J Clin Oncol 2008;26:1148-1159

36. Schröder FH, Hugosson J, Carlsson S, Tammela T, Määttänen L, Auvinen A, et al. Screening for Prostate Cancer Decreases the Risk of Developing Metastatic Disease: Findings from the European Randomized Study of Screening for Prostate Cancer (ERSPC). Eur Urol. 2012 Jun 7 Abstract.

37. Shi X, Xue L, ang J, Ma A, Zhao J, Xu M, et al. An androgen regulated miRNA suppresses Bak1 expression and induces androgen-independent growth of prostate cancer cells. PNAS 2007:19983-19988.

38. Sun T, Yang M, Wang Q, Chen S, Balk S, Brown M, et al. Castration-Resistant Prostate cancer-related microRNAs. ASCO Abstract 150 (2010 Genitourinary Cancers Symposium).

39. Szczyrba J, Löprich E, Wach S, Jung V, Unteregger G, Barth S, et al. The microRNA profile of prostate carcinoma obtained by deep sequencing. Mol Cancer Res. 2010;8(4):529-38.

40. Tangen CM, Faulkner JR, Crawford Ed, Thompson IM, Hirano D, Eisenberger M, Hussain M. Ten-year survival in patients with metastatic prostate cancer. Clin Prostate Cancer 2003;2:41-45.

41. Taplin ME, Montgomery RB, Logothetis C, Bubley GJ, Richie JP, Dalkin BL, et al. Effect of neoadjuvant abiraterone acetate (AA) plus leuprolide acetate (LHRHa) on PSA, pathological complete response (pCR) and near pCR in localized high-risk prostate cancer (LHRPC): Results of a randomized phase II study. J Clin Oncol 30;2012 (suppl; abstr 4521).

42. Tong A, Fulgham P, Jay C, Chen P, Khalil I, Liu S, et al. MicroRNA profile analysis of human prostate cancers. Cancer Gene Therapy 2009: 16;206-216

43. Wang L, Tang H, Thayanithy V, Subramanian S, Oberg A, Cunningham J, Cerhan J, Steer C, Thibodeau S. Gene networks and microRNAs implicated in aggressive prostate cancer. Cancer Res. 2009: 69:9490-7.

44. Wang SK and Tsiatis AA. Approximately optimal one-parameter boundaries for group sequential trials. Biometrics 1987; 43:193-99.

45. Yamaoka M, Hara T, Kusaka M. Overcoming persistent dependency on androgen signaling after progression to castration-resistant prostate cancer. Clin Cancer Res 2010;16:4319-24.

Attachment 1: Summary of RECIST Criteria Version 1.1

The following information was extracted from Section 3, Section 4, and Appendix I of the New response evaluation criteria in solid tumors: revised RECIST guideline (version 1.1) authored by Eisenhauer et al (2009). Refer to the European Journal of Cancer article (2009;45(2):228-247) for the complete publication.[8]

3. Measurability of tumor at baseline

3.1 Definitions

At baseline, tumor lesions/lymph nodes will be categorized measurable or non-measurable as follows:

3.1.1 Measurable

Tumor lesions: Must be accurately measured in at least one dimension (longest diameter in the plane of measurement is to be recorded) with a *minimum* size of:

- 10 mm by CT scan (CT scan slice thickness no greater than 5 mm)

The following two methods of measure are not allowed in this protocol:

- 10 mm caliper measurement by clinical exam (lesions which cannot be accurately measured with calipers should be recorded as non-measurable)
- 20 mm by chest X-ray
- *Malignant lymph nodes:* To be considered pathologically enlarged and measurable, a lymph node must be ≥15 mm in short axis when assessed by CT scan (CT scan slice thickness recommended to be no greater than 5 mm). At baseline and in follow-up, only the short axis will be measured and followed. See also ‘Baseline documentation of target and non-target lesions’ in section 4.2 of the RECIST guideline for information on lymph node measurement.

3.1.2 Non-measurable

All other lesions, including small lesions (longest diameter <10 mm or pathological lymph nodes with ≥10 to <15 mm short axis) as well as truly non-measurable lesions. Lesions considered truly non-measurable include: leptomeningeal disease, ascites, pleural or pericardial effusion, inflammatory breast disease, lymphangitic involvement of skin or lung, and abdominal masses/abdominal organomegaly identified by physical exam that is not measurable by reproducible imaging techniques.

3.2 Specifications by methods of measurements

3.2.1 Measurement of lesions

All measurements should be recorded in metric notation, using calipers if clinically assessed. All baseline evaluations should be performed as close as possible to the treatment start and never more than 4 weeks before the beginning of the treatment.

3.2.2 Method of assessment

The same method of assessment and the same technique should be used to characterize each identified and reported lesion at baseline and during follow-up. Imaging based evaluation should always be done rather than clinical examination.

4. Tumor response evaluation

4.1 Assessment of overall tumor burden and measurable disease

To assess objective response or future progression, it is necessary to estimate the *overall tumor burden at baseline* and use this as a comparator for subsequent measurements.

4.2 Baseline documentation of 'target' and 'non-target' lesions

When more than one measurable lesion is present at baseline all lesions up to a maximum of five lesions total (and a maximum of two lesions per organ) representative of all involved organs should be identified as *target lesions* and will be recorded and measured at baseline (this means in instances where patients have only one or two organ sites involved a *maximum* of two and four lesions respectively will be recorded). For evidence to support the selection of only five target lesions, see analyses on a large prospective database in the article by Bogaerts et al. (Reference #10 in Eisenhauer publication).

Target lesions should be selected on the basis of their size (lesions with the longest diameter), be representative of all involved organs, but in addition should be those that lend themselves to *reproducible repeated measurements*. It may be the case that, on occasion, the largest lesion does not lend itself to reproducible measurement in which circumstance the next largest lesion, which can be measured reproducibly, should be selected.

Lymph nodes merit special mention since they are normal anatomical structures, which may be visible by imaging even if not involved by tumor. As noted in Section 3, pathological nodes which are defined as measurable and may be identified as target lesions must meet the criterion of a short axis of ≥15 mm by CT scan. Only the short axis of these nodes will contribute to the baseline sum. The *short* axis of the node is the diameter normally used by radiologists to judge if a node is involved by solid tumor. Nodal size is normally reported as two dimensions in the plane in which the image is obtained (for CT scan this is almost always the axial plane; for MRI the plane of acquisition may be axial, sagittal or coronal). The smaller of these measures is the short axis. For example, an abdominal node which is reported as being 20 mm•30 mm has a short axis of 20 mm and qualifies as a malignant, measurable node. In this example, 20 mm should be recorded as the node measurement (See also the example in Fig. 4 in Appendix II of the Eisenhauer reference). All other pathological nodes (those with short axis ≥10 mm but <15 mm) should be considered non-target lesions. Nodes that have a short axis <10 mm are considered non-pathological and should not be recorded or followed.

A sum of the diameters (longest for non-nodal lesions, short axis for nodal lesions) for all target lesions will be calculated and reported as the *baseline sum diameters*. If lymph nodes are to be included in the sum, then as noted above, only the short axis is added into the sum. The baseline sum diameters will be used as reference to further characterize any objective tumor regression in the measurable dimension of the disease.

All other lesions (or sites of disease) including pathological lymph nodes should be identified as *non-target lesions* and should also be recorded at baseline. Measurements are not required and these lesions should be followed as 'present', 'absent', or in rare cases 'unequivocal progression' (more details to follow). In addition, it is possible to record multiple nontarget lesions involving the same organ as a single item on the case record form (e.g. 'multiple enlarged pelvic lymph nodes' or 'multiple liver metastases').

4.3 Response criteria

This section provides the definitions of the criteria used to determine objective tumor response for target lesions.

4.3.1 Evaluation of target lesions

Complete Response (CR): Disappearance of all target lesions. Any pathological lymph nodes (whether target or non-target) must have reduction in short axis to <10 mm.

Partial Response (PR): At least a 30% decrease in the sum of diameters of target lesions, taking as reference the baseline sum diameters.

Progressive Disease: At least a 20% increase in the sum of diameters of target lesions, taking as reference the smallest sum on study (this includes the baseline sum if that is the smallest on study). In addition to the relative increase of 20%, the sum must also demonstrate an absolute increase of at least 5 mm. (Note: the appearance of one or more new lesions is also considered progression).

Stable Disease (SD): Neither sufficient shrinkage to qualify for PR nor sufficient increase to qualify for progressive disease, taking as reference the smallest sum diameters while on study.

4.3.2 Special notes on the assessment of target lesions

Lymph nodes. Lymph nodes identified as target lesions should always have the actual short axis measurement recorded (measured in the same anatomical plane as the baseline examination), even if the nodes regress to below 10 mm on study. This means that when lymph nodes are included as target lesions, the 'sum' of lesions may not be zero even if complete response criteria are met, since a normal lymph node is defined as having a short axis of <10 mm. Case report forms or other data collection methods may therefore be designed to have target nodal lesions recorded in a separate section where, in order to qualify for CR, each node must achieve a short axis <10 mm. For PR, SD and progressive disease, the actual short axis measurement of the nodes is to be included in the sum of target lesions.

Target lesions that become 'too small to measure'. While on study, all lesions (nodal and non-nodal) recorded at baseline should have their actual measurements recorded at each subsequent evaluation, even when very small (eg 2 mm). However, sometimes lesions or lymph nodes which are recorded as target lesions at baseline become so faint on CT scan that the radiologist may not feel comfortable assigning an exact measure and may report them as being 'too small to measure'. When this occurs it is important that a value be recorded on the CRF. If it is the opinion of the radiologist that the lesion has likely disappeared, the measurement should be recorded as 0 mm. If the lesion is believed to be present and is faintly seen but too small to measure, a default value of 5 mm should be assigned (*Note*: It is less likely that this rule will be used for lymph nodes since they usually have a definable size when normal and are frequently surrounded by fat such as in the retroperitoneum; however, if a lymph node is believed to be present and is faintly seen but too small to measure, a default value of 5 mm should be assigned in this circumstance as well). This default value is derived from the 5 mm CT slice thickness (but should not be changed with varying CT slice thickness). The measurement of these lesions is potentially non-reproducible; therefore providing this default value will prevent false responses or progressions based upon measurement error. To reiterate, however, if the radiologist is able to provide an actual measure, that should be recorded, even if it is below 5 mm.

Lesions that split or coalesce on treatment. When non-nodal lesions 'fragment', the longest diameters of the fragmented portions should be added together to calculate the target lesion sum. Similarly, as lesions coalesce, a plane between them may be maintained that would aid in obtaining maximal diameter measurements of each individual lesion. If the lesions have truly coalesced such that they are no longer separable, the vector of the longest diameter in this instance should be the maximal longest diameter for the 'coalesced lesion'.

4.3.3 Evaluation of non-target lesions

This section provides the definitions of the criteria used to determine the tumor response for the group of non-target lesions. While some non-target lesions may actually be measurable, they need not be measured and instead should be assessed only qualitatively at the timepoints specified in the protocol.

Complete Response (CR): Disappearance of all non-target lesions and normalization of tumor marker level. All lymph nodes must be non-pathological in size (<10 mm short axis).

Non-CR/Non-progressive disease: Persistence of one or more non-target lesion(s) and/or maintenance of tumor marker level above the normal limits.

Progressive Disease: Unequivocal progression (see comments below) of existing non-target lesions. (*Note*: the appearance of one or more new lesions is also considered progression).

4.3.4 Special notes on assessment of progression of non-target disease

The concept of progression of non-target disease requires additional explanation as follows:

When the patient also has measurable disease. In this setting, to achieve ‘unequivocal progression’ on the basis of the non-target disease, there must be an overall level of substantial worsening in non-target disease such that, even in presence of SD or PR in target disease, the overall tumor burden has increased sufficiently to merit discontinuation of therapy. A modest ‘increase’ in the size of one or more non-target lesions is usually not sufficient to quality for unequivocal progression status. The designation of overall progression solely on the basis of change in non-target disease in the face of SD or PR of target disease will therefore be extremely rare.

When the patient has only non-measurable disease. This circumstance arises in some Phase studies when it is not a criterion of study entry to have measurable disease. The same general concepts apply here as noted above, however, in this instance there is no measurable disease assessment to factor into the interpretation of an increase in non-measurable disease burden. Because worsening in non-target disease cannot be easily quantified (by definition: if all lesions are truly non-measurable) a useful test that can be applied when assessing patients for unequivocal progression is to consider if the increase in overall disease burden based on the change in non-measurable disease is comparable in magnitude to the increase that would be required to declare progressive disease for measurable disease: i.e. an increase in tumor burden representing an additional 73% increase in ‘volume’ (which is equivalent to a 20% increase diameter in a measurable lesion). Examples include an increase in a pleural effusion from ‘trace’ to ‘large’, an increase in lymphangitic disease from localized to widespread, or may be described in protocols as ‘sufficient to require a change in therapy.’ If ‘unequivocal progression’ is seen, the patient should be considered to have had overall progressive disease at that point. While it would be ideal to have objective criteria to apply to non-measurable disease, the very nature of that disease makes it impossible to do so, therefore the increase must be substantial.

4.3.5 New lesions

The appearance of new malignant lesions denotes disease progression; therefore, some comments on detection of new lesions are important. There are no specific criteria for the identification of new radiographic lesions; however, the finding of a new lesion should be unequivocal: i.e. not attributable to differences in scanning technique, change in imaging modality or findings thought to represent something other than tumor (for example, some ‘new’ bone lesions may be simply healing or flare of pre-existing lesions). This is particularly important when the patient’s baseline lesions show partial or complete response. For example, necrosis of a liver lesion may be reported on a CT scan report as a ‘new’ cystic lesion, which it is not.

A lesion identified on a follow-up study in an anatomical location that was not scanned at baseline is considered a new lesion and will indicate disease progression. An example of this is the patient who has visceral disease at baseline and while on study has a CT or MRI brain ordered which reveals metastases. The patient’s brain metastases are considered to be evidence of progressive disease even if he/she did not have brain imaging at baseline.

If a new lesion is equivocal, for example because of its small size, continued therapy and follow-up evaluation will clarify if it represents truly new disease. If repeat scans confirm there is definitely a new lesion, then progression should be declared using the date of the initial scan.

4.4.1 Timepoint response

It is assumed that at each protocol specified timepoint, a response assessment occurs. Table 1 in this attachment provides a summary of the overall response status calculation at each timepoint for patients who have measurable disease at baseline.

When patients have non-measurable (therefore non-target) disease only, Table 2 in this attachment is to be used.

4.4.2 Missing assessments and inevaluable designation

When no imaging/measurement is done at all at a particular timepoint, the patient is not evaluable (NE) at that timepoint. If only a subset of lesion measurements are made at an assessment, usually the case is also considered NE at that timepoint, unless a convincing argument can be made that the contribution of the individual missing lesion(s) would not change the assigned timepoint response. This would be most likely to happen in the case of progressive disease. For example, if a patient had a baseline sum of 50 mm with three measured lesions and at follow-up only two lesions were assessed, but those gave a sum of 80 mm, the patient will have achieved progressive disease status, regardless of the contribution of the missing lesion.

4.4.3 Best overall response: all timepoints

The *best overall response* is determined once all the data for the patient is known.

Best response determination in studies where confirmation of complete or partial response IS NOT required: Best response in these studies is defined as the best response across all timepoints (for example, a patient who has SD at first assessment, PR at second assessment, and progressive disease on last assessment has a best overall response of PR). When SD is believed to be best response, it must also meet the protocol specified minimum time from baseline. If the minimum time is not met when SD is otherwise the best timepoint response, the patient's best response depends on the subsequent assessments. For example, a patient who has SD at first assessment, progressive disease at second and does not meet minimum duration for SD, will have a best response of progressive disease. The same patient lost to follow-up after the first SD assessment would be considered inevaluable.

Table 1 - Timepoint response: patients with Target (+/- non-target) disease

Target lesions	**Non-target lesions**	**New lesions**	**Overall response**
CR	CR	No	CR
CR	Non-CR/non-PD	No	PR
CR	Not evaluated	No	PR
PR	Non-PD or not all evaluated	No	PR
SD	Non-PD or not all evaluated	No	SD
Not all evaluated	Non-PD	No	NE
PD	Any	Yes or No	PD
Any	PD	Yes or No	PD
Any	Any	Yes	PD

CR = complete response; PR = partial response; SD = stable disease; PD = progressive disease; NE = not evaluable.

Table 2 - Timepoint response: patients with non-target disease only

Non-target lesions	New lesions	Overall response
CR	No	CR
Non-CR/non-PD	No	Non-CR/non-PD[a]
Not all evaluated	No	NE
Unequivocal PD	Yes or No	PD
Any	Yes	PD

CR = complete response; PD = progressive disease; NE = not evaluable.

a ‘Non-CR/non-PD’ is preferred over ‘stable disease’ for non-target disease since SD is increasingly used as endpoint for assessment of efficacy in some studies so to assign this category when no lesions can be measured is not advised.

Attachment 2: ECOG Performance Status

ECOG Grade Scale (with Karnofsky conversion)

0	Fully active, able to carry on all predisease performance without restriction. (Karnofsky 90-100)
1	Restricted in physically strenuous activity but ambulatory and able to carry out work on a light or sedentary nature, eg, light housework, office work. (Karnofsky 70-80)
2	Ambulatory and capable of all self-care but unable to carry out any work activities. Up and about more than 50% of waking hours. (Karnofsky 50-60)
3	Capable of only limited self-care; confined to bed or chair more than 50% of waking hours. (Karnofsky 30-40)
4	Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair. (Karnofsky 10-20)
5	Dead. (Karnofsky 0)

Attachment 3: New York Heart Failure Criteria

The following table presents the New York Heart Association classification of cardiac disease:

Class	Functional Capacity	Objective Assessment
I	Patients with cardiac disease but without resulting limitations of physical activity. Ordinary physical activity does not cause undue fatigue, palpitation, dyspnea, or anginal pain.	No objective evidence of cardiovascular disease
II	Patients with cardiac disease resulting in slight limitation of physical activity. They are comfortable at rest. Ordinary physical activity results in fatigue, palpitation, dyspnea, or anginal pain.	Objective evidence of minimal cardiovascular disease
III	Patients with cardiac disease resulting in marked limitation of physical activity. They are comfortable at rest. Less than ordinary activity causes fatigue, palpitation, dyspnea, or anginal pain.	Objective evidence of moderately severe cardiovascular disease
IV	Patients with cardiac disease resulting in inability to carry on any physical activity without discomfort. Symptoms of heart failure or the anginal syndrome may be present even at rest. If any physical activity is undertaken, discomfort is increased.	Objective evidence of severe cardiovascular disease

Attachment 4: Additional Information on CYP450 Drug Interactions

http://www.fda.gov/drugs/developmentapprovalprocess/developmentresources/druginteractionslabeling/ucm093664.htm

http://medicine.iupui.edu/clinpharm/ddis/table.aspx

INVESTIGATOR AGREEMENT

I have read this protocol and agree that it contains all necessary details for carrying out this study. I will conduct the study as outlined herein and will complete the study within the time designated.

I will provide copies of the protocol and all pertinent information to all individuals responsible to me who assist in the conduct of this study. I will discuss this material with them to ensure that they are fully informed regarding the study drug, the conduct of the study, and the obligations of confidentiality.

Coordinating Investigator (where required):

Name (typed or printed): ____________________

Institution and Address: ____________________

Signature: ____________________ Date: ____________________
(Day Month Year)

Principal (Site) Investigator:

Name (typed or printed): ____________________

Institution and Address: ____________________

Telephone Number: ____________________

Signature: ____________________ Date: ____________________
(Day Month Year)

Sponsor's Responsible Medical Officer:

Name (typed or printed): Peter De Porre, MD

Institution: Janssen Resear[redacted] Development, LLC

Signature: [redacted] Date: 24 March 2016
(Day Month Year)

Note: If the address or telephone number of the investigator changes during the course of the study, written notification will be provided by the investigator to the sponsor, and a protocol amendment will not be required.

LAST PAGE

Janssen Research & Development, LLC

Statistical Analysis Plan

A Randomized, Double-blind, Comparative Study of ZYTIGA® (Abiraterone Acetate) Plus Low-dose Prednisone Plus Androgen Deprivation Therapy (ADT) Versus ADT Alone in Newly Diagnosed Subjects With High-Risk, Metastatic Hormone-Naive Prostate Cancer (mHNPC)

Protocol 212082PCR3011; Phase 3

JNJ-212082 (abiraterone acetate)

Status: Approved

Date: 6 December 2016

Prepared by: Janssen Research & Development

Document No.: EDMS-ERI-134688744, 1.0

TABLE OF CONTENTS

LIST OF ATTACHMENTS

LIST OF IN-TEXT TABLES

LIST OF ABBREVIATIONS AND DEFINITION OF TERMS

AA-P	abiraterone acetate (plus prednisone)
AE	adverse event
ALB	albumin
ALK-P	alkaline phosphatase
ALT/SGPT	alanine aminotransferase
ANC	absolute neutrophil count
AST/SGOT	aspartate aminotransferase
bid	twice daily
BPI-SF	Brief Pain Inventory-Short Form
CRF	case report form
CRO	Contract Research Organization
CRPC	castration-resistant prostate cancer
CSR	clinical study report
CI	confidence interval
CT	computed tomography
CTCAE	Common Terminology Criteria for Adverse Events
DHEA-S	dihydroepiandrosterone sulphate
ECG	electrocardiogram
ECOG	Eastern Cooperative Oncology Group
FACT-P	Functional Assessment of Cancer Therapy – Prostate
HDL	high density lipoprotein
HR	hazard ratio
IDMC	Independent Data Monitoring Committee
IPCW	Inverse Probability of Censoring Weighted
ITT	intent-to-treat
IVRS	Interactive Voice Response System
IWRS	Interactive Web Response System
LDH	lactate dehydrogenase
LDL	low density lipoprotein
LHRH	luteinizing hormone-releasing hormone
MedDRA	Medical Dictionary for Regulatory Activities
MRI	magnetic resonance imaging
MRU	medical resource utilization
OS	overall survival
PBO	placebo (plus prednisone)
PK	pharmacokinetics

PS	performance status
PSA	prostate specific antigen
PCWG2	Prostate Cancer Working Group
PRO	subject-reported outcome
PT	preferred term
qd	once daily
RECIST	Response Evaluation Criteria in Solid Tumors
rPFS	radiographic progression-free survival
SAE	serious adverse event
SAP	statistical analysis plan
SOC	system organ class
SD	standard deviation
T	serum testosterone
TLGs	tables, listings, and graphs
ULN	upper limit of normal
WHO	World Health Organization

1 STUDY DESCRIPTION

1.1 Objectives

Primary objective:

- The primary objective is to determine whether ZYTIGA in combination with low-dose prednisone and ADT is superior to ADT alone in improving rPFS and OS in subjects with mHNPC with high-risk prognostic factors.

Secondary objectives:

- To evaluate the clinically relevant improvements as well as the safety of ZYTIGA plus low-dose prednisone and ADT compared to ADT alone.
- To identify microRNA (miRNA) and mRNA profiles predictive of ZYTIGA response or resistance.

1.2 Study Design

This is a multinational, multicenter, randomized, double-blind, active-controlled study designed to determine if newly diagnosed subjects with mHNPC who have high-risk prognostic factors will benefit from the addition of ZYTIGA and low-dose prednisone to ADT (LHRH agonists or orchiectomy). Approximately 1200 subjects were to be enrolled in this study. The study population includes newly diagnosed (within 3 months prior to randomization) adult men with high-risk mHNPC (having at least 2 of the following 3 risk factors: (1) Gleason score of ≥8; (2) presence of 3 or more lesions on bone scan; (3) presence of visceral metastasis). Subjects will be stratified by presence of visceral disease and ECOG performance grade of 0, 1 vs. 2 prior to randomization. Subjects must have distant metastatic disease as documented by positive bone scan or metastatic lesions on CT or MRI to be eligible. Eligible subjects may have received ADT or had an orchiectomy, with or without concurrent anti-androgens prior to Cycle 1 Day 1, within 3 months of randomization (for subjects receiving an LHRH agonist, antiandrogens are allowed for the first 2 weeks after Cycle 1 Day 1, any other use is prohibited). ZYTIGA and low-dose prednisone are considered as study drugs.

The study consists of a Screening Phase of up to 28 days before randomization to establish study eligibility and document baseline measurements; a Double-blind Treatment Phase; and a Follow-up Phase of up to 60 months to monitor survival status and subsequent prostate cancer therapy. Subject-reported outcomes (Euro-QoL [EQ-5D-5L], Brief Pain Inventory-Short Form [BPI-SF], Functional Assessment of Cancer Therapy-Prostate [FACT-P], and Brief Fatigue Inventory [BFI]) are collected. Each cycle is 28 days. Treatment is to continue until disease progression or the occurrence of unacceptable toxicity.

Subjects who meet all of the inclusion criteria and none of the exclusion criteria are to be centrally randomized in a 1:1 ratio to the active group [ZYTIGA (1000 mg once daily) plus low-dose prednisone (5 mg once daily) plus ADT] or the control group [ADT plus placebos]. See Section 1.5 for details on randomization.

1.3 Study Endpoints

1.3.1 Efficacy Endpoints

1.3.1.1 Primary Endpoint of rPFS

The co-primary endpoint of rPFS is defined as the time interval from randomization to the first date of radiographic progression or death. Radiographic progression included progression by bone scan (according to modified PCWG2 criteria) and progression of soft tissue lesions by CT or MRI (according to RECIST 1.1 criteria), both assessed by investigators. Subjects without radiographic progression or death will be censored at the last disease assessment.

Radiographic progression in this study is defined as one of the following:

I. Progression of soft tissue lesions measured by CT or MRI as defined in RECIST 1.1 (Protocol Attachment 1)

II. Progression by bone lesions observed on bone scan:
Each subject would fall into one of the following scenarios, depending on what is observed at Cycle 5 Day 1 scan and what is observed on the confirmatory scan. Note that Cycle 5 Day 1 scan needs to be at 16 weeks or later from Cycle 1 Day 1.

1. Subjects who are observed to have ≥2 new bone lesions on Cycle 5 Day 1 scans compared with baseline scans, will need to have a confirmatory scan performed within 42-120 days of Cycle 5 and would fall into one of the 2 categories below:
    a. Subjects with confirmatory scans that show at least 2 new lesions compared to the Cycle 5 Day 1 scans (so a total of at least 4 new lesions compared with baseline scan) will be considered to have disease progression by bone scan.
    b. Subjects who on confirmatory scan did not have at least 2 new lesions compared with the Cycle 5 Day 1 scan will not be considered to have disease progression at that time. For these subjects all subsequent scans would serve as “confirmatory scans” for the Cycle 5 Day 1 scan findings. In order to be considered to have disease progression, these subjects will need to have subsequent scans with at least 2 new lesions compared with Cycle 5 Day 1 scans (ie, at least 4 total new lesions compared to baseline scan).
2. Subjects who at Cycle 5 Day 1 do not have ≥2 new bone lesions compared with baseline would not need to have confirmatory scan. For any subsequent scan beyond the Cycle 5 scan time point, the FIRST scan that would show at least 2 new lesions compared with baseline scans, would be considered to demonstrate disease progression by bone scan.

In defining rPFS, the following censoring rules apply:

1. If the patient does not a have baseline scan or on-study scans, the patient’s data will be censored on the date of randomization;

2. If the patient does not show progression according to RECIST 1.1 or bone scan, the patient's data will be censored on the date of the last scan;

3. If the patient remains on study treatment and prior scans do not show radiographic progression, the patient's data will be censored on the date of the last scan showing no disease progression;

4. If the patient discontinues study treatment for any reason and progression was not observed in the scans prior to the discontinuation, the patient's data will be censored on the last scan showing no disease progression;

5. If the patient discontinues study treatment for any reason and additional new lesions were observed in the scan prior to the discontinuation, and there was no confirmatory bone scan, the patient's data will be censored on the date of the last scan that showed no disease progression;

6. Patient data will also be censored on the date of the last scan that shows no disease progression if:

    a. the patient receives subsequent therapy (ie, cytotoxic chemotherapy) known or intended for prostate cancer during the study prior to the radiographic progression;

    b. the patient misses ≥2 consecutive planned radiographic scans or has ≥2 consecutive unreadable scans (more than 35 weeks apart);

    c. the patient has unequivocal progression of non-bone non-target lesions (eg, appearance of nonmeasurable visceral metastases or pathologically confirmed malignant effusions).

1.3.1.2 Primary Endpoint of OS

The co-primary efficacy endpoint of overall survival is defined as the time interval from randomization to date of death from any cause. For subjects alive at the time of analysis, data will be censored on the last date the subject was known to be alive or lost to follow-up or withdraw consent.

1.3.1.3 Secondary Endpoints

- Time to skeletal-related event (earliest one of the following):
    - Clinical and pathological fracture
    - Spinal cord compression
    - Palliative radiation to bone
    - Surgery to bone
- Time to PSA progression (by PCWG2 criteria)
- Time to subsequent therapy for prostate cancer
- Time to initiation of chemotherapy

- Time to pain progression

1.3.1.4 Exploratory Endpoints

- PSA response rate
- Subject Reported Outcome (PRO) measures: EQ-5D-5L, BPI - SF, FACT - P, BFI
- Time to symptomatic local progression defined as occurrence of urethral obstruction or bladder outlet obstruction
- Prostate cancer specific survival
- Time to chronic opiate use
- Best Overall response

1.3.2 Safety Endpoints

Safety assessments include medical history, vital signs, physical exam, concomitant therapy and procedures, adverse events (AE) collection, SAEs, clinical laboratory parameters, serum lipids, and electrocardiograms (ECG).

1.3.3 Other Endpoints

Biomarkers and MRU are not part of this SAP and not part of the CSR.

1.4 Sample Size Justification

The overall level of significance for the study will be 0.05, with 0.049 allocated in the testing of OS, and 0.001 allocated in the testing of rPFS. Subjects will be randomized in a 1:1 ratio to receive abiraterone acetate plus low-dose prednisone plus ADT or ADT alone.

1.4.1 Hypothesized OS Hazard Ratio of 0.81

It is assumed that failure will follow an exponential distribution with a constant hazard rate. Assuming a median overall survival (OS) of 33 months for the control group (ADT alone), a planned sample size of approximately 1200 subjects will provide 85% power to detect a hazard ratio (HR) of 0.81 (33 months versus 40.75 months, or >7 months of improvement) at a 2-tailed significance level of 0.049 and an enrollment duration of approximately 24 months over a total study duration of 78 months to obtain the required 852 events.

The OS endpoint will incorporate group sequential design by including 2 interim analyses and one final analysis with alpha spending function calculated as Wang-Tsiatis power boundaries of shape parameter 0.2 (East® software).

1.4.2 Hypothesized rPFS Hazard Ratio of 0.667

Only one analysis will be conducted for rPFS. It is assumed that failure will follow an exponential distribution with a constant hazard rate. It is estimated that 565 rPFS events would be required to provide at least 94% power in detecting a HR of 0.667 (median rPFS of 20 months for the ADT alone group versus 30 months for the abiraterone acetate plus low-dose prednisone plus ADT group) at a 2-tailed level of significance of 0.001; the same enrollment assumptions as described for OS are assumed.

1.5 Method of Assigning Subjects to Treatment Groups

A country by country randomization scheme was performed using permuted block randomization. Subjects will be stratified by presence of visceral disease and ECOG performance grade of 0, 1 vs. 2. The stratification factors are thought to represent important prognostic factors that may potentially affect treatment outcome and are included to eliminate bias and to increase the precision of overall treatment effect estimates.

Subjects were assigned in a 1:1 ratio to receive either ZYTIGA plus low-dose prednisone plus ADT or ADT alone. The block size was chosen to minimize the chance of accidental unblinding while sufficiently controlling for potential imbalance between treatment groups, and was kept confidential as part of the randomization schedule. The randomization schedule will be prepared by an independent statistician not otherwise involved with this study, and will be implemented within a global interactive web or voice response system (IWRS/IVRS).

1.6 Blinding Procedures

It is the intent for all subjects, their families, study team members (at the study site, at Sponsor, or at the participating Clinical Research Organization [CRO]) to remain blinded to treatment group assignment until the completion of the study, with the following exceptions:

- The independent biostatisticians and independent statistical programmers responsible for preparing interim tables, listings, and graphs for IDMC review. Individuals assigned to this task will be the minimum required to generate the required tables, listings, and graphs (TLGs) to the IDMC. They will have no other responsibilities associated with this study, and they will be restricted from revealing treatment group assignments to anyone external to these assigned individuals.
- The IDMC members. Data with masked treatment arms will be reviewed by IDMC, meaning the treatment arms will be labeled as A and B instead of the true treatment arms. When it is necessary to fully evaluate whether the study should be stopped early for efficacy/futility or safety, only then unblinding with actual treatment label will be available at the request by the IDMC. Unblinding procedures and the control of the unblinded data are described in the IDMC Charter.
- Personnel performing blood serum concentration assays and analysis for pharmacokinetics.
- Unblinded safety representative and Ethics Committee for SAE reporting.

1.7 Audit Plan

Study 212082PCR3011 will use investigator assessment of rPFS for the primary analysis of rPFS. However, to confirm that the investigator review is not biased in favor of the experimental arm, an audit plan based on the publication of Amit et al[5] will be utilized. All scans will be collected and stored in a central location so that a valid audit can be performed using independent centralized review of blinded radiographic scans. A sample of approximately 160 subjects will be randomly selected for central review. These subjects have to be evaluable, defined as having both baseline disease assessment and at least one on study assessment.

The audit uses calculations of early discrepancy rates (EDR) and late discrepancy rates (LDR) for each treatment group to determine whether bias is present in the investigator assessments. Table 1 shows the agreement between Blinded Independent Central Review and Investigator Review.

Table 1: Blinded Independent Central Review versus Investigator Review of Disease Progression

	Blinded Independent Central Review	
	PD	No PD
Investigator PD	$a = a1 + a2 + a3$	b
No PD	c	d

$a1$: number of agreements on timing and occurrence of PD.
$a2$: number of times investigator declares PD later than Central Review.
$a3$: number of times investigator declares PD earlier than Central Review.

The early discrepancy rate (EDR) of the investigator review is defined as:

$$\mathrm{EDR} = \frac{\mathrm{b} + \mathrm{a}_3}{\mathrm{a} + \mathrm{b}}$$

The late discrepancy rate (LDR) of the investigator review is defined as:

$$LDR = \frac{c + a_2}{b + c + a_2 + a_3}$$

Note the EDR and LDR represent the frequency with which the investigator review declares progression earlier than independent review (EDR) and the frequency with which the investigator review declares progression later than independent review (LDR).

Bias will be considered present in the investigator review if

EDR (experimental arm) – EDR (control arm) < -10%

indicating the EDR in the experimental arm is lower than EDR in the control arm by more than 10%,

or

LDR (experimental arm) – LDR (control arm) > 10%

indicating LDR in the experimental arm is higher than LDR in the control arm by more than 10%.

The final determination of EDR and LDR for each treatment arm and interpretation of the findings will be coordinated by the independent SSG statistician for IDMC. Therefore, all investigators, Company personnel, and the central reviewer will remain blinded to treatment assignment until the IDMC reviews the audit findings and confirms that no bias is present in the investigator review. If bias is considered present, a complete review of all subjects' scans will commence.

2 STANDARD ANALYSIS CONVENTIONS

2.1 Population Analyzed

Based on noncompliance with the ICH GCP guidelines at 1 site in Russia (Site [redacted] in which the Company cannot ascertain trust in the data, all data from these subjects (N=10) at that site were excluded from the CSR primary analyses.

The following analysis populations are defined for this study:

- Intent-to-Treat (ITT) Population – subjects randomized into the study excluding Site [redacted] and who will be classified according to their assigned treatment group, regardless of the actual treatment received. This population will be used for all efficacy analyses, and all analyses of disposition, demographic, and baseline disease characteristics.
- Safety Population – all subjects excluding Site [redacted] who receive any part of the study drug.
- Subjects with measurable disease at baseline
- Safety Population including Site [redacted] – all subjects including Site [redacted] who receive any part of the study drug. This population will be used for safety sensitivity analyses.

2.2 Study Day

Subjects' time on study will be determined in Study Days. Study Day 1 will be defined as the first day of dosing. Positive study days will count forward from Study Day 1. Study Day -1 will be the day before Study Day 1, and all subsequent negative study days will be measured backward from Study Day -1. There will be no Study Day 0. A cycle is defined as a 28 days period in this protocol.

A value obtained on Study Day 1 before administration of study treatment will be considered the baseline value. If a value is not available on Study Day 1, the last available value collected prior to the first dose of study drug will be used as the baseline value.

The study date and corresponding study visit will be captured on each case report form (CRF) from which study day will be calculated.

Study Day will be used in the analysis of safety data. The protocol allows a 72-hour window (3 days) between randomization and first dose, all efficacy analyses on time-to-event endpoints will consider the date of randomization as Day 1.

2.3 Missing Data

Unless otherwise specified, the following general imputation rule will be used for missing data in the assessment of an event:

If all parts of the date are missing, the date will not be imputed. In the case where only the start day of an event is missing, it will be replaced by the start day of study treatment if the event occurs in the same month and year. Otherwise, it will be replaced by the first of the month. If the stop day is missing, the stop day of the event will be replaced by the stop day of study treatment. Otherwise, the last day of the month will be used to replace the missing stop day.

If both the start day and month of an event are missing, the start day and month will be replaced by the start day and month of study treatment if the event and the start of the treatment occur in the same year; otherwise, it will be replaced by 1^{st} of January.

All imputed dates have to be prior to the dates of withdrawal of consent, lost to follow-up, and death.

3 STATISTICAL ANALYSIS METHODS

All statistical analyses will be performed using SAS®. Summaries and listings will be provided by treatment group. Unless otherwise specified, all continuous endpoints will be summarized using descriptive statistics, which will include the number of subjects with a valid measurement (n), mean, standard deviation (SD), median, minimum, and maximum. All categorical endpoints will be summarized using frequencies and percentages. Time-to-event endpoints will be analyzed using Kaplan-Meier product limit methods to estimate the survival distributions and the median time-to-event. Inference for time-to-event endpoints will be assessed using a stratified log-rank statistic as the primary analysis. The proportional hazard assumption will be assessed graphically by plotting log (-log[estimated survival distribution function]) against log(survival time). The resulting graphs should have approximately parallel lines when the assumption holds. If the proportional hazards assumption is reasonably met, then the HR will be used as an estimate of treatment effect. If the proportional hazards assumption is violated, then the inference remains statistically valid for testing equality in survival distributions, but treatment effect will only be estimated using the median time to event in each treatment group.

3.1 Statistical Hypothesis for the Study Objectives

The primary study objective is to compare the clinical benefit of ZYTIGA plus low-dose prednisone plus ADT with ADT alone and will be assessed using rPFS and OS as the co-primary endpoints. The general hypotheses used to address this objective are as follows:

H_o: The survival distributions of the ZYTIGA plus low-dose prednisone plus ADT group, $S_Z(t)$, and the ADT alone group, $S_A(t)$, are equal at all time points t:

$S_Z(t) = S_A(t)$, for all $t > 0$

versus

H_1: The survival distributions are not equal for at least one time point t:

$S_Z(t) \neq S_A(t)$, for some $t > 0$

These hypotheses will be tested using the stratified log rank test. The statistical inference of OS will be carried out within the context of a group sequential testing design as described in Section 3.2.

3.2 Interim Analysis for OS

Interim analyses will be performed on the efficacy endpoint OS using the ITT Population; all subjects randomized into the study at the time of an interim analysis will be included. In this study, 2 interim analyses and 1 final analysis are planned after approximately 50% (~426 events) and approximately 65% (~554 events) and the total number of required (~852) events have occurred, respectively. The East® software was used to obtain stopping boundaries that controls the overall 2-tailed level of significance at alpha 0.049 and provide 85% power to detect a hazard ratio (AA-P:PBO) of 0.81 (corresponds to a 23% improvement in median OS for the AA-P group compared with the PBO group.)

The alpha spending function will be calculated as Wang-Tsiatis power boundaries of shape parameter 0.2. The operating characteristics for these boundaries are presented in Table 2.

Table 2: Statistical Operating Characteristics for OS

Variable	Analyses		
	Interim 1 (50% of Total Events)	**Interim 2** (65% of Total Events)	Final
Projected Observed OS Events	~426	~554	~852
Efficacy Boundary (HR)	0.78	0.81	0.87
Stopping p-value Under H_o (Cumulative)	0.011 (0.011)	0.018 (0.022)	0.038 (0.049)
HR=Hazard ratio; H_o = 0% improvement; H_1 = 23% improvement			

The purpose of the interim analyses will be to unblind the study early if superiority of the ZYTIGA plus low-dose prednisone plus ADT group is demonstrated. However, subjects will continue to be followed for survival as indicated in the protocol. An IDMC will be formed to monitor the safety at regular intervals, and to evaluate efficacy at the interim analysis. The decision to terminate the study will be based on a recommendation from the IDMC who will adhere to the prescribed statistical guidelines. The IDMC will be provided with blinded efficacy data at the time of the interim analysis, with access to the actual treatment group assignments, if this information is deemed necessary before recommending that the study be stopped. The IDMC activities and responsibilities are provided in a separate IDMC Charter.

3.3 Analysis Specifications

In general, all hypotheses testing except the co-primary efficacy endpoints will be performed at a 2-tailed significance level of $\alpha = 0.05$. All interval estimations will be reported using 95% confidence intervals.

Secondary efficacy endpoints given in Section 1.3.1 will be tested using the Hochberg test procedure to strongly control the familywise type I error rate; ie, testing will begin with p value $P_{(m)}$ with corresponding hypothesis $H_{(m)}$, where $P_{(m)}$ is the largest ordered p values amongst ‘m’ secondary endpoints and $H_{(m)}$ is the corresponding hypothesis. If $P_{(m)} \leq \alpha$, then all hypotheses are rejected. If not, then $P_{(j)}$, the (j)th largest p-value, is compared with $\alpha/(m-j+1)$. If smaller, then all hypotheses from $H_{(j)}$ to $H_{(1)}$ are rejected. The testing procedure continues in this manner until no significant result is found. This procedure controls the overall level of significance at the 2-tailed, 0.05 level.

Exploratory endpoints will also be analyzed. The list is provided in

in Section 3.5.3. No adjustment for multiple testing will be made. Each endpoint will be tested at a 2-tailed 0.05 level of significance.

Descriptive statistics will be reported for all safety data. Inferential statistics are not planned to be performed on safety data, however, selected safety parameters may be statistically analyzed to assist in data review.

3.4 Analysis of Subject Disposition and Treatment

3.4.1 Subject Disposition

Distribution of subjects by treatment group for each of the analysis population will be provided. In addition, the number of subjects in the ITT Population will be summarized by study site and treatment group.

Treatment discontinuation will be summarized according to reasons of discontinuation as documented in the CRF (dosing noncompliance, AE, prohibited medications, consent withdrawal, etc).

3.4.2 Protocol Exemptions and Deviations

3.4.2.1 Protocol Deviations

Protocol deviations are captured in the CRF and reviewed by a medical monitor. Protocol deviations will be listed by treatment group, subject number, and categorized according to the deviation reasons. If a significant number of deviations occur, a summary table will be produced showing the number of subjects for each deviation reason. Study withdrawal by reasons collected in the CRF (eg, eligibility criteria violation, prohibited concomitant treatment, etc) will also be summarized.

3.4.3 Study Treatment and Extent of Exposure

Treatment compliance will be summarized separately by treatment group at each cycle and at end of study (cumulative). Summaries will be reported using descriptive statistics for discrete variables and will be performed using the ITT Population. The summaries will consist of the number and percentage of subjects who completed the specified range of tablets, starting at ≤75% compliance to a maximum of 100% in increments of 5%. The calculation of the percentages will be the ratio of the number of tablets taken divided by the actual number of days for that visit times 4. In addition, the number of subjects starting each cycle will also be presented (regardless of the percent of dose compliance).

Since dose reduction is permitted in the study; the number and percentage of subjects with dose reduction will be recorded and summarized according to 1 dose reduction or 2 dose reductions (2 reductions are the maximum allowed per protocol) at end of study. A subject experiencing dose reduction multiple times during the study will be counted according to the lowest dose reduced to.

3.4.4 Demographics and Baseline Characteristics

Subject baseline demographics and disease status variables will be summarized by treatment group for the ITT Population as described in Table 3.

Table 3: Baseline Demographics and Disease Status

Variable	Description
Baseline Demographics	
Continuous Variables:	(n, mean, SD, median, minimum, and maximum)
• Age (yr)	
Categorical Variables:	
• Age group (<65, 65 to 69, 70 to 74, ≥ 75)	
• Sex (Male)	
• Ethnicity (Hispanic or Latino; Not Hispanic or Latino	(n, %)
• Race (White, Black, Asian, American Indian or Alaskan Native; Native Hawaiian or Other Pacific Islander; Other)	
Baseline Vital Signs	
Continuous Variables	
• Weight (kg)	
• Height (cm)	
• Body Temperature (°C)	
• Upright Blood Pressure (mmHg)	
• Heart Rate (beats/min)	
• Respiratory Rate (breaths/min)	

Table 3: Baseline Demographics and Disease Status

Variable	Description
Baseline Disease Status Continuous Variables: • Time from initial diagnosis to first dose (years) • Baseline Serum PSA(ng/mL) • Baseline pain score (worst pain over 24 hours) • Hgb (g/dL) • LDH (IU/L) •	(n, mean, SD, median, minimum, and maximum)
Categorical Variables • Stage at diagnosis (T, N, M) • Gleason score (<7, 7 , 8, 9, 10,…) • Prior cancer treatment (cancer-related surgery, radiotherapy, hormonal therapy, other) • Extent of disease (bone, soft tissue/node, other) • Number of bone lesions (0, 1-2, 3-10, and >10) • Measurable visceral status (yes, no) • High risk factors - GS≥8 + ≥3 bone lesions - GS≥8 + Measurable Visceral - ≥3 bone lesions + Measurable Visceral - All three	(n, %)
Hgb=hemoglobin; LDH=lactate dehydrogenase; LHRH; luteinizing hormone releasing hormone; PSA=prostate specific antigen	

The distribution of subjects within each level of the stratification factors ECOG PS (0-1 versus 2) and presence of visceral disease (yes/no) will also be presented.

Other than visual inspection, no formal tests to confirm homogeneity between treatment groups at baseline will be performed.

3.4.5 Prior and Concomitant Medications

For summarization purposes, medications will be coded to a generic term based on the World Health Organization (WHO) dictionary. Medications administered prior to the expected first dose will be considered prior medications. Concomitant therapies will be those that are taken on or after Study Day 1 through 30 days after the last dose of study drug. Medications started prior to Study Day 1 and continued into the treatment period will be included in the summary of concomitant therapies. Incidence of prior medication and concomitant therapies by generic term will be summarized.

Summaries of subjects' prior prostate cancer systemic therapy, prior prostate cancer radiotherapy, and prior prostate cancer related surgery will be presented as appropriate.

3.5 Analysis of Efficacy Endpoints

Analysis of efficacy endpoints will be conducted on the ITT Population. The primary analysis on the primary and secondary efficacy endpoints will be based on the stratified log rank test; sensitivity analyses using non-stratified log rank test and Cox proportional hazards model will also be performed as supportive analyses.

3.5.1 Efficacy Analysis for the Primary Endpoint of rPFS

The efficacy endpoint of rPFS is defined in Section 1.3.1.1. The rPFS distribution and median rPFS and the 95% confidence interval will be estimated using Kaplan-Meier method. Note that statistical inference of rPFS will not be conducted in the context of group sequential testing design. The following SAS® code will be used for the analysis:

```
proc lifetest data=DATASET_NAME;                                        (1)
time rPFSTIME*CENSOR(0);
strata ECOG VISCERAL
group=TREATMENT;
run;
```

where DATASET_NAME is the name of the dataset, rPFSTIME is a vector containing the OS measurement for each patient, CENSOR is its respective censoring vector, TREATMENT is a vector containing treatment group assignment, **and** ECOG and VISCERAL are vectors for the stratification factors of ECOG PS score category and visceral status at baseline.

The resulting statistic (stratified log rank test statistic) will be evaluated using East® to ensure that it is compared with the appropriate stopping boundary given the precise number of events observed at the time of the interim analysis. If the statistic crosses a stopping boundary, then the p value from the SAS® output will be used.

Supportive analysis using the Cox proportional hazards model will also be performed. The score statistic (which is equivalent to log rank test statistic) may be used to provide an adjusted estimate of the HR and corresponding 2-tailed 95% confidence interval. This information will be included in the IDMC package for review as detailed in the IDMC charter.

```
proc phreg data=DATASET_NAME; (2)
model rPFSTIME*CENSOR(0)=TREATMENT / ties=discrete;
strata ECOG VISCERAL; run;
```

where the variable names are as defined above. The score statistic (which is equivalent to log rank test statistic) may be used to provide an adjusted estimate of the HR and corresponding 2-tailed 95% confidence interval. This information will be included in the IDMC package for review as detailed in the IDMC charter and interim analysis plan.

3.5.2 Efficacy Analysis for the Primary Endpoint of OS

The efficacy endpoint of OS is defined in Section 1.3.1.2. The primary and supportive analyses of OS will utilize similar SAS® codes as in (1) and (2) above, replacing rPFSTIME with appropriate SAS® variable OSTIME. The OS distribution, median OS, and the 95% confidence interval will be estimated using the Kaplan-Meier method. Statistical inference will be evaluated in the context of the group sequential testing design described in Sections 3.2.

3.5.3 Efficacy Analyses for the Secondary Endpoints and Other Study Endpoints

Secondary endpoints are defined and listed in Table 4. Comparisons between treatment groups will be conducted according the Hochberg's test procedure as described in Section 3.3 at an overall 2-sided 0.05 level of significance.

Table 4: Secondary Endpoints

Variable	Description
TT_SRE	Time from randomization to next skeletal-related event (one of the following): □ Clinical or pathological fracture □ Spinal cord compression □ Palliative radiation to bone □ Surgery to bone Subjects with no event will be censored at the last known alive date.
TT_PSA	Time to PSA progression The time interval from the date of randomization to the date of the PSA progression as defined in the PCWG2 criteria. Subjects who have no PSA progression at the time of analysis will be censored at the later of last known date of no progression and randomization.
TT_FU_TRT	Time to next subsequent therapy for prostate cancer The time interval from the date of randomization to the date of initiation of subsequent therapy for prostate cancer. Subjects who have no subsequent therapy at the time of analysis will be censored at the last known alive date.
TT_CHEMO	Time to initiation of chemotherapy The time interval from the date of randomization to the date of initiation of cytotoxic chemotherapy for prostate cancer. Subjects who have no cytotoxic chemotherapy administration at the time of analysis will be censored at the last known alive date.
TT_PAIN	Time to pain progression The time interval from randomization to the first date a subject experience a BPI-SF increase by ≥30% from baseline in the BPI-SF worst pain intensity (items 3) observed at 2 consecutive evaluations ≥4 weeks apart. Subjects who have not experienced pain progression at the time of analysis will be censored on the last known date a subject was known to have not progressed. Subjects with no on-study assessment or no baseline assessment will be censored at date of randomization.
If subject received a subsequent anti-prostate cancer therapy prior to the event, then subject will be censored at the last assessment prior to the subsequent therapy.	

Exploratory endpoints to be analyzed are included in Table 5. No adjustments for multiple testing are planned; each comparison between treatment groups will be carried out at α = 0.05.

Table 5: Exploratory Endpoints

PSA_RR	PSA response rate Proportion of subjects achieving a PSA decline ≥50% according to PCWG2 criteria
PRO	Subject Reported Outcome (PRO) measures: EQ-5D-5L, BPI - SF, FACT - P, BFI Details will be in a separate SAP.
TT_SymLocProg	Time from randomization to first symptomatic local progression defined as occurrence of urethral obstruction or bladder outlet obstruction with symptoms that require medical intervention. Subjects with no event will be censored at the later of last dose date + 30 days or at randomization if the subject is never treated.
Prostate Cancer Specific Survival	Time from randomization to death date due to prostate cancer. Subjects alive or who died due to other reasons will be censored at last date of known alive or death not due to prostate cancer.
Time to chronic opiate use	Time from randomization to new opioid analgesics use, or increased dose or frequency of the existing opioid analgesics from Cycle 1 Day 1 for ≥ 3 weeks orally; 7 days parenterally, ie, non-oral formulation). Subjects with no event will be censored at last dose + 30 days or at randomization if the subject is never treated.
Best Overall response	Best overall response by RECIST 1.1 in Subjects with Measurable Disease at Baseline

Estimates of the time-to-event endpoints will be obtained using the Kaplan-Meier estimates of the survival distributions. Statistical inference will be evaluated similarly as in the primary analysis, except that they will not be evaluated using the group sequential testing design. Response endpoints will be summarized using descriptive statistics for categorical data by dose group. The relative risk (treatment:control) will be reported along with the associated 2-tailed 95% confidence intervals. Statistical inference will be evaluated using the Chi-square statistic; the Fisher's exact test may be used if the expected counts in some cells are small.

Endpoints with change in scores will use 2-independent t-test procedure.

3.5.4 Sensitivity Analysis and Subgroup Analysis

Sensitivity analyses on primary endpoints will be performed to assess the robustness and consistency of the endpoints. The results from the analyses will not be adjusted for multiplicity testing. Each analysis will be compared at significance level of 0.05.

3.5.4.1 Sensitivity Analysis for OS

A large number of subjects are expected to crossover to abiraterone acetate or receive life-extending subsequent therapies, the following analysis may be used in estimating the true treatment effect.

1. Inverse Probability of Censoring Weighted (IPCW) log-rank Tests by Robins et al[1,2,3,4] will be used to estimate the treatment effect and its associated confidence interval.

2. Censoring at the time of initiation of subsequent anticancer therapy.

3. Using a time-dependent Cox regression; the HR prior to receiving subsequent anticancer therapy and after receiving the subsequent anticancer therapy will be estimated, the associated 95%CI will also be calculated.

Additional analyses may be performed for OS if appropriate.

3.5.4.2 Subgroup Analysis for the Primary Endpoint

The non-stratified analysis on co-primary endpoints of rPFS and OS will be performed. Since this analysis is not being analyzed in the context of group sequential testing design, no adjustment will be made to the p values or estimated treatment effects.

Subgroup analysis is planned for the co-primary endpoints of rPFS and OS to investigate whether treatment effects are consistent within subgroups using nonstratified univariate model. Each subgroup will be analyzed separately. The subgroups are as follows:

1. Age (<65, ≥65, ≥75)
2. ECOG PS at randomization (0/1 vs. 2)
3. Gleason score (<8 vs. ≥8)
4. Number of baseline bone lesions (≤10 vs, >10)
5. Presence of visceral disease at randomization (Yes vs. No)
6. Subjects whose baseline PSA is greater than the median baseline value (Yes vs. No)
7. Subjects whose LDH value is greater than the median baseline value (Yes vs. No)
8. Region (Eastern EU, Western EU, Asia Pacific, Rest of World)

The hazard ratio within each subgroup will be estimated using a non-stratified Cox proportional hazard model. Results from these analyses will not be considered inconsistent with the primary analysis unless the 95% confidence interval for the hazard ratio within a subgroup does not include the point estimate for the primary analysis.

3.5.5 Other Exploratory Analysis

A post hoc multivariate analysis may be conducted to evaluate the treatment effect when controlling for clinically meaningful factors at baseline. Univariate analyses will be performed investigating key laboratory parameters and factors defining subgroups. The following significant prognostic factors may be included: ECOG performance status scale, baseline serum PSA, baseline LDH, measurable visceral, bone metastasis at baseline and age. Then a Cox regression model may be run with treatment and those factors as covariates. In this model, the HR for treatment effect will be estimated.

The strength of association between rPFS and OS will be evaluated using Spearman's correlation coefficient estimated through the Clayton copula[6], which takes censoring into account.

To evaluate the treatment outcome following the subsequent therapy for prostate cancer, analysis of PFS2 will be performed. PFS2 is defined as the time from randomization to the second disease progression during follow-up after the subsequent therapy, or death from any cause. For patients alive and for whom a second progression has not been observed, data should be censored at the last time known to be alive and without second disease progression.

3.6 Analysis of Safety Endpoints

The safety variables to be analyzed include vital sign measurements, AEs, ECGs, routine hematology, coagulation factors, serum chemistry panel, serum lipids, and deaths. Safety variables will be tabulated by descriptive statistics (n, mean, median, SD, minimum, and maximum; or n and percent) if appropriate.

3.6.1 Adverse Events

Adverse events are coded to System Organ Class (SOC) and preferred terms (PT) using the MedDRA® coding system (version 18.0 or higher).

The severity of AEs is graded on a scale of 1 to 5 according to the adult NCI Common Terminology Criteria for Adverse Events (CTCAE version 4.0) where higher grades indicate events of higher severity. Adverse events will be summarized by grade according to the worst grade experienced.

Table 6 describes the summaries of AEs that will be provided as incidence tables (number of subjects experiencing an event) by treatment group and overall. For summary purposes, AEs will be defined as all reported events with a start date on or after Study Day 1 or that increase in severity on or after Study Day 1. All AEs are collected through 30 days post last dose (treatment-emergent AEs).

All AEs have their relationship to study drug assessed by the investigator as not related, doubtful, possible, probable, very likely Adverse events will be categorized and summarized according to their highest relationship to study drug. Adverse events reported as possible, probable, very likely will be classified as treatment-related AEs.

<u>Adverse Events of Special Interest</u>

Adverse events of special interest (ie, important identified risks), including cardiac disorders; events related to mineralocorticoid excess (hypertension, hypokalemia, and fluid retention/edema, and hypokalemia); hepatotoxicity; cataract; osteoporosis, including osteoporosis-related fractures; rhabdomyolysis/myopathy; allergic alveolitis; and drug-drug interactions (CYP2D6) and food effect, were identified based on previous experience with abiraterone acetate. Preferred terms were grouped according to Modified MedDRA Queries

(SMQ) that were used across abiraterone acetate Studies 301 and 302. The incidence of AEs of special interest will be summarized by preferred term.

Adverse Events of Clinical Importance

Adverse events of clinical importance (ie, important potential risks) include anemia, diarrhea, sexual dysfunction, thrombocytopenia, dyspepsia, urinary tract infection, hematuria, and adrenal insufficiency, which were identified based on previous experience with abiraterone acetate. Preferred terms were grouped according to Modified MedDRA Queries (SMQ) that were used across abiraterone acetate Studies 301 and 302. The AEs of clinical importance are summarized by preferred term.

Modified MedDRA Queries (SMQ) for AE of special interest and of clinical importance can be found in Attachment 1.

Table 6: Summary of Adverse Events

Summary	Sorted By	All Study Events	Treatment-Related Events
AEs	SOC, PT	✔	✔
	SOC, PT	✔	
Most common (≥ 5%) AEs [1]	SOC, PT	✔	
Grade 3/4 AEs	SOC, PT	✔	✔
Most common (≥ 1%) SAEs	SOC, PT	✔	
SAEs by toxicity grade	SOC, PT	✔	
AEs, SAEs, and AEs of special interest, event rate per 100-subject years of treatment duration	SOC (no SOC for AEs of special interest), PT	✔	
AEs leading to discontinuation of study medication	SOC, PT	✔	
AEs leading to death	SOC, PT	✔	
AEs of special interest	PT	✔	
AEs of special interest by toxicity grade	PT	✔	
AEs of special interest leading to discontinuation of study medication	PT	✔	
SAEs of special interest	PT	✔	
AEs leading to dose modification or interruption of abiraterone acetate or prednisone/prednisolone	SOC, PT	✔	
AEs, leading to hospitalization	SOC, PT	✔	

[1] The 5% may be adjusted if warranted to better represent the safety data

To adjust for unequal lengths of study treatment duration among subjects, and potentially between treatment groups, an additional summary based on event rate per 100-subject years of treatment duration will also be summarized. The event rate per 100-subject years of treatment duration is calculated as the total number of treatment emergent events divided by the total treatment duration per 100-subject years. Adverse events of interest selected according to Appendix 1 will also be summarized.

Listings will be provided for subjects who experienced any Grade 3 or 4 AEs, All TEAEs, TEAEs of Special Interest, and others as described below.

3.6.2 Premature Discontinuations due to Adverse Events

A summary of reasons for discontinuation will be provided summarizing the number and percentage of subjects for each reason indicated. A listing of premature discontinuations due to AEs including the subject number, site, treatment group, start date and study day of AE, severity, relationship to study regimen, action taken, and outcome of AE will be provided. Additional summary table summarizing preferred term may be provided if warranted.

3.6.3 Serious Adverse Events (SAE)

A listing of SAEs including the subject number, site, treatment group, start date and study day of SAE, severity, relationship to study drug, action taken, and outcome of SAE will be provided. Additional summary table by preferred term may be provided if warranted.

3.6.4 Deaths

A listing of deaths including the subject number, site, treatment group, date and study day of death, and cause of death will be provided.

3.6.5 Clinical Laboratory Test Analyses

Laboratory results will be classified according to the NCI CTC. Baseline grade and the worst grade per subjects during postbaseline will be tabulated. Shift table analyses for select hematology variables are to be performed summarizing the number of subjects using baseline grade and worst grade at postbaseline. Additional laboratory tests such as serum testosterone (T) will be summarized similarly as above.

Subjects who Met Laboratory Criteria for eDISH (Evaluation of Drug Induced Serious Hepatotoxicity) will be listed. eDISH is defined as 1) Elevated ALT or AST at any time. That is, max (ALT/ULN) >3 or max (AST /ULN) >3. 2) Elevated TBL at any time. That is, max (TBL/ULN) >=2.

3.6.6 Vital Signs

Vital sign measurements collected include blood pressure, heart rate, respirations, and body temperature. Incidence of abnormalities in vital signs during treatment phase will be summarized.

3.6.7 Electrocardiograms (ECGs)

Descriptive statistics for the baseline QTc will be presented, both the Fredericia and Bazett corrections will be reported.

4 REFERENCES

1. Hernan M, Brumback B, Robins J. Marginal structural models to estimate the causal effect of zidovudine on the survival of HIV-positive men, Department of Epidemiology and Biostatistics, Harvard School of Public Health, Lippincott Williams & Wilkins, Inc. 2000, 561-570.
2. Robins J, Finkelstein D. Correcting for noncompliance and dependent censoring in an AIDS clinical trial with inverse probability of censoring wieighted (IPCW) log-rank tests. Biometrics. 2000; 56:779-788.
3. Robins J, Hernan N, Brumback B. Marginal structural models and causal inference in epidemiology, Epidemilogy 2010; 11:550-560.
4. Westreich D, Cole S et al. Time scale and adjusted survival curves for marginal structural cox models, Am J Epidemiology 2010;171: 691-700.
5. Amit O, Mannino F, Stone AM, et al. Blinded independent central review of progression in cancer clinical trials: Results from a meta-analysis. Eur J Cancer 2011;47:1772-1778.
6. Burzykowski T, Molenberghs G, Buyse M, et al. Validation of surrogate end points in multiple randomized clinical trials with failure time end points. Appl Statist 2001: 50; 405-422.

Attachment 1: Modified MeDDRA Queries as SEARCH CRITERIA for AE OF special INTEREST and AE of clinical importance

The search criteria of adverse event(s) of special interest (AEOSI) or of clinical importance (AEOCI) are based on adverse event preferred terms from MedDRA Version 18.0 dictionary.

Most categories are based on a MedDRA SMQ but if one does not exist a compilation of terms that reflect the event will be proposed for extraction and analysis of the data. Each of these events is defined below:

Adverse Event of Special Interest Category= Allergic alveolitis	
Search Criteria Category= PT Alveolitis allergic	
Alveolitis allergic	
Adverse Event of Clinical Importance Category= Anaemia	
Search Criteria Category= HLT Anaemia deficiencies	
Anaemia folate deficiency	Iron deficiency anaemia
Anaemia of pregnancy	Pernicious anaemia
Anaemia vitamin B12 deficiency	Protein deficiency anaemia
Anaemia vitamin B6 deficiency	Subacute combined cord degeneration
Deficiency anaemia	
Search Criteria Category= HLT Anaemias NEC	
Anaemia megaloblastic	Hyperchromic anaemia
Anaemia postoperative	Hypochromic anaemia
Anaemic hypoxia	Malaria
Babesiosis	Plasmodium falciparum infection
Blood loss anaemia neonatal	Plasmodium malariae infection
Congenital anaemia	Plasmodium ovale infection
Congenital malaria	Plasmodium vivax infection
Conjunctival pallor	Sideroblastic anaemia
Haemorrhagic anaemia	Tropical sprue
Hereditary sideroblastic anaemia	
Search Criteria Category= HLT Anaemias due to chronic disorders	
Anaemia of chronic disease	Nephrogenic anaemia
Anaemia of malignant disease	
Search Criteria Category= HLT Marrow depression and hypoplastic anaemias	
Aase syndrome	Febrile bone marrow aplasia
Autoimmune aplastic anaemia	Granulocytes maturation arrest
Autoimmune pancytopenia	Hypoplastic anaemia

Bicytopenia	Myeloid maturation arrest
Bone marrow failure	Pancytopenia
Bone marrow toxicity	Panmyelopathy
Congenital aplastic anaemia	Pearson's syndrome
Congenital dyskeratosis	Pure white cell aplasia
Cytopenia	Shwachman-Diamond syndrome
Dubowitz syndrome	
Search Criteria Category= Haematopoietic cytopenias affecting more than one type of blood cell (SMQ)	
Aspiration bone marrow abnormal	Full blood count decreased
Biopsy bone marrow abnormal	Haematotoxicity
Blood count abnormal	Myelodysplastic syndrome
Blood disorder	Myelodysplastic syndrome transformation
Bone marrow disorder	Myeloid metaplasia
Bone marrow infiltration	Plasmablast count decreased
Bone marrow myelogram abnormal	Scan bone marrow abnormal
Bone marrow necrosis	
Search Criteria Category= Haematopoietic erythropenia (SMQ)	
Anaemia	Haemoglobin decreased
Anaemia macrocytic	Leukoerythroblastic anaemia
Anaemia neonatal	Microcytic anaemia
Aplasia pure red cell	Normochromic normocytic anaemia
Aplastic anaemia	Proerythroblast count abnormal
Erythroblast count abnormal	Proerythroblast count decreased
Erythroblast count decreased	Red blood cell count abnormal
Erythroid maturation arrest	Red blood cell count decreased
Erythropenia	Reticulocyte count abnormal
Erythropoiesis abnormal	Reticulocyte count decreased
Haematocrit abnormal	Reticulocyte percentage decreased
Haematocrit decreased	Reticulocytopenia
Haemoglobin abnormal	
Adverse Event of Special Interest Category= Cardiac Disorders	
Search Criteria Category= Myocardial infarction (SMQ)	
Acute coronary syndrome	Electrocardiogram ST-T segment elevation
Acute myocardial infarction	Infarction
Blood creatine phosphokinase MB abnormal	Kounis syndrome

Blood creatine phosphokinase MB increased	Myocardial infarction
Blood creatine phosphokinase abnormal	Myocardial necrosis
Blood creatine phosphokinase increased	Myocardial necrosis marker increased
Coronary artery embolism	Myocardial reperfusion injury
Coronary artery occlusion	Myocardial stunning
Coronary artery reocclusion	Papillary muscle infarction
Coronary artery thrombosis	Post procedural myocardial infarction
Coronary bypass thrombosis	Postinfarction angina
Coronary vascular graft occlusion	Scan myocardial perfusion abnormal
ECG electrically inactive area	Silent myocardial infarction
ECG signs of myocardial infarction	Troponin I increased
Electrocardiogram Q wave abnormal	Troponin T increased
Electrocardiogram ST segment abnormal	Troponin increased
Electrocardiogram ST segment elevation	Vascular graft occlusion
Search Criteria Category= Other cardiac disorders (PT grouping)	
Atrioventricular block	Electrocardiogram QT prolonged
Atrioventricular block first degree	Hypertrophic cardiomyopathy
Bundle branch block left	Left ventricular hypertrophy
Cardiac disorder	Long QT syndrome
Cardiomyopathy	Metabolic cardiomyopathy
Cardiomyopathy acute	Mitral valve disease
Cardiomyopathy alcoholic	Non-obstructive cardiomyopathy
Chest pain	Restrictive cardiomyopathy
Conduction disorder	Sinus arrhythmia
Congestive cardiomyopathy	Sinus bradycardia
Cytotoxic cardiomyopathy	Sinus node dysfunction (SSS is not the PT in 18.0 but the LLT)
Diabetic cardiomyopathy	Viral cardiomyopathy
Electrocardiogram QT interval abnormal	Wandering pacemaker
Search Criteria Category= Other ischaemic heart disease (SMQ)	
Angina pectoris	Dissecting coronary artery aneurysm
Angina unstable	ECG signs of myocardial ischaemia
Anginal equivalent	Electrocardiogram ST segment depression
Arteriogram coronary abnormal	Electrocardiogram ST-T segment abnormal
Arteriosclerosis coronary artery	Electrocardiogram ST-T segment depression
Arteriospasm coronary	Electrocardiogram T wave abnormal
Cardiac stress test abnormal	Electrocardiogram T wave inversion

Cardiopulmonary exercise test abnormal	Exercise electrocardiogram abnormal
Computerised tomogram coronary artery abnormal	Exercise test abnormal
Coronary angioplasty	External counterpulsation
Coronary arterial stent insertion	Haemorrhage coronary artery
Coronary artery bypass	Ischaemic cardiomyopathy
Coronary artery disease	Microvascular coronary artery disease
Coronary artery dissection	Myocardial ischaemia
Coronary artery insufficiency	Percutaneous coronary intervention
Coronary artery restenosis	Prinzmetal angina
Coronary artery stenosis	Stress cardiomyopathy
Coronary endarterectomy	Stress echocardiogram abnormal
Coronary no-reflow phenomenon	Subclavian coronary steal syndrome
Coronary ostial stenosis	Subendocardial ischaemia
Coronary revascularisation	
Search Criteria Category= SMQ Arrhythmia related investigations, signs and symptoms (SMQ)	
Bradycardia	Heart rate abnormal
Cardiac arrest	Heart rate decreased
Cardiac death	Heart rate increased
Cardiac telemetry abnormal	Loss of consciousness
Cardio-respiratory arrest	Palpitations
Chronotropic incompetence	Rebound tachycardia
Electrocardiogram RR interval prolonged	Sudden cardiac death
Electrocardiogram U-wave abnormality	Sudden death
Electrocardiogram abnormal	Syncope
Electrocardiogram ambulatory abnormal	Tachycardia
Electrocardiogram change	Tachycardia paroxysmal
Electrocardiogram repolarisation abnormality	
Search Criteria Category= SMQ Cardiac failure (SMQ)	
Acute left ventricular failure	Dilatation ventricular
Acute pulmonary oedema	Dyspnoea paroxysmal nocturnal
Acute right ventricular failure	Ejection fraction decreased
Artificial heart implant	Heart transplant
Atrial natriuretic peptide abnormal	Hepatic vein dilatation
Atrial natriuretic peptide increased	Hepatojugular reflux
Brain natriuretic peptide abnormal	Jugular vein distension
Brain natriuretic peptide increased	Left ventricular dysfunction
Cardiac asthma	Left ventricular failure

Cardiac cirrhosis	Low cardiac output syndrome
Cardiac failure	Lower respiratory tract congestion
Cardiac failure acute	Myocardial depression
Cardiac failure chronic	N-terminal prohormone brain natriuretic peptide abnormal
Cardiac failure congestive	N-terminal prohormone brain natriuretic peptide increased
Cardiac failure high output	Neonatal cardiac failure
Cardiac index decreased	Nocturnal dyspnoea
Cardiac output decreased	Obstructive shock
Cardiac resynchronisation therapy	Oedema due to cardiac disease
Cardiac ventriculogram abnormal	Orthopnoea
Cardiac ventriculogram left abnormal	Pulmonary congestion
Cardiac ventriculogram right abnormal	Pulmonary oedema
Cardio-respiratory distress	Pulmonary oedema neonatal
Cardiogenic shock	Radiation associated cardiac failure
Cardiomegaly	Right ventricular dysfunction
Cardiopulmonary failure	Right ventricular failure
Cardiorenal syndrome	Stroke volume decreased
Cardiothoracic ratio increased	Systolic dysfunction
Central venous pressure increased	Venous pressure increased
Chronic left ventricular failure	Venous pressure jugular abnormal
Chronic right ventricular failure	Venous pressure jugular increased
Cor pulmonale	Ventricular assist device insertion
Cor pulmonale acute	Ventricular dysfunction
Cor pulmonale chronic	Ventricular dyssynchrony
Diastolic dysfunction	Ventricular failure
Search Criteria Category= SMQ Supraventricular tachyarrhythmias (SMQ)	
Arrhythmia supraventricular	Electrocardiogram P wave abnormal
Atrial fibrillation	Junctional ectopic tachycardia
Atrial flutter	Retrograde p-waves
Atrial flutter with 1:1 atrioventricular conduction	Sinus tachycardia
Atrial parasystole	Supraventricular extrasystoles
Atrial tachycardia	Supraventricular tachyarrhythmia
ECG P wave inverted	Supraventricular tachycardia
Search Criteria Category= Ventricular tachyarrhythmias (SMQ)	
Accelerated idioventricular rhythm	Ventricular fibrillation
Cardiac fibrillation	Ventricular flutter

Parasystole	Ventricular parasystole
Rhythm idioventricular	Ventricular pre-excitation
Torsade de pointes	Ventricular tachyarrhythmia
Ventricular arrhythmia	Ventricular tachycardia
Ventricular extrasystoles	
Adverse Event of Special Interest Category= Cataract (SMQ Lens disorders)	
Search Criteria Category= SMQ Lens disorders (SMQ)	
Acquired lenticonus	Lens capsulotomy
Anterior capsule contraction	Lens discolouration
Aphakia	Lens dislocation
Atopic cataract	Lens disorder
Capsular block syndrome	Lens extraction
Cataract	Lenticular opacities
Cataract cortical	Lenticular pigmentation
Cataract diabetic	Posterior capsule opacification
Cataract nuclear	Posterior lens capsulotomy
Cataract operation	Radiation cataract
Cataract operation complication	Red reflex abnormal
Cataract subcapsular	Suture fixation of intraocular lens
Colour blindness acquired	Toxic anterior segment syndrome
Deposit eye	Toxic cataract
Fibrin deposition on lens postoperative	Vision blurred
Floppy iris syndrome	Visual acuity reduced
Hepato-lenticular degeneration	Visual impairment
Adverse Event of Clinical Importance Category= Diarrhea	
Search Criteria Category= Diarrhea	
Diarrhoea	
Adverse Event of Special Interest Category= Drug-drug interaction	
Search Criteria Category= Drug-Drug Interaction (CYP2D6) (PT grouping) Note medical review will be performed to identify this category in reference to the searched terms	
Drug interaction	Labelled drug-drug interaction medication error
Inhibitory drug interaction	Potentiating drug interaction
Adverse Event of Clinical Importance Category= Dyspepsia	

Search Criteria Category= PT Dyspepsia	
Dyspepsia	
Adverse Event of Special Interest Category= Fluid retention/oedema	
Search Criteria Category= Haemodynamic oedema, effusions and fluid overload (SMQ)	
Administration site joint effusion	Injection site swelling
Administration site oedema	Instillation site oedema
Administration site swelling	Joint effusion
Amyloid related imaging abnormalities	Joint swelling
Application site joint effusion	Lipoedema
Application site joint swelling	Local swelling
Application site oedema	Localised oedema
Application site swelling	Lymphoedema
Ascites	Medical device site joint effusion
Bone marrow oedema	Medical device site joint swelling
Bone marrow oedema syndrome	Mouth swelling
Bone swelling	Muscle oedema
Brain oedema	Muscle swelling
Bronchial oedema	Myocardial oedema
Capillary leak syndrome	Non-cardiogenic pulmonary oedema
Catheter site oedema	Oedema
Cerebral oedema management	Oedema due to renal disease
Cervix oedema	Oedema mucosal
Compression stockings application	Oedema neonatal
Cytotoxic oedema	Oedema peripheral
Effusion	Oedematous kidney
Elephantiasis nostras verrucosa	Oesophageal oedema
Extensive swelling of vaccinated limb	Pelvic fluid collection
Fluid overload	Pericardial effusion
Fluid retention	Perinephric effusion
Gallbladder oedema	Peripheral oedema neonatal
Gastrointestinal oedema	Peripheral swelling
Generalised oedema	Pleural effusion
Gestational oedema	Prevertebral soft tissue swelling of cervical space
Gravitational oedema	Puncture site oedema
Heat oedema	Reexpansion pulmonary oedema
Hydraemia	Retroperitoneal effusion

Hydrothorax	Retroperitoneal oedema
Hypervolaemia	Scleroedema
Hypoosmolar state	Skin oedema
Implant site oedema	Skin swelling
Implant site swelling	Spinal cord oedema
Incision site oedema	Subdural effusion
Incision site swelling	Swelling
Infusion site joint effusion	Testicular swelling
Infusion site joint swelling	Vaccination site joint effusion
Infusion site oedema	Vaccination site joint swelling
Infusion site swelling	Vasogenic cerebral oedema
Injection site joint swelling	Visceral oedema
Injection site oedema	
Adverse Event of Special Interest Category= Food interaction	
Search Criteria Category= Increased Exposure with Food (PT grouping) Note medical review will be performed to identify this category in reference to the searched terms	
Food interaction	Labelled drug-food interaction medication error
Adverse Event of Clinical Importance Category= Haematuria	
Search Criteria Category= PT Haematuria	
Haematuria	
Adverse Event of Special Interest Category= Hepatotoxicity	
Search Criteria Category= Biliary system related investigations, signs and symptoms (SMQ)	
Abnormal faeces	Cholecystogram oral abnormal
Bile culture positive	Deficiency of bile secretion
Bile duct pressure increased	Endoscopic retrograde cholangiopancreatography abnormal
Bile output abnormal	Endoscopy biliary tract abnormal
Bile output decreased	Faeces pale
Bile output increased	Gallbladder palpable
Biliary ascites	Jaundice
Bilirubin conjugated abnormal	Jaundice cholestatic
Bilirubin conjugated increased	Jaundice extrahepatic obstructive
Bilirubinuria	Limy bile syndrome
Biopsy bile duct abnormal	Ultrasound biliary tract abnormal

Cholangiogram abnormal	Urobilinogen urine decreased
Cholecystogram intravenous abnormal	X-ray hepatobiliary abnormal
Search Criteria Category= Cholestasis and jaundice of hepatic origin (SMQ)	
Bilirubin excretion disorder	Icterus index increased
Cholaemia	Jaundice hepatocellular
Cholestasis	Ocular icterus
Cholestatic pruritus	Parenteral nutrition associated liver disease
Drug-induced liver injury	Yellow skin
Search Criteria Category= Hepatic failure, fibrosis and cirrhosis and other liver damage-related conditions (SMQ)	
Acute hepatic failure	Hepatotoxicity
Acute yellow liver atrophy	Intestinal varices
Anorectal varices	Intrahepatic portal hepatic venous fistula
Anorectal varices haemorrhage	Liver and small intestine transplant
Asterixis	Liver disorder
Bacterascites	Liver injury
Biliary cirrhosis	Liver operation
Biliary cirrhosis primary	Liver transplant
Biliary fibrosis	Lupoid hepatic cirrhosis
Cholestatic liver injury	Minimal hepatic encephalopathy
Chronic hepatic failure	Mixed liver injury
Coma hepatic	Nodular regenerative hyperplasia
Cryptogenic cirrhosis	Non-alcoholic steatohepatitis
Diabetic hepatopathy	Oedema due to hepatic disease
Duodenal varices	Oesophageal varices haemorrhage
Gallbladder varices	Peripancreatic varices
Gastric variceal injection	Peritoneovenous shunt
Gastric variceal ligation	Portal fibrosis
Gastric varices	Portal hypertension
Gastric varices haemorrhage	Portal hypertensive enteropathy
Hepatectomy	Portal hypertensive gastropathy
Hepatic atrophy	Portal shunt
Hepatic calcification	Portal vein cavernous transformation
Hepatic cirrhosis	Portal vein dilatation
Hepatic encephalopathy	Portopulmonary hypertension
Hepatic encephalopathy prophylaxis	Renal and liver transplant
Hepatic failure	Reye's syndrome

Hepatic fibrosis	Reynold's syndrome
Hepatic infiltration eosinophilic	Small-for-size liver syndrome
Hepatic lesion	Spider naevus
Hepatic necrosis	Splenic varices
Hepatic steatosis	Splenic varices haemorrhage
Hepatobiliary disease	Spontaneous intrahepatic portosystemic venous shunt
Hepatocellular foamy cell syndrome	Stomal varices
Hepatocellular injury	Subacute hepatic failure
Hepatopulmonary syndrome	Varices oesophageal
Hepatorenal failure	Varicose veins of abdominal wall
Hepatorenal syndrome	
Search Criteria Category= Hepatitis, non-infectious (SMQ)	
Acute graft versus host disease in liver	Hepatitis chronic active
Allergic hepatitis	Hepatitis chronic persistent
Autoimmune hepatitis	Hepatitis fulminant
Chronic graft versus host disease in liver	Hepatitis toxic
Chronic hepatitis	Ischaemic hepatitis
Graft versus host disease in liver	Liver sarcoidosis
Granulomatous liver disease	Lupus hepatitis
Hepatitis	Portal tract inflammation
Hepatitis acute	Radiation hepatitis
Hepatitis cholestatic	Steatohepatitis
Search Criteria Category= Liver related investigations, signs and symptoms (SMQ)	
5'nucleotidase increased	Hepatic pain
Alanine aminotransferase abnormal	Hepatic sequestration
Alanine aminotransferase increased	Hepatic vascular resistance increased
Ammonia abnormal	Hepatobiliary scan abnormal
Ammonia increased	Hepatomegaly
Aspartate aminotransferase abnormal	Hepatosplenomegaly
Aspartate aminotransferase increased	Hyperammonaemia
Biopsy liver abnormal	Hyperbilirubinaemia
Blood alkaline phosphatase abnormal	Hypercholia
Blood alkaline phosphatase increased	Hypertransaminasaemia
Blood bilirubin abnormal	Hypoalbuminaemia
Blood bilirubin increased	Kayser-Fleischer ring
Blood bilirubin unconjugated increased	Leucine aminopeptidase increased

Blood cholinesterase abnormal	Liver function test abnormal
Blood cholinesterase decreased	Liver induration
Bromosulphthalein test abnormal	Liver iron concentration abnormal
Child-Pugh-Turcotte score increased	Liver iron concentration increased
Computerised tomogram liver	Liver palpable
Foetor hepaticus	Liver scan abnormal
Galactose elimination capacity test abnormal	Liver tenderness
Galactose elimination capacity test decreased	Mitochondrial aspartate aminotransferase increased
Gamma-glutamyltransferase abnormal	Molar ratio of total branched-chain amino acid to tyrosine
Gamma-glutamyltransferase increased	Perihepatic discomfort
Glutamate dehydrogenase increased	Periportal oedema
Guanase increased	Peritoneal fluid protein abnormal
Haemorrhagic ascites	Peritoneal fluid protein decreased
Hepaplastin abnormal	Peritoneal fluid protein increased
Hepaplastin decreased	Pneumobilia
Hepatic artery flow decreased	Portal vein flow decreased
Hepatic congestion	Portal vein pressure increased
Hepatic enzyme abnormal	Retinol binding protein decreased
Hepatic enzyme decreased	Retrograde portal vein flow
Hepatic enzyme increased	Total bile acids increased
Hepatic fibrosis marker abnormal	Transaminases abnormal
Hepatic fibrosis marker increased	Transaminases increased
Hepatic function abnormal	Ultrasound liver abnormal
Hepatic hydrothorax	Urine bilirubin increased
Hepatic hypertrophy	Urobilinogen urine increased
Hepatic mass	
Adverse Event of Special Interest Category= Hypertension	
Search Criteria Category= Hypertension (SMQ)	
Accelerated hypertension	Hypertensive cardiomegaly
Aldosterone urine abnormal	Hypertensive cardiomyopathy
Aldosterone urine increased	Hypertensive crisis
Angiotensin I increased	Hypertensive emergency
Angiotensin II increased	Hypertensive encephalopathy
Angiotensin converting enzyme increased	Hypertensive heart disease
Blood aldosterone abnormal	Hypertensive nephropathy

Blood aldosterone increased	Labile blood pressure
Blood catecholamines abnormal	Labile hypertension
Blood catecholamines increased	Malignant hypertension
Blood pressure abnormal	Malignant hypertensive heart disease
Blood pressure ambulatory abnormal	Malignant renal hypertension
Blood pressure ambulatory increased	Maternal hypertension affecting foetus
Blood pressure diastolic abnormal	Mean arterial pressure increased
Blood pressure diastolic increased	Metabolic syndrome
Blood pressure fluctuation	Metanephrine urine abnormal
Blood pressure inadequately controlled	Metanephrine urine increased
Blood pressure increased	Neurogenic hypertension
Blood pressure management	Non-dipping
Blood pressure orthostatic abnormal	Norepinephrine abnormal
Blood pressure orthostatic increased	Norepinephrine increased
Blood pressure systolic abnormal	Normetanephrine urine increased
Blood pressure systolic increased	Orthostatic hypertension
Catecholamines urine abnormal	Pre-eclampsia
Catecholamines urine increased	Prehypertension
Diastolic hypertension	Primary hyperaldosteronism
Diuretic therapy	Procedural hypertension
Eclampsia	Pseudoaldosteronism
Ectopic aldosterone secretion	Renal hypertension
Ectopic renin secretion	Renal sympathetic nerve ablation
Endocrine hypertension	Renin abnormal
Epinephrine abnormal	Renin increased
Epinephrine increased	Renin-angiotensin system inhibition
Essential hypertension	Renovascular hypertension
Gestational hypertension	Retinopathy hypertensive
HELLP syndrome	Secondary aldosteronism
Hyperaldosteronism	Secondary hypertension
Hypertension	Systolic hypertension
Hypertension neonatal	Tyramine reaction
Hypertensive angiopathy	Withdrawal hypertension
Adverse Event of Special Interest Category= Hypokalemia	
Search Criteria Category= Hypokalemia (PT grouping) Verified that these PT terms are available	
Blood potassium	Blood potassium decreased

Blood potassium abnormal	Hypokalaemia
Adverse Event of Special Interest Category= Osteoporosis including osteoporosis-related fractures	
Search Criteria Category= SMQ Osteoporosis/osteopenia (SMQ)	
Acetabulum fracture	Internal fixation of fracture
Atypical femur fracture	Kyphoscoliosis
Body height abnormal	Kyphosis
Body height below normal	Lumbar vertebral fracture
Body height decreased	Multiple fractures
Bone density abnormal	N-telopeptide urine increased
Bone density decreased	Open reduction of fracture
Bone formation decreased	Open reduction of spinal fracture
Bone formation test	Osteocalcin increased
Bone formation test abnormal	Osteopenia
Bone loss	Osteoporosis
Bone metabolism disorder	Osteoporosis postmenopausal
Bone resorption test	Osteoporosis prophylaxis
Bone resorption test abnormal	Osteoporotic fracture
C-telopeptide	Pelvic fracture
C-telopeptide increased	Post-traumatic osteoporosis
Cervical vertebral fracture	Pubis fracture
Closed fracture manipulation	Pyridinoline urine increased
Deoxypyridinoline urine increased	Radius fracture
External fixation of fracture	Resorption bone increased
Femoral neck fracture	Rib fracture
Femur fracture	Sacroiliac fracture
Forearm fracture	Senile osteoporosis
Fracture	Spinal compression fracture
Fracture treatment	Spinal deformity
Fractured ischium	Spinal fracture
Fractured sacrum	Thoracic vertebral fracture
Hip arthroplasty	Vertebral body replacement
Hip fracture	Vertebroplasty
Hip surgery	Wrist fracture
Ilium fracture	Wrist surgery
Search Criteria Category= Verified that these PT terms are still current	
Ankle fracture	Jaw fracture

Atypical fracture	Lower limb fracture
Avulsion fracture	Open fracture
Comminuted fracture	Patella fracture
Complicated fracture	Scapula fracture
Compression fracture	Skull fracture
Facial bones fracture	Skull fractured base
Femoral neck fracture	Sternal fracture
Fibula fracture	Stress fracture
Foot fracture	Tibia fracture
Fracture displacement	Tooth fracture
Fractured coccyx	Torus fracture
Fractured skull depressed	Traumatic fracture
Hand fracture	Ulna fracture
Humerus fracture	Upper limb fracture
Impacted fracture	
Adverse Event of Special Interest Category= Rhabdomyolysis/myopathy	
Search Criteria Category= Rhabdomyolysis/myopathy (SMQ Narrow)	
Muscle necrosis	Myoglobinuria
Myoglobin blood increased	Myopathy
Myoglobin blood present	Myopathy toxic
Myoglobin urine present	Rhabdomyolysis
Myoglobinaemia	
Adverse Event of Clinical Importance Category= Sexual dysfunction, decreased libido, and impotence	
Search Criteria Category= Verified that these terms are still current	
Ejaculation disorder	Loss of libido
Erectile dysfunction	Sexual dysfunction
Libido decreased	
Adverse Event of Clinical Importance Category= Thrombocytopenia	
Search Criteria Category= Haematopoietic thrombocytopenia (SMQ Narrow)	
Megakaryocytes decreased	Platelet production decreased
Platelet count decreased	Platelet toxicity
Platelet maturation arrest	Thrombocytopenia

Adverse Event of Clinical Importance Category= Urinary Tract Infection	
Search Criteria Category= Urinary Tract Infection (PT grouping)	
Escherichia Urinary Tract Infection	Urinary Tract Infection Enterococcal
Genitourinary Tract Infection	Urinary Tract Infection Pseudomonal
Urinary Tract Infection	Urinary Tract Infection Staphylococcal
Urinary Tract Infection Bacterial	